

QFT

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Part N

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Part I

Math

1.1. Notation (Read everything or you won't be able to understand later, or come back to it later)

- (Loose) Einstein Summation notation is used in which if an index appears twice in both top and bottom, such as $\partial_\mu \partial^\mu$, it is equivalent to summing over said indices such as $\sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
 - The reason that this is loose is due to the fact that if two indices appear twice in an expression but they are both on the bottom, it is implied that the reader should raise the second appearance of the index. The following is an example so the reader doesn't get confused: $\partial_\mu \partial_\mu = \partial_\mu \partial^\mu = \sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
- The *Dirac Delta Function* is defined as

$$\delta(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0 \end{cases} \quad (1)$$

with

$$\int dx \delta(x) = 1 \quad (2)$$

This is not, in fact, a function, due to its unorthodox definition. Instead, it is a distribution (it can be defined as the limit of several different distributions, such as the *normal distribution*, a *uniform distribution*, etc.).

- $\vec{x} = x^i$, $i \in \{1, 2, 3\}$ (spatial components of x)
- The *d'Alembertian* operator $\square \equiv \partial_\mu \partial^\mu$

This operator is useful in *Lagrangians* and the *Klein-Gordon equation*.

- The *Laplacian* operator $\triangle \equiv \partial_i \partial^i$ where $i \in \{1, 2, 3\}$
- Natural Units are used in which $\hbar = c = 1$
- Both Sided Derivative: $\phi^* \vec{\partial}_\mu \phi \equiv (\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*)$
- Feynman Slash: $\not{p} \equiv p_\mu \gamma^\mu$
- Pauli matrices

$$\sigma^0 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \sigma^1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, \sigma^2 = \begin{pmatrix} & -i \\ i & \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \quad (3)$$

and $\sigma^\mu \equiv (\sigma^0, \vec{\sigma})$, $\bar{\sigma}^\mu \equiv (\sigma^0, -\vec{\sigma})$

- $$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix} \quad (4)$$

in the Weyl representation

- $\bar{\psi} \equiv \psi^\dagger \gamma^0$ where ψ is a Dirac spinor
- Left Handed Derivative

$$\begin{aligned}
& \bar{\psi}(-i\vec{\not{\partial}} - m) \\
&= -i\partial_\mu \bar{\psi}\gamma^\mu - m\bar{\psi}
\end{aligned} \tag{5}$$

(Derivative acts on the left)

- $k_\mu^2 \equiv k_\mu k^\mu$
- ψ^α left handed Weyl spinor
- $\tilde{\psi}_{\dot{\alpha}}$ right handed Weyl spinor
- $\psi = \begin{pmatrix} \psi^\alpha \\ \tilde{\chi}_{\dot{\beta}} \end{pmatrix}$ Dirac spinor (Representation of $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$)

All of these expressions regarding spinors will be explained or will quickly become evident in Section 2, due to the complexity of their structure.

1.2. Integrals that are useful

$$\bullet \quad \int dx f(x)\delta(x^2 - a^2) = \frac{1}{2|a|}(f(a) + f(-a)) \tag{6}$$

$$\bullet \quad \int d^3k \theta(k^0)\delta(k_\mu^2 - m^2) = \frac{1}{2\omega_k}[\theta(\omega_k) + \theta(-\omega_k)] = \frac{1}{2\omega_k} \tag{7}$$

where $\omega_k = \sqrt{\vec{k}^2 + m^2}$

$$\bullet \quad \int \frac{d^4k}{(2\pi)^4} \exp(ik(x - y)) = \delta^4(x - y) \tag{8}$$

$$\bullet \quad \int \frac{d^Nk}{(2\pi)^N} \exp(ik(x - y)) = \delta^N(x - y) \tag{9}$$

$$\bullet \quad \int dx \exp\left(-\frac{1}{2}ax^2 + Jx\right) = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \exp\left(\frac{J^2}{2a}\right) \tag{10}$$

1.3. Group Theory

Definition 1.3.1: A Group G is a set of elements $\{g_i\} \in G$ and a rule $g_i \times g_j = g_k$ giving how they “multiply.”

Group Theory Axioms

- For each element $g_i \in G$ and $g_j \in G$, their product $g_i \times g_j = g_k \in G$ is an element of G
- Products must be Associative:

$$(g_i \times g_j) \times g_k = g_i \times (g_j \times g_k) \tag{11}$$

- There must be an Identity:

$$\mathbb{1} \times g_i = g_i \times \mathbb{1} = g_i \quad (12)$$

- Elements have an Inverse:

$$g_i^{-1} \times g_i = \mathbb{1} \quad (13)$$

Definition 1.3.2 (Representation): An embedding of g_i as operators in a vector space

- Matrices are finite-dimensional representations of a group

Definition 1.3.3 (Faithful Representation): Each element in a group gets its own matrix

1.3.1. Lie Groups

- A Group G whose elements can be written as $g_i = \exp(ic_i^g \lambda_i)$ for some number c_i^g and some matrix λ_i (known as a *generator*)
- In an algebra, you can add and multiply
- In a Group, you have to follow the previously mentioned rules (just multiplication is allowed)

Definition 1.3.1.1 (Lie Group): A *Lie Group* is a type of group with an infinite number of elements and a finite number of generators.

Definition 1.3.1.2 (Lie Algebra): The Generators of a *Lie group* form a *Lie Algebra*.

Example 1.3.1.1 (Lie Algebras):

- QED : $U(1)$ (unitarity)
- Weak: $SU(2)$ (special unitarity)
- Strong: $SU(3)$
- Lorent Group: $O(1, 3)$ (orthogonal and preserving (1,3) metric) (see §2.2)

Part II

Scalars

- Scalar Field Theory is a theory of quantum fields. It is mostly used as a toy theory to base other theories off of, since out of the many *many* particles that exist (dozens counting baryons), there is an entirety of 1 scalar particle (the Higgs boson). Therefore, Scalar Field Theory primarily is only useful to base other models of the universe off of it as opposed to modeling the universe after it.

2.1. Classical Field Theory

- The idea behind **classical field theory** is to utilize the idea of Lagrangians that will be presented in *Lagrangians* and in *Classical Symmetries* to obtain a “toy theory” without the quantum effects that will later appear such as Loops in diagrams, etc. In practice, this means that it is essentially the *Lagrangian Formalism* of Classical Mechanics where *position* and *momentum* coordinates (p, q) are replaced by fields (ϕ, π) , as seen before in *Lagrangians* without having been explicitly stated.
- Classical Field Theory, much like *Classical Mechanics*, functions as the unquantized equivalent to Quantum Field Theory. Quantization does not require much to achieve as simply putting quantization conditions on the operators being used achieves it with relative ease.

Example 2.1.1 (Klein Gordon Equation):

- A very standard equation for a field in Classical Field theory is the *Klein Gordon Equation*. (This, in fact, describes a *free* field in which there are no interactions, allowing for a very good “toy theory” to use and potentially apply later).

$$(\square + m^2)\phi = 0 \quad (14)$$

or

$$(\partial^2 + m^2)\phi = 0 \quad (15)$$

is the *Klein Gordon Equation*.

2.1.1. Lagrangians

Definition 2.1.1.1 (Lagrangian): The *Lagrangian* (Density) $\mathcal{L}$, similar to the *Hamiltonian* H (or $\mathcal{H}$ for Hamiltonian density, defined as $H = \int d^3x \mathcal{H}$) as they both can be used to describe the evolution of a system. In Field Theories, densities are used more commonly. The Density of either quantity, as stated before, is a more useful quantity in Field theories due to Lorentz Invariance and other such things.

Definition 2.1.1.2 (Action): Action S is defined as

$$S = \int dt L = \int d^4x \mathcal{L} \quad (16)$$

which should be familiar from studying classical mechanics (Lagrangian mechanics specifically). This quantity is minimized in order to obtain the evolution of a system (**Equations of Motion**).

Theorem 2.1.1.1 (Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu \phi]$, the system must satisfy the **Euler Lagrange Equations**:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0 \quad (17)$$

A differential equation which ends up being extremely useful in physics for describing the time evolution of a system.

Proof: We aim to equate the variation of the action S with respect to the field ϕ to zero. That is, we require

$$\frac{\delta S}{\delta \phi} = 0 \quad (18)$$

Note that

$$S = \int dt L = \int d^4x \mathcal{L}[\phi, \partial_\mu \phi] \quad (19)$$

in order to obtain δS , we must perform a transformation $\phi \rightarrow \phi + \delta \phi$ in the Lagrangian. This provides

$$\delta S = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right) \quad (20)$$

Via the product rule and the fact that infinitesimal variations commute with partial derivatives¹, we have that

¹Look into variational calculus if you want to learn more about this

$$\begin{aligned}
\delta S &= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \\
\delta S &= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \right) \\
\delta S &= \int d^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \right)
\end{aligned} \tag{21}$$

Now we need to prove that the final term (the one without the shared common factor of $\delta \phi$) vanishes as our integration bounds become infinite (as we are integrating over all spacetime of course). Since this term is a total derivative of some quantity that we will define to be

$$A^\mu \equiv \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \tag{22}$$

(Something similar with the form of $\frac{A^\mu}{\delta \alpha}$ is a physically important quantity but this A^μ is not of any significance besides being useful for math). With the derivative being of course $\partial_\mu A^\mu$. The integral of this quantity is

$$\int d^4x \partial_\mu A^\mu \tag{23}$$

and by the **4 dimensional divergence theorem**, we obtain

$$\int_V d^4x \partial_\mu A^\mu = \oint_{\partial V} d^3\sigma_\mu A^\mu \tag{24}$$

This must vanish, given that the $\delta \phi$ is forced to vanish at the boundaries by the principle of least action. Therefore, we have that

$$\frac{\delta S}{\delta \phi} = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0 \tag{25}$$

For the integral to vanish, the integrand must vanish, therefore

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0} \tag{26}$$

■

Theorem 2.1.1.2 (Extended Euler Lagrange Equations): In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu \phi, \partial_\nu \partial_\mu \phi, \dots]$, the system must satisfy the **Extended Euler Lagrange Equations**:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} - \dots = 0 \quad (27)$$

(This was actually a problem in the Schwartz QFT textbook, so if you are reading that, potentially skip this part.)

Proof: We begin in a similar way to the proof to the *Euler-Lagrange equations*.

$$\delta S = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) + \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\nu \partial_\mu \phi) + \dots \right) \quad (28)$$

We then utilize the product rule (or the reverse of it) to obtain

$$\begin{aligned} \partial_\nu \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta \phi \right) &= \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta \phi \\ &+ \partial_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\mu \phi) \\ &+ \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\nu \phi) \\ &+ \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\nu \partial_\mu \phi) \end{aligned} \quad (29)$$

We can rewrite $\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\nu \phi)$ and $\partial_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta (\partial_\mu \phi)$ via similar methods and we can continue to do so with more terms forming a pattern of the n th (with n partial derivatives) term has a coefficient of $(-1)^n$ (this method similar to the *principle of inclusion-exclusion*)

$$\begin{aligned} \delta S &= \int d^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} - \dots \right) \delta \phi \right. \\ &+ \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) + \partial_\mu \left(\partial_\nu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta \phi \right) + \partial_\nu \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} \delta \phi \right) \\ &\left. + \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \partial_\nu \phi)} \delta \phi \right) + \dots \right) \end{aligned} \quad (30)$$

Every term besides those with the shared factor of $\delta \phi$ disappears due to similar reasons mentioned in the proof of the *Euler-Lagrange equations* (i.e. they are total derivatives) and therefore, since we want $\frac{\delta S}{\delta \phi} = 0$, we have that

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\nu \partial_\mu \phi)} - \dots = 0} \quad (31)$$

■

Example 2.1.1.1 (Lagrangian): An example of a Lagrangian that can apply to some general scalar field ϕ^2 is the Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial_\mu \phi) - V[\phi] \quad (32)$$

This would give us the equation of motion

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} &= 0 \\ -V'[\phi] - \square \phi &= 0 \end{aligned} \quad (33)$$

so

$$(\square + V'[\phi])\phi = 0 \quad (34)$$

which is related to the *Klein-Gordon equation* $(\square + m^2)\phi = 0$.

- One can split the terms of a Lagrangian $\mathcal{L}$ into *Kinetic* terms and *Interacting* terms.
 - *Kinetic* terms of a Lagrangian are ones which have at most 2 fields in it

$$\mathcal{L}_{\text{kin}} \supset \frac{1}{2}\phi\square\phi, \bar{\psi}\not{\partial}\psi, \frac{1}{4}F_{\mu\nu}^2, \frac{1}{2}m^2\phi^2, \dots \quad (35)$$

- Interacting terms are terms with more than 2 fields

$$\mathcal{L}_{\text{int}} \supset \lambda\phi^3, g\bar{\psi}\not{A}\psi, g\partial_\mu\phi A_\mu\phi^*, \dots \quad (36)$$

Note: Interacting terms in the lagrangian all have a factor λ or g associated with them. This is due to the fact that it is a coupling constant related to the interaction.

2.1.2. Classical Symmetries

2.1.2.1. Noether's Theorem

- This theorem is the core in the utilizing symmetries in classical and non-classical field theory.

²That is, it doesn't apply universally, but just in this circumstance

Theorem 2.1.2.1.1 (Noether's Theorem): If a Lagrangian has a continuous symmetry (i.e. it is parameterizable by some continuous α), there exists a conserved current J^μ associated with said symmetry when the equations of motion of the Lagrangian are satisfied, i.e. the **Euler-Lagrange** equations. In other words, *symmetries imply conservation laws*.

Proof: The Proof for *Noether's Theorem* is fairly short and clever.

We define $\mathcal{L}$ as the Lagrangian of the system, ϕ_n as the field(s) involved in the system, and α as the parameter of the continuous symmetry.

Similar to the derivation of the *Euler-Lagrange* equations,

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left(\left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \right) \quad (37)$$

This expression must be equivalent to zero due to α parametrizing a symmetry (i.e. the Lagrangian is invariant wrt some variation in the parameter). It is necessary to define a

$$J^\mu \equiv \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \quad (38)$$

Since we are only considering the case in which the equations of motion hold true, (i.e. the *Euler-Lagrange equations* are true), we have that

$$\partial_\mu J^\mu = \partial_\mu \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = \sum_n \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = 0 \quad (39)$$

Therefore due to some continuous symmetry parametrized by some α , there exists a conserved current J^μ . ■

- The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ is the conserved current for spacetime translation.

Definition 2.1.2.1.1 (Energy Momentum Tensor): The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ for some Lagrangian $\mathcal{L}[\phi_1, \partial_\mu \phi_1, \dots, \phi_n, \partial_\mu \phi_n]$ and some spacetime translation $x^\nu \rightarrow x^\nu + \xi^\nu$ is defined as

$$\mathcal{T}_{\mu\nu} \equiv \sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \quad (40)$$

Derivation below.

Starting with the translation

$$\phi(x) \rightarrow \phi(x + \xi) = \phi(x) + \partial_\nu \xi^\nu \phi(x) + \dots \quad (41)$$

This provides $\frac{\delta\phi}{\delta\xi^\nu} = \partial_\nu\phi$ which applies universally regardless of the field ϕ (and in fact regardless of the physical object, as it applies to the Lagrangian $\mathcal{L}$, a consequence of the fact that $\delta S = 0$).

We then have that

$$\begin{aligned}\frac{\delta\mathcal{L}}{\delta\xi^\nu} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)} \frac{\delta\phi_n}{\delta\xi^\nu} \right) \\ &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)} \partial_\nu\phi_n \right) \\ \partial_\nu\mathcal{L} &= \partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)} \partial_\nu\phi_n \right)\end{aligned}\tag{42}$$

We then have that

$$\partial_\mu \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)} \partial_\nu\phi_n - g_{\mu\nu}\mathcal{L} \right) = 0\tag{43}$$

Where we then obtain

$$\mathcal{T}_{\mu\nu} \equiv \left(\sum_n \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)} \partial_\nu\phi_n - g_{\mu\nu}\mathcal{L} \right)\tag{44}$$

and $\partial_\mu\mathcal{T}_{\mu\nu} = 0$.

- Energy density:

$$\mathcal{E} = \mathcal{T}_{00} = \sum_n \frac{\partial\mathcal{L}}{\partial(\dot{\phi}_n)} \dot{\phi}_n - \mathcal{L}\tag{45}$$

- $\dot{\phi}_n = \partial_t\phi_n$
- In the case of Lagrangians with $\dot{\phi} = \pi = \frac{\partial\mathcal{L}}{\partial\phi}$, $\mathcal{E} = \mathcal{H}$ due to the *Legendre Transform*

Definition 2.1.2.1.2 (4-Momentum): In terms of the energy-momentum tensor, 4-momentum p^μ is defined as

$$p^\nu \equiv \int d^3x \mathcal{T}^{0\nu} = (E, \vec{p})\tag{46}$$

2.1.2.2. More information on currents

- A current can refer to a Noether current (such as the ones previously seen)
- They can be external, such as a moving electron in a wire
- They can be sources for fields
 - They are never fields with kinetic terms
- They can be placeholders in Lagrangians

Example 2.1.2.2.1 (Currents in Lagrangians): In

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\square\phi - ieA_\mu(\tilde{\phi}\vec{\partial}_\mu\phi), \quad (47)$$

we can write it as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\square\phi - A_\mu J^\mu \quad (48)$$

- Clearly, you can see how currents are used to represent quantities in lagrangians that we do not care for.

2.2. Scalar Field Theory

- Scalar Field Theory is fairly similar to Classical Field Theory with the exception that it is quantized via the methods that will be presented shortly. This ensures that it obeys the laws of quantum mechanics, while still being a “Toy Theory” due to the simplicity of the objects involved in it ((potentially) complex fields ϕ as opposed to *vectors* A_μ or *spinors* ψ which are closer to real, physical particles).

2.2.1. Quantization

- As seen in ‘s Quantum Field Theory Notes, one could try to naively try to quantize the Schrodinger Equation (via the use of a series to have the equation be on-shell), which would bring rise to some problems

$$\begin{aligned} H|\psi\rangle &= i\partial_t|\psi\rangle \\ H &= \sqrt{\hat{p}^2 + m^2} \end{aligned} \quad (49)$$

2.2.1.1. Harmonic Oscillator/First Quantization

- The standard Hamiltonian for a simple harmonic oscillator is

$$H = \frac{1}{2} \frac{p^2}{m} + \frac{1}{2} m\omega^2 x^2 \quad (50)$$

. The relevance to quantization will become apparent.

Definition 2.2.1.1.1 (First Quantization): The idea behind first quantization is to apply the commutation relation $[\hat{x}, \hat{p}] = i$ to a system. This commutation relation is known as the *canonical commutation relation*, and it can be used to obtain quantum mechanics from a classical theory (which we are doing now).

- Utilizing a change of variables to a and $a^\dagger$, one can derive an alternative form for the Hamiltonian

$$a \equiv \sqrt{\frac{m\omega}{2}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger \equiv \sqrt{\frac{m\omega}{2}} \left(x - \frac{ip}{m\omega} \right) \quad (51)$$

$$\triangleright \quad H = \omega \left(a^\dagger a + \frac{1}{2} \right) \quad (52)$$

is written in a very suggestive form. Such $a^\dagger$ and a can be thought of as “exciting” the harmonic oscillator, as you may know from quantum mechanics.

- The eigenstates of this Hamiltonian are eigenstates of $\hat{N} \equiv a^\dagger a$
- Notice that this hamiltonian’s eigenvalues (the energies of the eigenstates) are discrete. This is a result from quantum mechanics (or, rather, defining quantum mechanics). Such a result is the reason behind both the name *Quantum Mechanics* and the idea of *First Quantization*.
- $\hat{N}$ operator:
 - $\hat{N}|n\rangle = n|n\rangle$
 - $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$
 - $a|n\rangle = \sqrt{n}|n-1\rangle$

2.2.1.2. Heisenberg Picture

- $i \frac{d}{dt} a = [a, H] = \left[a, a \left(a^\dagger a + \frac{1}{2} \right) \right] = \omega (a a^\dagger a - a^\dagger a a) = \omega [a, a^\dagger] a = \omega a \quad (53)$
- Solution is

$$a(t) = e^{-i\omega t} a(0) \quad (54)$$

2.2.1.3. Relativistic Fields

- A common relativistic equation (heavy foreshadowing) for a (free) field is the equation

$$\square \phi = 0. \quad (55)$$

- Solution to this is

$$\phi(x, t) = \int \frac{d^3 p}{(2\pi)^3} [a_p(t) e^{i\vec{p} \cdot \vec{x}} + \tilde{a}_p(t) e^{-i\vec{p} \cdot \vec{x}}] \quad (56)$$

which is written in a fairly suggestive manner

2.2.1.4. 2nd Quantization

Definition 2.2.1.4.1 (Second Quantization): To obtain 2nd quantization (and hence quantum field theory), one can use an annihilation operator a_p and a creation operator $a_p^\dagger$ which exist per wave number as opposed to being universal since now, we utilize infinite harmonic oscillators which exist per p

- Free Hamiltonian is obtained via integration over these

$$H = \int \frac{d^3 p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} \right) \quad (57)$$

with $\omega_p = |\vec{p}|$

- 2nd Quantization new things:
 - Each (of the infinitely many) point of momentum $\vec{p}$ gets its own harmonic oscillator (which we integrate over to obtain our result)
 - The n th excitation of the harmonic oscillator $\vec{p}$ as being a state with n particles

2.2.1.5. Fock Space

- Instead of Hilbert space $\mathcal{H}$, like in Quantum Mechanics/First Quantization, we have a *Fock Space* $\mathcal{F}$.
- A Fock space $\mathcal{F}$ is the direct sum of Hilbert spaces $\mathcal{H}_n$, that is,

$$\mathcal{F} = \bigoplus_n \mathcal{H}_n \quad (58)$$

- States in $\mathcal{H}_n$ are linear combinations of $\{|p_1^\mu \dots p_n^\mu\rangle\}$
 - Since we are using *Fock Space*, this means that there can be however many particles due to the direct sum over Hilbert Spaces

Note (Momentum Notation): $p^0 = \sqrt{\vec{p}^2 + m^2}$ so $|p\rangle = |\vec{p}\rangle = |p^\mu\rangle$ due to no loss of generality (this only works if it is on-shell).

2.2.1.6. Field Expansion

- Previously, we noted that $[a, a^\dagger] = 1$. This generalizes to $[a_k, a_p^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$
- $a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$
- $\langle 0|0\rangle = 1$ since it is the vacuum ground state
 - $\langle p|k\rangle = 2\sqrt{\omega_p \omega_k} \langle 0|a_p a_o^\dagger|0\rangle = 2\omega_p (2\pi)^3 \delta^3(\vec{p} - \vec{k})$
- Identity operator

$$\mathbb{1} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle \langle p| \quad (59)$$

- Free field definition

$$\phi_0(\vec{x}) \equiv \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}) \quad (60)$$

which is why I said $\phi(\vec{x}, t)$ from 4.1.2 was written in a suggestive manner (although this is still technically just a definition)

Note (Momentum state inner product with free field):

$$\begin{aligned}
\langle \vec{p} | \phi_0(\vec{x}) | 0 \rangle &= \langle 0 | \sqrt{2\omega_p} a_p \int \frac{d^3k}{(2\pi)^3} (a_k e^{i\vec{k}\cdot\vec{x}} + a_k^\dagger e^{-i\vec{k}\cdot\vec{x}}) | 0 \rangle \\
&= \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_p}{\omega_k}} \left[e^{i\vec{k}\cdot\vec{x}} \langle 0 | a_p a_k | 0 \rangle + e^{-i\vec{k}\cdot\vec{x}} \langle 0 | a_p a_k^\dagger | 0 \rangle \right] = e^{-i\vec{p}\cdot\vec{x}} \\
&= e^{-i\vec{p}\cdot\vec{x}}
\end{aligned} \tag{61}$$

This is the expected result of $\langle \vec{p} | \vec{x} \rangle$, and so we conclude $\phi_0(\vec{x}) | 0 \rangle = | \vec{x} \rangle$

2.2.1.7. Time Dependence

- Because of Heisenberg's equations of motion, in the Heisenberg perspective, $a_p(t)$ and $a_p^\dagger(t)$ are time dependent
 - $a_p(t) = e^{-i\omega_p t} a_p$, $a_p^\dagger(t) = e^{i\omega_p t} a_p^\dagger$
- Heisenberg equations of motion

$$\begin{aligned}
\text{▸ } [H_0, \phi_0(\vec{x}, t)] &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\omega_p \left(a^\dagger a_p + \frac{1}{2} \right), a_k e^{-i\vec{k}\cdot\vec{x}} + a_k^\dagger e^{i\vec{k}\cdot\vec{x}} \right] \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [-\omega_p a_p e^{-ipx} + \omega_p a_p^\dagger e^{ipx}] \\
&= -i\partial_t \phi_0(\vec{x}, t)
\end{aligned} \tag{62}$$

- Should have that, for any field ϕ ,

$$[H, \phi(\vec{x}, t)] = i\partial_t \phi(\vec{x}, t) \tag{63}$$

$$- \phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p(t) e^{-ipx} + a_p^\dagger(t) e^{ipx}] \tag{64}$$

is the solution to the *Heisenberg equations of motion* for ϕ , our field which does not necessarily have to be free in this case

2.3. Cross Sections

- Cross sections are how physicists analyze collider data (such as the exabytes of LHC data that is collected every year)

Important Result of this Section

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2 \tag{65}$$

and the Born approximation

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} [\tilde{V}(\vec{k})]^2 \tag{66}$$

- Cross sections used to make sense of particle accelerator data

- Cross sectional area:

$$\begin{aligned}\sigma &= \frac{\# \text{ particles}}{\text{time} \times \text{number density} \times \text{velocity of beam}} \\ &= \frac{1}{T} \frac{1}{\Phi} N, \Phi = \# \text{ density} \times \text{beam vel.} = \frac{1}{V} \times |\vec{v}|\end{aligned}\quad (67)$$

- More useful/natural is differential cross section

$$\frac{d\sigma}{d\Omega} \quad (68)$$

for some angle/change in angle Ω .

- Info about shape of object with this

2.3.0.1. QM

- Generalize to Differential Cross Sections due to their changing based on angle in colliders
- In QM, there is a probability of scattering

$$P = \frac{N}{N_{\text{inc}}} \quad (69)$$

with N_{inc} number of incident particles

- Cross section

$$d\sigma = \frac{1}{T} \frac{1}{\Phi} dP \quad (70)$$

(normalize flux)

- Differential scattering number

$$dN = L \times d\sigma \quad (71)$$

- L is defined as the *Luminosity* through this equation

Note (Differential solid angle): The cross section $d\sigma$ is generally measured via $\frac{d\sigma}{d\Omega}$ as seen earlier. This $d\Omega$ is equivalent to

$$d\Omega = \sin \theta d\theta d\phi = -d \cos \theta d\phi \quad (72)$$

2.3.1. Cross Section to Matrix Element

- $2 \rightarrow N$ Particle Interaction:

$$p_1 + p_2 \rightarrow \{p_j\} \quad (73)$$

$$\begin{aligned}\Phi &= \frac{|\vec{v}_1 - \vec{v}_2|}{V} \\ \Rightarrow d\sigma &= \frac{V}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} dP\end{aligned}\quad (74)$$

- Have that

$$dP = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle} d\Pi \quad (75)$$

► with

$$d\Pi = \prod_j \frac{N}{(2\pi)^3} d^3p_j \quad (76)$$

►

$$\int d\Pi = 1 \quad (77)$$

since

$$\int \frac{dP}{(2\pi)} = \frac{1}{V} \quad (78)$$

•

$$[a_p, a_q^\dagger] = (2\pi)^3 \delta^3(p - q) \quad (79)$$

► Implies that $\langle p|p\rangle = (2\pi)^3 \delta^3(0)$

► Normally $\delta^3(0)$ is infinite but

$$\begin{aligned} (2\pi)^3 \delta^3(p) &= \int_V d^3x e^{i\vec{p}\cdot\vec{x}} \\ \Rightarrow \delta^3(0) &= \frac{V}{(2\pi)^3} \end{aligned} \quad (80)$$

$$\text{and } \delta^4(0) = \frac{TV}{(2\pi)^4}$$

•

$$\langle p|p\rangle = 2\omega_p V = 2E_p V \quad (81)$$

• Since

$$|i\rangle = |p_1\rangle |p_2\rangle \quad (82)$$

and

$$|f\rangle = \prod_j |p_j\rangle \quad (83)$$

► It is implied that

$$\langle i|i\rangle = (2E_1 V)(2E_2 V) \quad (84)$$

► It is also implied that

$$\langle f|f\rangle = \prod_j 2E_j V \quad (85)$$

2.3.2. Actually Calculating the S-matrix (perturbatively)

• Notice that the *first order* of a S-matrix is just $\mathbb{1}$ since that is the case in which nothing happens

• Write $S = \mathbb{1} + i\mathcal{T}$

► $\mathcal{T} \equiv$ **transfer matrix**

- $\mathcal{T}$ implies momentum conservation since it vanishes unless $|i\rangle$ and $|f\rangle$ have the same 4-momentum
- $\mathcal{T} = (2\pi)^4 \delta^4(\Sigma p) \mathcal{M}$

2.3.2.1. Matrix Elements

- $\langle f|S - \mathbb{1}|i\rangle = i(2\pi)^4 \delta^4(\Sigma P) \langle f|\mathcal{M}|i\rangle$
 - $$\begin{aligned} |\langle f|S|i\rangle|^2 &= \delta^4(0) \delta^4(\Sigma p) (2\pi)^8 |\langle f|M|i\rangle|^2 \\ &= \delta^4(\Sigma p) TV (2\pi)^4 |\mathcal{M}|^2 \end{aligned} \quad (86)$$

2.3.3. Lorentz Invariant Phase Space

- We denote the differential $d\Pi_{\text{LIPS}}$ as being defined as the product over final momenta differentials times delta function factors, that is

$$d\Pi_{\text{LIPS}} \equiv \prod_{\text{final } j} \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_{p_j}} (2\pi)^4 \delta^4(\Sigma p) \quad (87)$$

- Utilizing this, (as well as the derived S-matrix elements from earlier) we can write the differential probability dP via

$$\begin{aligned} dP &= \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle \langle i|i\rangle} d\Pi = \frac{(2\pi)^4 \delta^4(\Sigma p) \times TV}{2E_1 V \times 2E_2 V} \\ &\times \frac{1}{\Pi_j (2E_j V)} |\mathcal{M}|^2 \prod_j \frac{d^3 p_j}{(2\pi)^3} V \\ &= T \frac{1}{2E_1 \times 2E_2} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \end{aligned} \quad (88)$$

- This easily gives us the cross section for $2 \rightarrow j$ decay:

$$d\sigma = \frac{1}{(2E_1)(2E_2)|\vec{v}_1 - \vec{v}_2|} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (89)$$

2.3.4. Particle Decay Rate

- This is another useful quantity in analyzing accelerator data denoted by $d\Gamma$
- The definition for this decay rate is $d\Gamma \equiv \frac{1}{T} dP$ (probability of decay in certain amount of time)
 - Specifically it is the probability that a particle of momentum p_1 decays into particles of momenta p_j in a time T
 - S-matrix theory cannot be used to describe these decays (?)
- $d\Gamma = \frac{1}{2E_1} |\mathcal{M}|^2 d\Pi_{\text{LIPS}}$

2.3.5. Cross Section formula

- Consider $2 \rightarrow 2$ Scattering
 - This means that the momenta are $p_1 + p_2 \rightarrow p_3 + p_4$
- In the center of mass frame, the net momentum for each is zero. Therefore $\mathbf{p}_1 = -\mathbf{p}_2, \mathbf{p}_3 = -\mathbf{p}_4$ which implies that $E_1 + E_2 = E_3 + E_4 \equiv E_{\text{CM}}$

- Then, for the LIPS differential, we get that

$$d\Pi_{\text{LIPS}} = (2\pi)^4 \delta^4(\Sigma p) \frac{d^3 p_3}{(2\pi)^3} \frac{1}{2E_3} \frac{d^3 p_4}{(2\pi)^3} \frac{1}{2E_4} \quad (90)$$

- This is equivalent to the integral

$$d\Pi_{\text{LIPS}} = \frac{1}{16\pi^2} d\Omega \int dp_f \frac{p_f^2}{E_3} \frac{1}{E_4} \delta(E_3 + E_4 - E_{\text{CM}}) \quad (91)$$

(I don't understand how this is true)

- Where $p_f = |\mathbf{p}_3| = |\mathbf{p}_4|$ and $E_3 = \sqrt{m_3^2 + p_f^2}$, $E_4 = \sqrt{m_4^2 + p_f^2}$
- Notice that the energies are dependent on the final momentum p_f , and so this integral is in fact, not trivial.
- Now we conduct a change of variables from p_f to $x(p_f) \equiv E_3(p_f) + E_4(p_f) - E_{\text{CM}}$
- Calculating the change of variables, we get

$$\frac{dx}{dp_f} = \frac{d}{dp_f}(E_3 + E_4 - E_{\text{CM}}) = \frac{p_f}{E_3} + \frac{p_f}{E_4} = \frac{E_3 + E_4}{E_3 E_4} p_f \quad (92)$$

- We also have that

$$E_3 + E_4 = E_{\text{CM}} \quad (93)$$

- Then,

$$\begin{aligned} d\Pi_{\text{LIPS}} &= \frac{1}{16\pi^2} d\Omega \int_{m_3+m_4-E_{\text{CM}}}^{\infty} dx \frac{p_f}{E_{\text{CM}}} \delta(x) \\ &= \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} \theta(E_{\text{CM}} - m_3 - m_4) \end{aligned} \quad (94)$$

where $\theta(x)$ is the Heaviside step function due to the integral of the delta function.

- Putting this all together, we obtain that

$$d\sigma = \frac{1}{(2E_1)(2E_2)|\mathbf{v}_1 - \mathbf{v}_2|} \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} |\mathcal{M}|^2 \theta(E_{\text{CM}} - m_3 - m_4) \quad (95)$$

- Using

$$|\mathbf{v}_1 - \mathbf{v}_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = |\mathbf{p}_i| \frac{E_{\text{CM}}}{E_1 E_2} \quad (96)$$

, we obtain that, under the condition that the masses are equivalent,

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2 \quad (97)$$

as desired, allowing for computation of cross sections for simple scattering based off of quantum field theory predictions for $|\mathcal{M}|^2$

2.3.6. Born Approximation

- The Born approximation is a special case where a scalar electron ϕ_e with mass m_e scatters off of a scalar proton ϕ_p with mass m_p
 - It is considered the nonrelativistic limit
- I will not derive the born approximation, however, it is given by the formula

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} \equiv \frac{m_e^2}{4\pi^2} [\tilde{V}(\vec{k})]^2 \quad (98)$$

where $\tilde{V}$ is the fourier transformed potential/the potential in fourier space.

$$\tilde{V}(\vec{K}) = \int d^3x e^{-i\vec{k}\cdot\vec{x}} V(\vec{x}) \quad (99)$$

Note (Dimension of Matrix Elements):

- Utilizing the formula

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2 \quad (100)$$

, we can fairly clearly see that since $[\frac{d\sigma}{d\Omega}] = -2$ and that $[E_{\text{CM}}^2] = -2$, $\mathcal{M}$ must be dimensionless which should logically make sense.

2.4. Propagators (TODO)

2.4.1. Green's Functions

Definition 2.4.1.1 (Green's Functions): A Green's function is a function of the form A^{-1} where A is an operator. A standard way to evaluate these green function operators is via a Fourier transform into k -space³. This is a type of *propagator*.

Definition 2.4.1.2 (Propagator): Defines or is used to define how a field propagates via applying the propagator Π to some other field or current. Can be used perturbatively in order to describe more complicated fields.

- An example of a propagator is the 2-point Green's function $\Pi = -\frac{1}{\square}$ which is defined by

$$\square_x \Pi(x, y) = -\delta^4(x - y) \quad (101)$$

- With $\square_x = g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu}$ This gives the solution (via the Fourier transform), that

$$\Pi(x, y) = \int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \frac{1}{k^2} \quad (102)$$

³ k -space is effectively momentum space due to the relation $p = \hbar k$ and the fact that we are using natural units

Note (Fourier Transform of Operators): Under *Fourier Transforms*, certain operators are effectively interchangeable with some counterpart that often simplifies calculations. An example of this is how $\Pi(x, y)$ is interchangeable with $\frac{1}{k^2}$.

Example 2.4.1.1 (Delta Functions): For an instance of this occurrence, we can utilize delta functions.

Note: The Fourier transform of any delta function is

$$\begin{aligned}\tilde{\delta}(\vec{k}) &= 1 \\ \Rightarrow \delta(\vec{x}) &= \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}}\end{aligned}\tag{103}$$

$$\begin{aligned}\Delta^n \delta^3(\vec{x}) &= \int \frac{d^3k}{(2\pi)^3} \Delta^n e^{i\vec{k}\cdot\vec{x}} \\ &= \int \frac{d^3k}{(2\pi)^3} (-\vec{k}^2)^n e^{i\vec{k}\cdot\vec{x}}\end{aligned}\tag{104}$$

- $\Delta^n \leftrightarrow (-\vec{k}^2)^n$ (this means that in a *Fourier Transform*, these are interchangeable).
- Additionally, we can do the same for the relativistic $\square = \partial_\mu \partial^\mu$

$$\square^n (\delta^4(x)) = \int \frac{d^4k}{(2\pi)^4} \square^n e^{ik\cdot x} = \int \frac{d^4k}{(2\pi)^4} (-k^2)^n e^{ik\cdot x}\tag{105}$$

- Have that $[\widetilde{\square^n \delta^4}](k) = (-k^2)^n$ and $[\widetilde{\Delta^n \delta^3}](k) = (-\vec{k}^2)^n$

2.4.2. Perturbation Theory

Definition 2.4.2.1 (Perturbation Theory): This topic is introduced in Quantum mechanics (and often before-hand) where we consider an initial state that we can solve and then higher order terms that we continue to be able to solve. Perturbation theory only works when the leading term is the “dominant” force in play. Often, we can utilize an initial term which is *non-interacting* and extra terms that are *interacting*.

Example 2.4.2.1 (Gravity): This example was provided from the Schwartz textbook⁴
We start with the Lagrangian

$$\mathcal{L} = -\frac{1}{2}h\Box h + \frac{1}{3}\lambda h^3 + Jh \quad (106)$$

where J is some *current* and h is our field. From the *Euler-Lagrange equations*, our equation of motion is

$$\Box h - \lambda h^2 - J = 0 \quad (107)$$

First order approx ($\lambda = 0$): $h_0 = \frac{1}{\Box}J$. We let $h = h_0 + h_1$.

$$\Box(h_0 + h_1) - \lambda(h_0 + h_1)^2 - J = 0 \quad (108)$$

implies

$$\begin{aligned} \Box h_1 &= \lambda h_0^2 + \mathcal{O}(\lambda^2) \\ h &= \frac{1}{\Box}J + \lambda \frac{1}{\Box} \left(\left(\frac{1}{\Box}J \right) \left(\frac{1}{\Box}J \right) \right) + \mathcal{O}(\lambda^2) \end{aligned} \quad (109)$$

which we now have in terms of our Green's Functions.

Note: In fact, something being defined as $\frac{1}{\Box}J$ is not that uncommon, as it also happens with the Electromagnetic field.

2.4.3. OFPT

- For **Old Fashioned Perturbation Theory** we utilize perturbation by setting the Hamiltonian to

$$H = H_0 + V \quad (110)$$

- The idea is that we have some $|\phi\rangle$ and some $|\psi\rangle$ such that

$$H_0|\phi\rangle = E|\phi\rangle \quad (111)$$

and

$$H|\psi\rangle = E|\psi\rangle \quad (112)$$

- From this we can derive

$$|\psi\rangle = |\phi\rangle + \frac{1}{E - H_0}V|\phi\rangle, \quad (113)$$

the *Lippmann-Schwinger* equation

- This allows us to define the propagator⁵

⁴This is the primary textbook that I am learning Quantum Field Theory through, as seen in the credits section.

⁵In fact, this is not well-defined, but the limit

$$\Pi_{\text{LS}} \equiv \frac{1}{E - H_0} \quad (115)$$

- We define an operator T as

$$\begin{aligned} V|\psi\rangle &= T|\phi\rangle \\ |\psi\rangle &= |\phi\rangle + \Pi_{\text{LS}}T|\phi\rangle \\ V|\psi\rangle &= V|\phi\rangle + V\Pi_{\text{LS}}T|\phi\rangle \end{aligned} \quad (116)$$

- Note that $V|\psi\rangle = T|\phi\rangle$ so

$$T|\phi\rangle = V|\phi\rangle + V\Pi_{\text{LS}}T|\phi\rangle = (V + V\Pi_{\text{LS}}T)|\phi\rangle. \quad (117)$$

(This holds when contracted with any eigenstate of H_0 , $|\phi_j\rangle$)

- This implies that $T = V + V\Pi_{\text{LS}}T$.
- Recursion provides

$$T = V + V\Pi_{\text{LS}}V + V\Pi_{\text{LS}}V\Pi_{\text{LS}}V + \dots \quad (118)$$

If we want a transition amplitude for this, we want to calculate

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \langle f|V\Pi_{\text{LS}}V|i\rangle + \dots \quad (119)$$

Note: We can insert the eigenstates within this equation due to the fact that

$$\sum_j |\phi_j\rangle\langle\phi_j| = \mathbb{1} \quad (120)$$

This provides

$$\langle\phi_f|T|\phi_i\rangle = \langle\phi_f|V|\phi_i\rangle + \langle\phi_f|V\Pi_{\text{LS}}|\phi_j\rangle\langle\phi_j|V|\phi_i\rangle + \dots \quad (121)$$

- In an alternative/more convenient notation,

$$T_{fi} = V_{fi} + \left(\sum_j\right) V_{fj}\Pi_{\text{LS}}(j)V_{ji} + \left(\sum_j\right) V_{fi}\Pi_{\text{LS}}(j)V_{jk}\Pi_{\text{LS}}(k)V_{ki} + \dots \quad (122)$$

Where $T_{ij} = \langle\phi_i|T|\phi_j\rangle$, $V_{ij} = \langle\phi_i|V|\phi_j\rangle$, $\Pi_{\text{LS}}(j) = \frac{1}{E - E_j}$ with $E_f = E_i$ and the sums tend to be omitted

Note (OFPT With particle interactions): We can with a certain particle state $|i\rangle$ and end with a final certain particle state $|f\rangle$ and use *OFPT* to calculate what happens in-between such states. An example of this is the case of an electron scattering off an electron.

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{E - H_0 + i\varepsilon} \quad (114)$$

is.

Example 2.4.3.1 (Electron Scattering): This is an example from the textbook that I was reading which allows for the derivation of the *Advanced* and *Retarded* propagators.

- The transition matrix element is

$$T_{fi} = V_{fi} + \sum_n V_{fn} \frac{1}{E_i - E_n} V_{ni} + \dots \quad (123)$$

- For electron scattering, the initial and final states are $|i\rangle = |\psi_e^1 \psi_e^2\rangle$ and $\langle f| = \langle \psi_e^3 \psi_e^4|$
- Due to the fact that Coulomb's law is replaced by photon exchange we need a photon field $\phi(x)$.
- This provides

$$V = \frac{1}{2} e \int d^3x \psi_e(x) \phi(x) \psi_e(x) \quad (124)$$

with the $\frac{1}{2}$ from spin and the e as some coupling constant (which happens to be the electron charge).

Note:

- This interaction turns a $\vec{p}_1$ electron into a $\vec{p}_3$ electron and a $\vec{p}_\gamma$ photon.
- Since both $|i\rangle$ and $|f\rangle$ have 2 electrons and no photons, $V_{ij} = 0$

2.4.3.1. Retarded and Advanced Propagators

- This gets split into two states. (propagated by the *Retarded* Propagator and the *Advanced* Propagator)
- The propagators that this creates are foundational to quantum field theory, (the advanced and retarded ones), which is the primary reason behind this methodology.

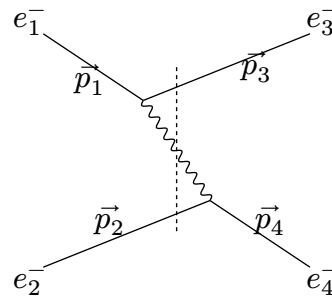


Figure 1: Retarded state

- In the Retarded state, electrons interact via the *Retarded Propagator*.
- We have that

$$|n\rangle = |\psi_e^3 \phi^\gamma \psi_e^2\rangle \quad (125)$$

- Which implies that

$$\begin{aligned}
V_{ni}^R &= \langle \psi_e^3 \phi^\gamma \psi_e^2 | V | \psi_e^1 \psi_e^2 \rangle \\
&= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle \langle \psi_e^2 | \psi_e^2 \rangle \\
&= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle
\end{aligned} \tag{126}$$

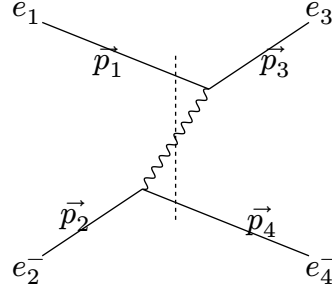


Figure 2: Advanced State

- For the *advanced state*, a similar result is derived

$$V_{ni}^A = \langle \psi_e^4 \phi^\gamma | V | \psi_e^2 \rangle \tag{127}$$

2.4.3.2. Evaluating The Propagators

$$V_{ni}^R = \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle = \frac{e}{2} \int d^3x \langle \psi_e^3 \phi^\gamma | \psi_e(x) \phi(x) \psi_e(x) | \psi_e^1 \rangle \tag{128}$$

Note:

- Free field (like the photon and electron fields inside of the integral)

$$\langle \phi^\gamma | \phi(x) | 0 \rangle = e^{-i\vec{p} \cdot \vec{x}} \tag{129}$$

- This implies:

$$\begin{aligned}
V_{ni}^R &= e \int d^3x e^{i(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \cdot \vec{x}} \\
&= e(2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_3)
\end{aligned} \tag{130}$$

- This is similar to other matrix elements
- First order:

$$T_{fi} = \int d^3\vec{p}_\gamma (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_\gamma) (2\pi)^3 \delta^3(\vec{p}_2 - \vec{p}_4 + \vec{p}_\gamma) \frac{e^2}{E_i - E_n} \tag{131}$$

- Energy is not conserved in such intermediate states (otherwise $E_i = E_n$ would blow u

2.4.3.3. Energy Calculations

- We want to calculate the energy of the states

Denotations

- Momenta of particles are

$$\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4, \vec{p}_\gamma \quad (132)$$

- The energy of each respective particle is

2.5. More Information on the S-matrix

- The S -matrix describes the interaction between particles in the system, and as previously seen, it is defined as

$$\langle f|S|i\rangle = \mathbb{1} + (2\pi)^4 \delta^4(p_f - p_i) \mathcal{M} \quad (133)$$

- This S-matrix is the primary goal of most calculations in quantum field theory (as it describes the interactions of the particles)

2.5.1. LSZ Reduction

- The way to obtain the S-matrix from the *Time-Ordered Product* involves the **LSZ Reduction Formula** which is extremely important, and allows for the derivation of the extremely simple momentum-space Feynman Rules, which you will see later.
- The LSZ reduction formula is as follows:

$$\begin{aligned} \langle f|S|i\rangle = & \left[i \int d^4x_1 e^{-ip_1x_1} (\square + m^2) \right] \dots \left[i \int d^4x_n e^{ip_nx_n} (\square + m^2) \right] \\ & \times \langle \Omega | T \phi(x_1) \phi(x_2) \phi(x_3) \dots \phi(x_n) | \Omega \rangle \end{aligned} \quad (134)$$

Definition 2.5.1.1 (LSZ Reduction): The *LSZ Reduction* is a formula providing S -matrix elements from the time ordered product (in the ground state for the interacting theory Ω).

- Additionally, this formula can allow for the calculation of transfer matrix elements $\mathcal{M}$, but it is more convenient to utilize Feynman diagrams derived from this formulation.
 - The cross sections are calculatable from this as well
- This formula effectively deletes terms in the time ordered product except for those with poles of $\frac{1}{p^2 - m^2}$
- The ground state in the interacting theory, $|\Omega\rangle$ is a very inconvenient way to calculate, due to lack of knowledge about the ground state

2.5.1.1. Correlation Functions

- A **correlation function** $\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle$ is the time ordered product of fields which appears in the LSZ reduction formula after the integrals
- This is specifically used in the derivation of the LSZ reduction formula

2.5.1.2. Derivation of LSZ Reduction

- Suppose that the initial state $|i\rangle$ and the final state $|f\rangle$ are of the form

$$|i\rangle = \sqrt{2\omega_1}\sqrt{2\omega_2}a_{p_1}^\dagger(-\infty)a_{p_2}^\dagger(\infty)|\Omega\rangle \quad (135)$$

and

$$|f\rangle = \sqrt{2\omega_1}\cdots\sqrt{2\omega_j}a_{p_3}^\dagger(\infty)\cdots a_{p_j}^\dagger(\infty)|\Omega\rangle \quad (136)$$

- Additionally there is the constraint that $|i\rangle \neq |f\rangle$ due to scattering happening.
- The S-matrix element is then

$$\langle f|S|i\rangle = 2^{\frac{j}{2}}\sqrt{\omega_1\cdots\omega_n} \times \langle\Omega|a_{p_3}(\infty)\cdots a_{p_j}(\infty)a_{p_1}^\dagger(-\infty)a_{p_2}^\dagger(-\infty)|\Omega\rangle \quad (137)$$

- Now it is desirable to convert this to scalar fields, ϕ
- From quantization, they are defined by

$$\phi = \int \frac{d^3p}{(2\pi)^3} (a_p(t)e^{-ipx} + a_p^\dagger(t)e^{ipx}) \quad (138)$$

2.5.1.2.1. Heisenberg picture to obtain operators

- Consider the Heisenberg picture
 - $a_p(t) = e^{iH(t-t_0)}a_p(t_0)e^{-iH(t-t_0)}$ and $\phi(x) = e^{iH(t-t_0)}\phi(\mathbf{x}, t_0)e^{-iH(t-t_0)}$ via unitary time evolution
 - The only useful thing to know about $\phi(x, t)$ and $a_p(t)$ is

$$\langle\Omega|\phi(\mathbf{x}, t = \pm\infty)|p\rangle = Ce^{-i\mathbf{p}\cdot\mathbf{x}} \quad (139)$$

- A useful identity:

$$i \int d^4x e^{ipx} (\square + m^2)\phi(x) = \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)] \quad (140)$$

- Proof:

$$\begin{aligned} i \int d^4x (\square + m^2)\phi(x) &= i \int d^4x e^{ipx} (\partial_t^2 - \vec{\partial}_x^2 + m^2)\phi \\ &= i \int d^4x e^{ipx} (\partial_t^2 + \vec{p}^2 + m^2)\phi(x) \\ &= i \int d^4x e^{ipx} (\partial_t^2 + \omega_p^2)\phi(x) \end{aligned} \quad (141)$$

- Identity for proof:

$$\begin{aligned} &\partial_t [e^{ipx} (i\partial_t + \omega_p)\phi(x)] \\ &= [i\omega_p e^{ipx} (i\partial_t + \omega_p) + e^{ipx} (i\partial_t^2 + \omega_p \partial_t^2)]\phi(x) \\ &= ie^{ipx} (\partial_t^2 + \omega^2)\phi(x) \end{aligned} \quad (142)$$

- Putting this into the integral,

$$\begin{aligned} i \int d^4x e^{ipx} (\square + m^2)\phi(x) &= \int d^4x \partial_t [e^{ipx} (i\partial_t + \omega_p)\phi(x)] \\ &= \int dt \partial_t \left[e^{i\omega_p t} \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p)\phi(x) \right] \end{aligned} \quad (143)$$

- This is true for all $\phi(x)$
- Quantum Field Case:

$$\begin{aligned}
& \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \phi(x) \\
&= \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \\
&\quad \times \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (a_k(t) e^{-ikx} a_k^\dagger(t) e^{ikx}) \\
&= \int \frac{d^3k}{(2\pi)^3} \int d^3x \left(\left(\frac{\omega_k + \omega_p}{\sqrt{2\omega_k}} \right) a_k(t) e^{-ikx} e^{-i\mathbf{p}\cdot\mathbf{x}} \right. \\
&\quad \left. + \left(\frac{-\omega_k + \omega_p}{\sqrt{2\omega_k}} \right) a_k^\dagger(t) e^{ikx} e^{-i\mathbf{p}\cdot\mathbf{x}} \right)
\end{aligned} \tag{144}$$

Note (Math Note):

- $\partial_t a_k(t) = 0$ at $t = \pm\infty$
- The d^3x integral enforces $\omega_k = \omega_p$ via delta functions
- Therefore, we get

$$\int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \phi(x) = \sqrt{2\omega_p} a_p(t) e^{-i\omega_p t} \tag{145}$$

- This implies

$$\begin{aligned}
i \int d^4x e^{ipx} (\square + m^2) \phi(x) &= \int dt \partial_t \left((e^{i\omega_p t}) (\sqrt{2\omega_p} a_p(t) e^{-i\omega_p t}) \right) \\
&= \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)]
\end{aligned} \tag{146}$$

as desired

2.5.1.2.2. S matrix element computing

- Now we can compute

$$\begin{aligned}
\langle f|S|i\rangle &= 2^{\frac{i}{2}} \sqrt{\omega_1 \omega_2 \dots \omega_j} \\
&\times \langle \Omega | a_{p_3}(\infty) \dots a_{p_j}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle
\end{aligned} \tag{147}$$

Definition 2.5.1.2.2.1 (Time Ordered Product):

- The time ordered product $T \mathcal{O}(x_1) \mathcal{O}(x_2) \dots \mathcal{O}(x_n)$ picks up the operators $\mathcal{O}$ and literally places them so that they are correctly ordered in time. This is the only thing that the operator does, as it keeps the products there.

- Notice that

$$\begin{aligned} \langle f|S|i\rangle &= \sqrt{2^j \omega_1 \dots \omega_j} \langle \Omega | T [a_{p_3}(\infty) - a_{p_3}(-\infty)] \dots [a_{p_j}(\infty) - a_{p_j}(-\infty)] \\ &\quad \times [a_{p_1}^\dagger(-\infty) - a_{p_1}^\dagger(\infty)] [a_{p_2}^\dagger(-\infty) - a_{p_2}^\dagger(\infty)] | \Omega \rangle \end{aligned} \quad (148)$$

- Therefore, we have the LSZ Reduction Formula

$$\begin{aligned} \langle p_3 \dots p_j | S | p_1 p_2 \rangle &= \left[i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) \right] \\ &\quad \times \dots \times \left[i \int d^4 x_j e^{ip_j x_j} (\square_j + m^2) \right] \\ &\quad \times \langle \Omega | T \phi(x_1) \phi(x_2) \dots \phi(x_j) | \Omega \rangle \end{aligned} \quad (149)$$

2.5.1.3. LSZ For Operators

- The LSZ Reduction formula is relatively powerful, as it applies for all operators $\mathcal{O}_j$
- This implies that

$$\begin{aligned} \langle p_3 \dots p_j | S | p_1 p_2 \rangle &= \left[i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) \right] \\ &\quad \times \dots \times \left[i \int d^4 x_j e^{ip_j x_j} (\square_j + m^2) \right] \\ &\quad \times \langle \Omega | T \mathcal{O}_1(x_1) \dots \mathcal{O}_j(x_j) | \Omega \rangle \end{aligned} \quad (150)$$

2.6. Feynman Diagrams and the Path Integral (TODO)

2.6.1. Feynman Propagators

- We can define the feynman propagator $G_F(x, y)$ to be implicitly defined by

$$(\square_x + m^2) G_F(x, y) = -i \delta^4(x - y) \quad (151)$$

- This provides us, in position space, with

$$\begin{aligned} G_F(x, y) &= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{ip \cdot (x-y)} \\ &\equiv \int \frac{d^4 p}{(2\pi)^4} G_F(p) e^{-ipx - y} \end{aligned} \quad (152)$$

- The momentum space Feynman Propagator $G_F(p)$ is defined as

$$G_F(p) \equiv \frac{i}{p^2 - m^2 + i\varepsilon} \quad (153)$$

- The Feynman propagator, which we will elaborate upon, is a very fundamental quantity that is useful in evaluating Feynman diagrams (to be more explicit, it could be one of the only ways to evaluate them).
 - In conclusion, Richard Feynman is the GOAT

Definition 2.6.1.1 (Feynman Propagator):

- The Feynman propagator $G_F(x, y)$ or $D_F(x, y)$ or $G_F(x - y)$ or D_{xy} depending on the situation is defined as the time ordered product of free fields.
- That is,

$$\begin{aligned} G_F(x, y) &\equiv \langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle \\ &= \lim_{\varepsilon \rightarrow 0} \int \frac{d^4 p}{(2\pi)^4} \frac{1}{k^2 - m^2 + i\varepsilon} e^{ik(x-y)} \end{aligned} \quad (154)$$

Definition 2.6.1.2 (Feynman Diagram): A Feynman diagram is something that is used to represent particle interactions, or simplifications of them which are summed. generally to calculate the LSZ reduction for matrix elements. These matrix elements are ultimately used to calculate cross sections as seen in *Cross Sections*.

Definition 2.6.1.3 (Feynman Rules): Feynman rules are a set of rules for a theory which is used to evaluate Feynman diagrams.

- A standard set of Feynman rules that can generally apply everywhere are the following

General Position Space Feynman Rules

- At each vertex, a factor of the interaction coupling is given
- All lines are integrated over $d^3 x$ after applying the Feynman propagator
- The LSZ reduction formula is applied to convert this to an S-matrix element
- Additional factors may be added if they are incoming/outgoing electrons/photons due to polarization and spin

General Momentum Space Feynman Rules (Arguably much more useful)

- Each vertex of a Feynman Diagram gets a factor of whatever the interaction coupling is.
- Vertices can only occur at points that satisfy interactions in the Lagrangian⁶
- For each edge whose momentum is known (that does not connect to an external vertex), a factor of the propagator for the theory Π is gained
- For each edge whose momentum k is unknown (i.e., a loop), a factor of Π is gained and must be integrated over

⁶For instance, if $\mathcal{L}_{\text{int}} = \frac{g}{3!} \phi^3$, only interactions which involve fields intersecting at three points are included in the set of tree-level diagram

$$\frac{d^4k}{(2\pi)^4}. \quad (155)$$

- The final product gives $i\mathcal{M}$ for the diagram
- Additional factors may be added if they are incoming/outgoing electrons/photons due to polarization and spin

2.6.2. Wick's Theorem and Feynman Diagram Derivation

- Before diagrams, it is useful to do some calculations
 - There are two different ways to derive Feynman rules for a theory, but the most common two are through the use of lagrangians and through the use of 2nd quantization

2.6.2.1. Lagrangian Method

- Consider Free Fields with

$$\begin{aligned} [\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)] &= 0 \\ [\phi(\mathbf{x}, t), \partial_t \phi(\mathbf{x}', t)] &= i\hbar \delta^3(\mathbf{x} - \mathbf{x}') \\ (\square + m^2)\phi &= 0 \end{aligned} \quad (156)$$

- We only know how to evaluate the free theory correlation functions at this point

$$\langle 0|T\phi(x)\phi(x')|0\rangle \quad (157)$$

Theorem 2.6.2.1.1 (Schwinger-Dyson Equations):

- An equation that will appear later via the use of Feynman path integrals
- The Schwinger Dyson Equations state that

$$\begin{aligned} (\square + m^2)\langle \phi_x \phi_1 \phi_2 \dots \phi_n \rangle &= \langle \mathcal{L}'_{\text{int}}[\phi_x] \phi_1 \dots \phi_n \rangle \\ -i\hbar \sum_j \delta^4(x - x_j) \langle \phi_1 \dots \phi_{j-1} \phi_{j+1} \dots \phi_n \rangle & \end{aligned} \quad (158)$$

- This holds if

$$\mathcal{L} = -\frac{1}{2}(\square + m^2)\phi + \mathcal{L}_{\text{int}}[\phi] \quad (159)$$

- They differentiate classical mechanics and quantum mechanics via the $\hbar$ term.

Proof Schwinger-Dyson Equation:

- This should barely constitute a proof but nevertheless I have to call it something.
- Consider

$$\partial_t \langle \Omega|T\phi(x)\phi(x')|\Omega\rangle \quad (160)$$

- This is equal to

$$\begin{aligned}
& \partial_t [\langle \Omega | \phi(x) \phi(x') | \Omega \rangle \Theta(t - t') + \langle \Omega | \phi(x') \phi(x) | \Omega \rangle \Theta(t' - t)] \\
&= \langle \Omega | T \partial_t \phi(x) \phi(x') | \Omega \rangle + \langle \Omega | \phi(x) \phi(x') | \Omega \rangle \partial_t \Theta(t - t') \\
&\quad + \langle \Omega | \phi(x') \phi(x) | \Omega \rangle \partial_t \Theta(t' - t)
\end{aligned} \tag{161}$$

- Using the fact that $\partial_T \Theta(t) = \delta(t)$, we obtain that the above expression is equal to

$$\langle \Omega | T \partial_t \phi(x) \phi(x') | \Omega \rangle + \delta(t - t') \langle \Omega | [\phi(x), \phi(x')] | \Omega \rangle \tag{162}$$

- Taking the second derivative, we obtain

$$\begin{aligned}
\partial_t^2 \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle &= \langle \Omega | T \partial_t^2 \phi(x) \phi(x') | \Omega \rangle \\
&\quad + \delta(t - t') \langle \Omega | [\partial_t \phi(x), \phi(x')] | \Omega \rangle
\end{aligned} \tag{163}$$

- Since the delta function forces time equivalence ($t = t'$) in the second term, we have that

$$[\partial_t \phi(x), \phi(x')] = -i\hbar \delta^3(\mathbf{x} - \mathbf{x}') \tag{164}$$

and therefore additionally have that

$$\partial_t^2 \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle = \langle \Omega | T \partial_t^2 \phi(x) \phi(x') | \Omega \rangle - i\hbar \delta^4(x - x') \tag{165}$$

Note (Chevron Notation):

- As you may have seen earlier, we have introduced notation $\langle \phi \rangle$ in order to make notation more condensed. As you may be able to guess, $\langle \phi \rangle \equiv \langle \Omega | \phi | \Omega \rangle$

- Therefore, the equation with $\square + m^2$ follows and implies our Feynman propagator,

$$(\square_x + m^2) G_F(x, y) = -i\hbar \delta^4(x - y) \tag{166}$$

- That is,

$$(\square + m^2) \langle \phi(x) \phi(x') \rangle = \langle (\square + m^2) \phi(x) \phi(x') \rangle - i\hbar \delta^4(x - x') \tag{167}$$

- Applying Euler-Lagrange equations gives

$$\langle \mathcal{L}'_{\text{int}[\phi]} \phi(x) \phi(x') \rangle - i\hbar \delta^4(x - x') \tag{168}$$

- Generalization provides the final equation,

$$(\square_x + m^2) \phi = \langle \mathcal{L}'_{\text{int}} \phi_x \phi_1 \dots \phi_n \rangle - i\hbar \sum_j \langle \phi_1 \dots \phi_j \phi_{j+1} \dots \phi_n \rangle \tag{169}$$

■

- Now we can simply evaluate the two point functions
- For instance,

$$\langle \phi_1 \phi_2 \rangle = \int d^4 \delta_{x_1} \langle \phi_x \phi_2 \rangle \tag{170}$$

(this is simply using the identity that $\int d^4 x \delta(x) f(x) = f(0)$).

- The above expression is then equivalent to

$$i \int d^4x (\square_x D_{x1}) \langle \phi_x \phi_2 \rangle = i \int d^4x D_{x1} \square_x \langle \phi_x \phi_2 \rangle \quad (171)$$

- Some condensed notation was used where $D_{ab} = G_F(x_a, x_b)$ and $\delta_{xb} = \delta(x - x_b)$
- In a free theory $\square_x \langle \phi_x \phi_Y \rangle = -i\delta_{xy}$, and so

$$\langle \phi_1 \phi_2 \rangle = \int d^4x D_{x1} \delta_{x2} = D_{12} \quad (172)$$

which is not very interesting (it is literally a particle going from point x to point y without anything happening)

2.6.2.2. Wick's Theorem/Contractions/QM Derivation

- Let us move to an interacting theory ($\mathcal{L}_{\text{int}} \neq 0$)

2.6.2.2.1. Consider the Heisenberg picture

- We have the Heisenberg equations of motion $i\partial_t \phi(x) = [\phi, H]$
 - These necessarily imply that $\phi(x, t)$ evolves via an operator S . i.e.

$$\phi(x, t) = S(t, t_0)^\dagger \phi(x) S(t, t_0) \quad (173)$$

, noting that S is the S-matrix.

- S satisfies $i\partial_t S(t, t_0) = H(t)S(t, t_0)$ (the schrodinger equation)
- In the Heisenberg picture, we do not need the explicit hamiltonian, just the time dependent part
 - This is because the states are time independent and the Hamiltonian is not
- $H(t) = H_0 + V(t)$

2.6.2.2.2. Interaction Picture

- In the interaction picture, fields only evolve via H_0
- Free fields:

$$\phi_0(x, t) = e^{iH_0(t-t_0)} \phi(x) e^{-iH_0(t-t_0)} \quad (174)$$

Note:

- This is equivalent to the plane wave solution for scalar fields. That is,

$$\phi_0(x, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx}) \quad (175)$$

- In order to get the real field for this, we have that

$$\begin{aligned} \phi(x, t) &= S^\dagger(t, t_0) e^{-iH(t-t_0)} \phi_0(x, t) e^{iH_0(t-t_0)} S(t, t_0) \\ &\equiv U^\dagger(t, t_0) \phi_0(x, t) U(t, t_0) \end{aligned} \quad (176)$$

- This $U(t, t_0) \equiv e^{iH_0(t-t_0)} S(t, t_0)$ is unitary time evolution operator that we can utilize
 - It relates the Heisenberg picture to free fields at some time t

- Notice that it satisfies (a form of) the Schrodinger equation. Namely,

$$\begin{aligned}
i\partial_t U(t, t_0) &= -e^{iH_0(t-t_0)} H_0 S(t, t_0) + e^{iH_0(t-t_0)} H(t) S(t, t_0) \\
&= e^{iH_0(t-t_0)} [-H_0 + H(t)] e^{-iH_0(t-t_0)} e^{iH_0(t-t_0)} S(t, t_0) \\
&= H_I(t) U(t, t_0)
\end{aligned} \tag{177}$$

with $H_I(t) \equiv e^{iH_0(t-t_0)} V(t) e^{-iH_0(t-t_0)}$

- This $H_I(t)$ is the Heisenberg interaction/potential in the interacting picture
- The solution to this is

$$U(t, t_0) = T \exp \left(-i \int_{t_0}^t H_I(t') dt' \right) \tag{178}$$

which is known as the **Dyson Equation**

2.6.2.3. More Feynman rules derivation

- Consider the operator

$$U_{12} \equiv U(t_1, t_2) = T \exp \left(-i \int_{t_1}^{t_2} dt' H_I(t') \right) \tag{179}$$

- In this notation, the later time is on the left
- We have the following identities:
 - $U_{21} U_{12} = 1$
 - $U^{-1}_{21} = U_{21}^\dagger = U_{12}$
 - For $t_1 < t_2 < t_3$, $U_{32} U_{21} = U_{31}$
 - $U_{32} U_{12} = U_{32}$
- Since $\phi(x_1) = \phi(x_1, t) = U_{10}^\dagger \phi_0(x_1, t) U_{10} = U_{01} \phi_0(x_1, t) U_{10}$

2.6.3. Vacuum Matrix Elements

- We want to evolve the ground state $|\Omega\rangle$ to a time t_0 such that the interacting picture and the free picture are equivalent
- In the interacting picture, states $|\Omega\rangle$ evolve via $S(t, t_0)$ and are annihilated by $a_p(t)$ at $t = -\infty$
- In the heisenberg picture, $a_p(t)$ evolves as $S(t, t_0)^\dagger a_p(t_0) S(t, t_0)$ and annihilates $|\Omega\rangle$ at $t = -\infty$
- In the free theory, $|0\rangle$ is annihilated by a_p
 - a_p evolves only via phase rotation
 - $a_p(t_0) e^{iH_0(t-t_0)} |0\rangle = 0$ at $t = -\infty$ (annihilation)
- We must have that $e^{iH_0(t-t_0)} |0\rangle$ and $S(t, t_0) |\Omega\rangle$ must be proportional
 - Then,

$$|\Omega\rangle = \mathcal{N}_i \lim_{t \rightarrow -\infty} S^\dagger(t, t_0) e^{iH_0(t-t_0)} |0\rangle = \mathcal{N}_i U_{0, -\infty} |0\rangle \tag{180}$$

$-\mathcal{N}_i$ is just a number

- Similarly, for the final time,

$$\langle \Omega | = \mathcal{N}_F \langle 0 | U_{\infty,0} \quad (181)$$

for some $\mathcal{N}_f$

- Then, for our time ordered product, assuming that $t_1 > t_2 > \dots t_n$ (WLOG), we have that

$$\begin{aligned} \langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle &= \langle \Omega | \phi(x_1) \dots \phi(x_n) | \Omega \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | U_{\infty,0} U_{01} \phi_0(x_1) U_{10} U_{02} \phi_0(x_2) U_{20} \dots U_{0n} \phi_0(x_n) U_{n0} U_{0,-\infty} | 0 \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | U_{\infty,1} \phi(x_1) U_{12} \phi(x_2) U_{23} \dots U_{(n-1)n} \phi_0(x_n) U_{n,-\infty} | 0 \rangle \end{aligned} \quad (182)$$

- Now we can simply apply the time ordering, giving

$$\begin{aligned} \langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle &= \mathcal{N}_i \mathcal{N}_f \langle 0 | T U_{\infty,1} \phi_0(x_1) U_{12} \phi_0(x_2) U_{23} \dots \phi_0(x_n) U_{n,-\infty} | 0 \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | T \phi_0(x_1) \dots \phi_0(x_n) U_{\infty,-\infty} | 0 \rangle \end{aligned} \quad (183)$$

- Additionally, $\langle \Omega | \Omega \rangle = 1$ which implies that, by normalization, $\mathcal{N}_i \mathcal{N}_f = \langle 0 | U_{\infty,-\infty} | 0 \rangle^{-1}$

Note (Oscillations): We have been taking $t \rightarrow \infty$ this whole time, which is incorrect. Utilizing $t \rightarrow \infty$ without any damping will result in oscillations. That is, if we insert the energies

$$e^{-iHt} | 0 \rangle \sum_n e^{iE_n t} | n \rangle \langle n | 0 \rangle \quad (184)$$

, taking $t \rightarrow \infty$ yields e^{-itE} for an energy E which at large times causes unpredictable phase oscillations. Thus, we must use an analytic continuation by taking $t \rightarrow \infty - i\varepsilon$ (which would provide a dampening $e^{-\varepsilon E}$)

- We can finally have the time ordered product

$$\begin{aligned} \langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle &= \lim_{T \rightarrow \infty - i\varepsilon} \frac{\langle 0 | T \phi_0(x_1) \dots \phi_0(x_n) \exp \left[-i \int_{-T}^T dt H_I(t) \right] | 0 \rangle}{\langle 0 | T \exp \left[-i \int_{-T}^T dt H_I(t) \right] | 0 \rangle} \end{aligned} \quad (185)$$

2.6.4. Wick's Theorem and Contractions (real)

- A contraction in a Feynman diagram is when

$$\begin{aligned} &= 9D_{12}D_{xx}D_{xy}P_{yy} + 6D_{12}D_{xy}^3 \\ &+ 18D_{1x}D_{2x}D_{xz}P_{yy} + 9D_{12}D_{xy}D_{xx}P_{yy} + \\ &18D_{12}D_{xy}D_{xy}^2 + 18D_{12}D_{xy}P_{yy}D_{xy}D_{xx} \\ &+ 9D_{1y}D_{xx}D_{xx}D_{yy} + 18D_{1y}D_{2x}D_{xy}^2 \end{aligned} \quad (186)$$

Part III

Vectors

3.1. Gauge Invariance

- Gauge invariance is a principle that photons obey (specifically, $A^\mu \rightarrow A^\mu + \partial_\mu \alpha$ must keep the lagrangian invariant)
 - This section will elaborate on what specifically gauge invariance is supposed to mean and the effects that it has on QFT.
- Additionally, it is useful to start with integer spins (and hence, we are using vector bosons like photons) due to simplicity of their behavior.

3.1.1. Unitary Representation of the Poincare Group

- Before reading this, note that there is no unitary matrix representation of the Poincare group in finite dimensions.

Poincare Group:

- For any field, call it $\psi(x)$, if $x^\nu \rightarrow x^\nu + a^\nu$, the observables of $\psi(x + a)$ should remain the same.
- Additionally, standard laws of lorentz invariance should apply, (objects transform under lorentz transformations of course).

Definition 3.1.1.1 (Poincare Group):

- The Group of coordinate translations and Lorentz transformations
 - It is denoted by $ISO(1, 3)$, where ISO means ISOmetric (ISOmetric minkowski space)
-
- Notice that under a rotation or a boost, only the spin projection of a particle and momenta can change
 - Poincare group determines this
 - A particle is defined as a set of states that mix among themselves under Poincare transformations

3.1.1.1. Representations of the Poincare Group

- Generally, it is expected that particles transform under the law

$$|\psi\rangle \rightarrow P|\psi\rangle \quad (187)$$

- A representation of a group is a set of objects that mix under some the transformation (this will be formalized later)
- Suppose that we have some basis for $|\psi\rangle$. Call it $\{|\psi_i\rangle\}$
 - Then, we have that

$$|\psi_i\rangle = P_{ij}|\psi_j\rangle \quad (188)$$

- We also want the representation of the Poincare group to be **irreducible**. That is, no subspace exists that is invariant under the group's actions. Or, as Schwartz puts it “no subset of states transform among themselves.”
- The final condition that we want to impose onto the representation is *Unitarity*

- This is the idea that $P^\dagger P = 1$ since we want the inner product to be Poincare invariant.
 $M = \langle \psi_1 | \psi_2 \rangle = \langle \psi_1 | P^\dagger P | \psi_2 \rangle$
- Hence, *Particles transform under irreducible representations of the Poincare Group*
- There is a final, stronger requirement, which is that the S -matrix must be unitary.
- The problem with unitarity is that all representations (provided unitarity is true) must be infinite dimensional.
- All irrep of the Poincare group are parametrized by a mass m and a spin j
 - Mass is a quantum number $m \geq 0, \in \mathbb{R}$
 - Spin is a quantum number $j \in 0, \frac{1}{2}, 1, \dots$
 - If $j > 0$, for all values of $p^2 = m^2$, there are $2j + 1$ states in the representations (and 2 states if $m = 0$)
- We want to force spin states into objects $\phi(x), V_\mu(x), T_{\mu\nu}(x), \dots$
 - The problem lies in the degrees of freedom
- As mentioned, the DOF in spin states are 1, 3, 5, 7, ..., $2j + 1$ and the DOF in tensors are 4^n for 4 values of μ in each index. Hence putting representations into objects is difficult.
- This is what leads to gauge invariance as a requirement

3.1.1.2. Unitarity VS LI disagreement

- Consider the state $|\psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$
- The inner product/magnitude is $\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2$
 - We want this quantity to remain invariant under Lorentz transformations
- Let's try a Lorentz rotation:

$$\begin{aligned} |\uparrow\rangle &\rightarrow \cos\theta|\uparrow\rangle + \sin\theta|\downarrow\rangle \\ |\downarrow\rangle &\rightarrow -\sin\theta|\uparrow\rangle + \cos\theta|\downarrow\rangle \end{aligned} \tag{189}$$

- In fact, this is invariant.
- Suppose now that we use a vector basis $|V_\mu\rangle$
- Let our state $|\psi\rangle = |V_0\rangle$. Then we apply a boost

$$|\psi'\rangle = \cosh\beta|V_0\rangle + \sinh\beta|V_1\rangle \tag{190}$$

- The inner product is $\langle\psi'|\psi'\rangle = \cosh^2\beta + \sinh^2\beta \neq 1 = \langle\psi|\psi\rangle$. Hence, this is not Lorentz invariant.
- This brings us to the heart of the problem; The hilbert space uses a $\delta^{\mu\nu}$ inner product whereas relativity mandates a $g^{\mu\nu}$ inner product.
- The problem, for a vector boson representation, is that

$$\begin{aligned} v_\mu^2 &= v_0^2 - v_1^2 - v_2^2 - v_3^2 \\ &= \text{direct sum of spin 0 and spin 1} \end{aligned} \tag{191}$$

- There are other problems, such as having infinite dimensions causing there to be no const.

$$\epsilon = (1, 0, 0, 0), \dots \quad (192)$$

3.1.2. Particles in Fields

- Let us start with forcing energy to be positive, and define

$$E = \int d^3x \mathcal{E} \quad (193)$$

$$\mathcal{E} = T_{00} = \sum_n \frac{\partial \mathcal{L}}{\partial \dot{\phi}_n} \dot{\phi}_n - \mathcal{L}$$

- Spin 0 causes the lagrangian to be

$$\mathcal{L}(x) = \frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{1}{2} m^2 \phi(x)^2 \quad (194)$$

- Therefore, we have the Klein Gordon equation

$$(\square + m^2)\phi = 0 \quad (195)$$

with solutions $\phi = e^{\pm i p x}$

- The energy density is then

$$\frac{\partial L}{\partial \dot{\phi}} \dot{\phi} - L = \frac{1}{2} [(\partial_t \phi)^2 + (\vec{\nabla} \phi)^2 + m^2 \phi^2] \quad (196)$$

- This is positive definite

3.1.2.1. Massive Spin 1

- In $m > 0$, there are 3 DOF that we need
- In contrast, a vector which we intend on using has $4 = 3 \oplus 1$ degrees of freedom.
 - The lorentz group is a division of 3d spin 1 and 1d spin 0 of SO(3) (3d rotation group)
- We can guess the lagrangian to be (for a free spin 1 field)

$$\mathcal{L} = -\frac{1}{2} \partial_\nu A_\mu \partial^\nu A^\mu + \frac{1}{2} m^2 A_\mu A^\mu \quad (197)$$

- This would imply that

$$(\square + m^2)A_\mu = 0 \quad (198)$$

- Hence, we have an energy density

$$\mathcal{E} = \frac{\partial \mathcal{L}}{\partial (\partial_t A_\mu)} \partial_t A_\mu - \mathcal{L} \quad (199)$$

$$= -\frac{1}{2} [(\partial_t A_0)^2 + (\vec{\nabla} A_0)^2 + m^2 A_0^2] + \frac{1}{2} [(\partial_t \vec{A})^2 + (\partial_i A_j)^2 + m^2 \vec{A}^2]$$

- However, there is a problem. Notice that there are fields within this quantity that are allowed to be negative. Hence, this is not the lagrangian. However, we had obtained 4 (or 3+1) scalars instead of a vector which implies that the lagrangian chooses everything.
- There is 1 more term that can be added to this lagrangian. This term is

$$A_\mu \partial_\mu \partial_\nu A_\nu \quad (200)$$

- The fully generalized lagrangian for a massive vector field is

$$\mathcal{L} = \frac{a}{2} A_\mu \square A^\mu + \frac{b}{2} A_\mu \partial^\mu \partial^\nu A_\nu + \frac{1}{2} m^2 A_\mu^2 \quad (201)$$

for some values a and b

- The EOM of this lagrangian are

$$\begin{aligned} a \square A_m + b \partial_\mu \partial_\nu A_\nu + m^2 A_m &= 0 \\ \xRightarrow{\partial_\mu} [(a+b) \square + m^2] (\partial_\mu A^\mu) &= 0 \end{aligned} \quad (202)$$

- If $a = -b$, this removes one DOF via reducing the EOM to $\partial_\mu A^\mu = 0$ and choose $a = 1$ & $b = -1$

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} A_\mu \square A^\mu - \frac{1}{2} A_\mu \partial^\mu \partial_\nu A^\nu + \frac{1}{2} m^2 A_\mu A^\mu \\ &= -\frac{1}{4} F_{\mu\nu}^2 + \frac{1}{2} m^2 A^2 \end{aligned} \quad (203)$$

with

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (204)$$

is the EM field tensor.

- The EOM of this lagrangian are

$$(\square + m^2) A_\mu = 0 \quad \& \quad \partial^\mu A_\mu = 0 \quad (205)$$

- The stress energy tensor for this is (for the first line rendering cal messed up unfortunately)

$$\begin{aligned} \mathcal{T}_{\mu\nu} &= \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\alpha)} \partial_\nu A_\alpha = g_{\mu\nu} \mathcal{L} \\ &= -F_{\mu\alpha} \partial_\nu A_\alpha + g_{\mu\nu} \left(\frac{1}{4} F_{\alpha\beta}^2 - \frac{1}{2} m^2 A_\alpha^2 \right) \end{aligned} \quad (206)$$

- Now, we can write this in terms of em fields.

$$\frac{F_{\mu\nu}^2}{4} = \frac{1}{2} (E^2 - B^2) \quad (207)$$

with

$$\vec{E} = \partial_t \vec{A} - \nabla A_0, \quad \vec{B} = \nabla \times \vec{A} \quad (208)$$

- Therefore, writing in terms of EM fields, we obtain that

$$\begin{aligned} \mathcal{E} = T_{00} &= -(\partial_t A_\alpha - \partial_\alpha A_0) \partial_t A_\alpha + \frac{1}{2} B^2 - \frac{1}{2} E^2 - \frac{1}{2} m^2 A_\alpha A_\alpha \\ &= \frac{1}{2} (\vec{B}^2 + \vec{E}^2) + \partial_j A_0 (\partial_j A_i - \partial_i A_j) - \frac{1}{2} m^2 A_0^2 + \frac{1}{2} m^2 \vec{A}^2 \end{aligned} \quad (209)$$

- It looks as if there can be negative. However, some terms cancel out. We can rewrite it as

$$\begin{aligned} \mathcal{E} &= \frac{1}{2} (\vec{E}^2 + \vec{B}^2) + \frac{1}{2} m^2 (A_0^2 + A_i^2) + \\ &\quad \underbrace{A_0 \partial_t (\partial_\mu A_\mu) - A_0 (\square + m^2) A_0}_{\text{cancels due to EOM}} + \underbrace{\partial_i (A_0 F_{0i})}_{\text{Total Derivative so doesn't contribute to } E} \end{aligned} \quad (210)$$

- Now our energy is positive and we have reduced the DOF to 3 due to a Lorentz invariant condition

3.1.2.2. Free Massive Spin 1 Field Solution

- The solution to the EOM is VERY similar to the solution for other fields. However, there is another factor that we must include which is the polarizations ε_μ^i .
- The most general solution to the EOM is by utilizing a fourier transform to obtain that

$$\begin{aligned} (\square + m^2) A_\mu &= 0 \\ \Rightarrow A_\mu(x) &= \sum_i \int \frac{d^3 p}{(2\pi)^3} \tilde{a}_i(\vec{p}) \varepsilon_\mu^i(p) e^{ipx} \end{aligned} \quad (211)$$

with $p_0 = \omega_p = \sqrt{\vec{p}^2 + m^2}$ as always.

- We want to choose the basis in a smart way.
- Notice that, since there is no Unitary finite dimension representation of the Lorentz group, $\varepsilon_\mu^i(p)$ must depend on p (and hence they are “infinite dimensions”)
- Suppose that $p^\mu = (E, 0, 0, p_z)$ with $E^2 - p_z^2 = m^2$
- In order to have $\varepsilon_\mu^i(p)$ s.t. $\partial_\mu A_\mu = 0$, it is a requirement that $p^\mu \varepsilon_\mu(p) = 0$
- The trivial ε_μ^i are

$$\epsilon_\mu^1 = (0, 1, 0, 0), \quad \epsilon_\mu^2 = (0, 0, 1, 0) \quad (212)$$

- These are known as transverse polarizations.
- The final polarization is

$$\epsilon_L^\mu = \left(\frac{p_z}{m}, 0, 0, \frac{E}{m} \right) \quad (213)$$

- Notice that we only need 3 of these *polarization vectors* in order to have a basis for our field.
- We are now able to generate an irr. unitary representation of the Poincare group

- Notice that this is infinite dimensional due to momentum dependence
- Now, we have a vector space that is created by integrating these basis vectors by $\tilde{a}_i(\vec{p})$
 - This vector space is the space of fields satisfying the EOM of A_μ of course

3.1.3. Massless Spin 1

- It is not impossible for Massive Spin 1 to exist, but it is not the most common of particle
 - However, photons are an *extremely* common particle which is where it is convenient to start
- In the $m \rightarrow 0$ limit of massive spin 1,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 \quad (214)$$

- Note: we no longer have

$$\partial_\mu A_\mu = 0 \quad (215)$$

since that was determined by

$$m^2(\partial_\mu A^\mu) = 0 \quad (216)$$

- Additionally, Longitudinal polarization blows up

$$\epsilon_\mu^L = (p_z/m, 0, 0, E/m) \quad (217)$$

- Momentum also goes to

$$p \rightarrow (E, 0, 0, E) \quad (218)$$

- In *Quantum Field Theory*, when one problem is solved, two more appear in its place. Much like a hydra.
- Thus, we must start from the beginning
- Let us start with

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 \quad (219)$$

Definition 3.1.3.1 (Gauge Invariance):

- Gauge invariance is the idea that an addition of some total derivative of a function keeps the lagrangian invariant. Explicitly, $\mathcal{L}$ is invariant under

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \alpha(x) \quad (220)$$

- Massless Vector Field theory has this gauge invariance of the Lagrangian.
- Equations of motion for Photon theory are

$$\square A_\mu - \partial_\mu(\partial_\nu A_\nu) = 0 \quad (221)$$

- Our goal again is to separate the temporal and spatial components.

$$\begin{aligned} -\partial_j^2 A_i + \partial_t \partial_j A_0 &= 0 \\ \square A_i - \partial_i (\partial_t A_0 - \partial_j A_j) &= 0 \end{aligned} \quad (222)$$

- Next, we utilize the technique of Gauge fixing.
- We apply a transformation such that

$$\partial_j A_j \rightarrow \partial_j A_j + \partial_j^2 \alpha \quad (223)$$

- In the **Coulomb Gauge**, we can choose this α such that $\partial_j A_j = 0$
- This therefore implies that

$$\partial_j A^j = 0 \quad (224)$$

- Under $\partial_i A_i \rightarrow \partial_i A_i + \partial_i^2 \alpha$, the coulomb gauge is preserved under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ such that $\partial_i^2 \alpha = 0$
- Transforming $A_0 \rightarrow A_0 + \partial_t \alpha$ forces A_0 to satisfy $\partial_i^2 A_0 = 0$
- This allows us to set $A_0 = 0$ via the Coulomb gauge (and get rid of a DOF)
- The EOM for the coulomb gauge is

$$\square A_i = 0 \quad (225)$$

- This gives a free field solution

$$A_\mu(x) = \int \frac{d^4 p}{(2\pi)^4} \varepsilon_\mu(p) e^{ipx} \quad (226)$$

- This provides some equations:

$$p^2 = 0, \quad p_i \varepsilon_i = 0 \text{ due to gauge, } \varepsilon_0 = 0 \text{ due to gauge} \quad (227)$$

- Now we can write $p_\mu = (E, 0, 0, E)$, giving

$$\varepsilon_\mu^1 = (0, 1, 0, 0), \varepsilon_\mu^2 = (0, 0, 1, 0) \quad (228)$$

- This has only degrees of freedom
- This is known as Linear polarization
- We can also use

$$\varepsilon_\mu^R = \frac{1}{\sqrt{2}}(0, 1, i, 0), \quad \varepsilon_\mu^L = \frac{1}{\sqrt{2}}(0, 1, -i, 0) \quad (229)$$

- These are known as circular polarizations
 - They are helicity eigenstates
- Additionally, there is the lorentz gauge

$$\partial_\mu A_\mu = 0 \quad (230)$$

- In this gauge,

$$\varepsilon_\mu^1 = (0, 1, 0, 0), \varepsilon_\mu^2 = (0, 0, 1, 0), \varepsilon_\mu^f = (1, 0, 0, 1) \quad (231)$$

- ε_μ^f is known as forward polarization

Note (Forward Polarization):

- Forward polarization is proportional to momentum
 - Hence, it corresponds to $A_\mu = \partial_\mu \phi$
 - Therefore, $A_\mu = 0$ due to gauge invariance
 - Therefore, forward polarization is not physical

3.1.4. Covariant Derivatives

- Interactions in the Lagrangian must retain Gauge invariance (since they involve A_μ)
 - For instance, we cannot add an interaction term such as $A_\mu \phi \partial_\mu \phi$ due to lack of gauge invariance

$$A_\mu \phi \partial_\mu \phi \rightarrow A_\mu \phi \partial_\mu \phi + (\partial_\mu \alpha) \phi \partial_\mu \phi \quad (232)$$

- Since scalar fields only have 1 DOF, it is not possible to couple any A_μ with ϕ
 - Hence, we need at least 2 fields
- We can write $\phi = \phi_1 + i\phi_2$ and have it transform as

$$\phi \rightarrow e^{-i\alpha} \phi \quad (233)$$

- Therefore $m^2 |\phi|^2$ is gauge invariant

3.1.4.1. Derivatives

- Notice that $|\partial_\mu \phi|^2$ is not gauge invariant
- Hence, we note that in the transformation

$$A_\mu \rightarrow A_\mu + \frac{1}{e} \partial_\mu \alpha \quad (234)$$

(electric charge is convention)

$$(\partial_m + ieA_m)\phi \rightarrow (\partial_m + ieA_m + i\partial_m \alpha)e^{-i\alpha(x)}\phi \quad (235)$$

. By the product rule,

$$e^{-i\alpha(x)}(\partial_\mu + ieA_\mu)\phi \quad (236)$$

- Thus, we can define the *Covariant Derivative*:

$$D_\mu \phi \equiv (\partial_\mu + ieA_\mu)\phi \rightarrow e^{-i\alpha(x)} D_\mu \phi \quad (237)$$

- The Covariant derivative provides us the *Scalar QED* lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (D_\mu \phi)^\dagger (D^\mu \phi) - m^2 \phi^\dagger \phi \quad (238)$$

- In general, for fields ϕ_n , they transform under

$$\phi_n \rightarrow e^{Q_n i\alpha(x)} \phi_n \quad (239)$$

- The covariant derivative would be

$$D_\mu \phi_n = (\partial_\mu - ieQ_n A_\mu) \phi_n \quad (240)$$

- Q represents the sign of the charge (for an electron, $Q = -1$, for quarks, $Q = \frac{2}{3}, -\frac{1}{3}$, etc.)
- This is an important relation that will be seen in more detail later.

3.1.4.2. Conserved Current of Scalar QED

- The scalar QED lagrangian can be written as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \partial_\mu \phi^* \partial^\mu \phi + ieA_\mu (\phi \overleftrightarrow{\partial}^\mu \phi^*) + e^2 A_\mu^2 \phi^* \phi - m^2 \phi^* \phi \quad (241)$$

- The equations of motion for this lagrangian can be written as

$$\begin{aligned} (\square + m^2)\phi &= -2ieA^\mu \partial_\mu \phi + e^2 A_\mu^2 \phi \\ (\square + m^2)\phi^* &= 2ieA^\mu \partial_\mu \phi^* + e^2 A_\mu^2 \phi^* \end{aligned} \quad (242)$$

- This allows us to obtain the Noether currents for this (associated with complex phase rotation):

$$\frac{\delta \phi}{\delta \alpha} = -i\phi, \quad \frac{\delta \phi^*}{\delta \alpha} = i\phi^* \quad (243)$$

- The current is

$$J_\mu = \sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} = -i \underbrace{(\phi \overleftrightarrow{\partial}_\mu \phi^*)}_{\text{Free Theory current}} - 2eA_\mu \phi^* \phi \quad (244)$$

Note (Term in lagrangian):

- The term in $\mathcal{L}$ that is linear in A_μ is $-eA_\mu J_\mu$
- There is a reason behind this.
- Define $\mathcal{L}_0$ as the limit of Gauge invariant lagrangians as $A_\mu \rightarrow 0$ or $e \rightarrow 0$
- Then, $\mathcal{L}_0$ will still be invariant under the global symmetry such that A_μ is the gauge field
 - If α is constant, then A will not transform
- If $\alpha = \alpha(x)$, $\mathcal{L}_0$ transforms based on $\partial_\mu \alpha$
- Suppose this is an infinitesimal transformation. Then,

$$\delta \mathcal{L}_0 = (\partial_\mu \alpha) J^\mu + \mathcal{O}(\alpha^2) \quad (245)$$

- In scalar QED with $A_\mu = 0$,

$$\mathcal{L}_0 = (\partial_\mu \phi)^* (\partial^\mu \phi) - m^2 |\phi|^2 \quad (246)$$

- This implies that an infinitesimal transformation is

$$\delta \mathcal{L}_0 = (\partial_\mu \alpha) J^\mu + (\partial_\mu \alpha)^2 \phi \quad (247)$$

- Integrating by parts and taking the terms linear in α , we obtain

$$\delta\mathcal{L}_0 = \alpha\partial_\mu J_\mu \quad (248)$$

- Then, under the EOM (since any variation must vanish), $\partial_\mu J_\mu = 0$
- There is another way without using the EOM.
- Suppose that we define $\mathcal{L} = \mathcal{L}_0 - A_\mu J_\mu$ s.t.

$$\delta\mathcal{L} = \mathcal{L}_0 - \delta A_\mu J^\mu = (\partial_\mu \alpha) J^\mu - (\partial_\mu \alpha) J^\mu = 0 \quad (249)$$

- This implies that the $A_\mu J_\mu$ term is universal

3.1.5. Quantization (and Ward Identity)

- Quantization of fields with more than 1 degrees of freedom works similarly to quantization of fields with 1 degree of freedom. For instance, quantizing complex scalar fields.

Example 3.1.5.1 (Quantizing complex Scalar fields):

- Suppose we have a scalar field $\phi = \phi_1 + i\phi_2$. We can write

$$\begin{aligned} \phi_1(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_{p,1} e^{-ipx} + a_{p,1}^\dagger e^{ipx}) \\ \phi_2(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_{p,2} e^{-ipx} + a_{p,2}^\dagger e^{ipx}) \end{aligned} \quad (250)$$

- Writing

$$\vec{\phi}(x) = \phi_1 + i\phi_2 = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \quad (251)$$

- We obtain

$$\begin{aligned} \vec{\phi}(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[\begin{pmatrix} a_{p,1} \\ a_{p,2} \end{pmatrix} e^{-ipx} + \begin{pmatrix} a_{p,1}^\dagger \\ a_{p,2}^\dagger \end{pmatrix} e^{ipx} \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{j \in \{1,2,3\}} (\vec{e}_j a_{p,j} + \vec{e}_j^* a_{p,j}^\dagger e^{ipx}) \end{aligned} \quad (252)$$

$$\text{with } \vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- For quantization of vector fields, we must allow for 4 components for basis polarizations instead of 2 components
 - $\vec{\epsilon}_i \rightarrow \epsilon_j^\mu(p)$
- As seen before in *Massive Spin 1*,

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{j \in \{1,2,3\}} (\epsilon_\mu^j(p) a_{p,j} e^{-ipx} + \epsilon_\mu^{j*}(p) a_{p,j}^\dagger e^{ipx}) \quad (253)$$

is how we quantized such fields

- $$\sum_{\mu} \varepsilon_{\mu}^0(p) \quad (254)$$

is known as the canonical set of bases

3.1.5.1. Creation and Annihilation operators for photons

- For creation operators,

$$a_{p,j}^{\dagger}|0\rangle = \frac{1}{\sqrt{2\omega_p}}|p \quad \varepsilon^j\rangle \quad (255)$$

- This implies that

$$\langle 0 | A_{\mu}(x) | p, \varepsilon^j \rangle = \varepsilon_{\mu}^j e^{-ipx} \quad (256)$$

- The vector field creates a particle at x with the polarization able to be projected through contraction
- Notice that the basis depends on p^{μ}
 - If ε_{μ} were not dependent on p (i.e. const.), a Lorentz transform $\varepsilon_{\mu}^i \rightarrow \Lambda_{\mu\nu} \varepsilon_{\nu}^i$ would mix timelike

3.1.5.2. Polarizations

- Now what happens if $p^{\mu} = (E, 0, 0, p_z)$
 - This would imply that

$$\varepsilon_{\mu}^1(p) = (0, 1, 0, 0) \varepsilon_{\mu}^2(p) = (0, 0, 1, 0) \varepsilon_{\mu}^L(p) = (p_z/m, 0, 0, E/m) \quad (257)$$

- Notice that it satisfies the conditions

$$\varepsilon_m^i \varepsilon_m^{i*} = 1 \varepsilon_m^i p_m = 0 \quad (258)$$

- However, one must wonder. Where did the fourth degree of freedom go? Clearly, our vector field is functional with only these three as bases but it still begs the question.
- There is still one more polarization that is perpendicular to all of these. This polarization is expressed as

$$\varepsilon^{\mu}(p) = \frac{1}{m} p^{\mu} = \left(\frac{E}{m}, 0, 0, \frac{p}{m} \right) \quad (259)$$

- In position space, it can be expressed as

$$\varepsilon_{\mu}^S = \frac{1}{m} \partial_{\mu} \alpha(x) \quad (260)$$

- This means that it is invariant under lorentz transforms
 - Hence, ε_{μ}^i do not mix with it
- SO(3) rotations only mix ε_{μ}^i , however there exist other lorentz transformations that can mix the others
- Suppose that we fix $p^{\mu} = q^{\mu}$
- Our polarization bectores are then $\varepsilon_{\nu} = c_j e_{\mu}^j(q)$

- Now suppose that we have a state $|\varepsilon\rangle = c_j|j\rangle$ such that $\langle j|j\rangle = 1$
- This implies that $\langle\varepsilon|\varepsilon\rangle = |c_1|^2 + |c_2|^2 + |c_3|^2 = \text{positive definite, L.I.}$
- The matrix element for our theory is $\mathcal{M} = \varepsilon_\mu M_\mu$ where ε_μ is a linear combination of ε_μ^j

3.1.6. Massless Spin 1

- We now have enough to establish our quantization for massless spin 1 (a photon)
- Massless spin 1 is the same as massive spin 1 except for the fact that we only need 2 polarizations instead of 3

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{i \in \{1,2\}} \left(\varepsilon_\mu^i(p) a_{p,i} e^{-ipx} + \varepsilon_\mu^{i*}(p) a_{p,i}^\dagger e^{ipx} \right) \quad (261)$$

- A sample basis is again if p points in $\hat{z}$
- Then, we have that

$$\begin{aligned} p^\mu &= (E, 0, 0, E) \\ \varepsilon_m^1(p) &= (0, 1, 0, 0) \\ \varepsilon_m^2(p) &= (0, 0, 1, 0) \end{aligned} \quad (262)$$

- This satisfies the conditions $(\varepsilon_\mu^i)^2 = -1; \varepsilon_\mu^i p^\mu = 0$
- There are, of course, orthogonal polarizations like we have seen before.

$$\varepsilon_\mu^f(p) = (1, 0, 0, 1), \varepsilon_\mu^b(p) = (1, 0, 0, -1) \quad (263)$$

- These are known as forward and backwards polarizations
- Notice now that we cannot create ε_μ due to mixing with longitudinal polarization under $\text{SO}(3)$
- In this case, longitudinal polarization becomes forward polarization
- Hence, $\text{SO}(3) \rightarrow \text{ISO}(2)$ or

$$\lim_{m \rightarrow 0} \varepsilon_\mu^i(p) = \varepsilon_\mu^f(p) \propto p_\mu \quad (264)$$

- Under $\text{ISO}(2)$, these polarization bases transform as

$$\begin{aligned} \varepsilon_\mu^1(p) &\rightarrow C_{11}(\Lambda) \varepsilon_\mu^1(p) + C_{12}(\Lambda) \varepsilon_\mu^2(p) + C_{13}(\Lambda) p_\mu \\ \varepsilon_\mu^2(p) &\rightarrow C_{21}(\Lambda) \varepsilon_\mu^1(p) + C_{22}(\Lambda) \varepsilon_\mu^2(p) + C_{23}(\Lambda) p_\mu \end{aligned} \quad (265)$$

3.1.7. Photon Propagator

- We denote the photon propagator by $\Pi^{\mu\nu}$ which is definite by

$$\langle 0 | T \{ A^\mu(x) A^\nu(y) \} | 0 \rangle = i \int \frac{d^4p}{(2\pi)^4} e^{ip(x-y)} \Pi^{\mu\nu}(p) \quad (266)$$

evaluated in the free theory

3.1.7.1. Photon Propagator as a Green's Function

- Suppose we have the lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 - A_\mu J^\mu \quad (267)$$

- Then, the equations of motion are

$$\partial_\mu F_{\mu\nu} = J_\nu \quad (268)$$

- This implies that

$$\partial_\mu \partial_\mu A_\nu - \partial_\mu \partial_\nu A_\mu = J_\nu \quad (269)$$

and hence, via a fourier transform,

$$(-p^2 \delta_{\mu\nu} + p_\mu p_\nu) A^\mu = J_\nu \quad (270)$$

- We want $A_\mu = \Pi_{\mu\nu} J_\nu$. Therefore, as a green's function equation,

$$(-p^2 \delta_{\mu\nu} + p_\mu p_\nu) \Pi_{\mu\alpha} = g_{\mu\alpha} \quad (271)$$

- However, there is a drawback. Notice that

$$\det(-p^2 g_{\mu\nu} + p_\mu p_\nu) = 0 \quad (272)$$

- Therefore, we must choose a gauge or add an auxiliary Field
 - Choosing a gauge is inconvenient and has 4 DOF instead of 2
 - Adding an auxiliary field is similar to utilizing Lagrange multipliers
- We let our lagrangian be

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 - J_\mu A^\mu \quad (273)$$

- Small ξ enforces $\partial_\mu A_\mu = 0$
- Thus we have that

$$\left[-p^2 g_{\mu\nu} + \left(1 - \frac{1}{\xi} \right) p_\mu p_\nu \right] \Pi_{\mu\nu} = g_{\mu\nu} \quad (274)$$

- Therefore, in terms of our new parameter ξ ,

$$\Pi_{\mu\nu} = \frac{g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2}}{p^2} \quad (275)$$

- It is simple to check this.

$$\begin{aligned}
& \left[-q^2 g_{\mu\alpha} + \left(1 - \frac{1}{\xi}\right) p_\mu p_\alpha \right] \Pi_{\alpha\nu} = \\
& \left[p^2 g_{\mu\nu} - \left(1 - \frac{1}{\xi}\right) p_\mu p_\nu \right] [p^2 g_{\alpha\nu} - (1 - \xi) p_\alpha p_\nu] \frac{1}{p^4} \\
& = g_{\mu\nu} + \left[-\left(1 - \frac{1}{\xi}\right) - (1 - \xi) + \left(1 - \frac{1}{\xi}\right)(1 - \xi) \right] \times \frac{p_\mu p_\nu}{p^2} = g_{\mu\nu}
\end{aligned} \tag{276}$$

- Hence, we have our result for the photon propagator:

$$i\Pi_{\mu\nu} = \frac{-i}{p^2 + i\epsilon} \left[g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right] \tag{277}$$

3.1.7.2. Covariant Gauges

- Finally, there are some gauges that are useful to enforce via ξ which make calculations simpler. Notice that the final result for any physical observable must be independent of ξ due to gauge invariance which implies that ξ is to be chosen to make our lives easier.
- There is the feynman guage where $\xi = 1$:

$$i\Pi^{\mu\nu}(p) = \frac{-ig^{\mu\nu}}{p^2 + i\epsilon} \tag{278}$$

- There is the Lorenz gauge where $\xi = 0$:

$$i\Pi^{\mu\nu}(p) = -i \frac{g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}}{p^2 + i\epsilon} \tag{279}$$

- Notice that as $\xi \rightarrow 0$, $\partial_\mu A_\mu \rightarrow 0$

3.2. Scalar QED in More Detail

- Scalar Quantum Electrodynamics describes a photon A_μ and another particle (presumably the electron) ϕ
- Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 + |D_\mu \phi|^2 - m^2 \phi^* \phi \tag{280}$$

- $D_\mu \phi = \partial_\mu \phi + ieA_\mu$
- $(D_\mu \phi)^* = \partial_\mu \phi^* - ieA_\mu$

3.2.1. Quantizing Scalar Fields

- The equations of motion for Scalar Quantum Electrodynamics are similar to the Klein Gordon equation

$$\square \phi + m^2 \phi = -i(-eA_\mu) \partial^\mu \phi + i\partial_\mu (-eA^\mu \phi) + (-eA_\mu)^2 \phi \tag{281}$$

$$\square \phi^* + m^2 \phi^* = i(eA_\mu) \partial^\mu \phi^* + i\partial_\mu (eA_\mu \phi^*) + (eA_\mu)^2 \phi^* \tag{282}$$

- A the solution to a free scalar field (a real one) is given by:

$$\phi_0(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx}) \quad (283)$$

- Generally, scalar fields are complex. Hence, it is convention to treat ϕ and ϕ^* separately

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{ipx} + b_p^\dagger e^{-ipx}) \quad (284)$$

$$\phi^*(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p^\dagger e^{ipx} + b_p e^{-ipx}) \quad (285)$$

3.2.2. Charge

- In scalar QED, as seen before, charge is associated with the U(1) gauge symmetry $\phi \rightarrow e^{-i\alpha} \phi$ which becomes important later

3.2.3. Scalar QED Feynman Rules

- The lagrangian for Scalar QED is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 - \phi^* (\square + m^2) \phi - ie A_\mu (\phi^* \vec{\partial}_\mu \phi) + e^2 A_\mu^2 |\phi|^2 \quad (286)$$

- The complex scalar propagator is

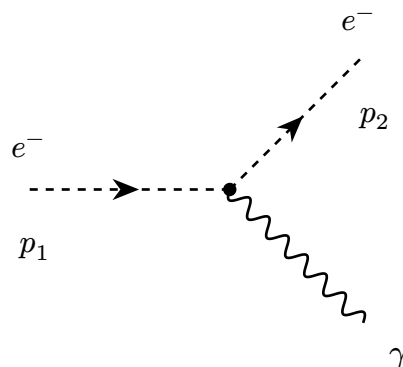
$$\text{---} \rightarrow \text{---} = \frac{1}{p^2 - m^2 + i\epsilon} \quad (287)$$

- The photon propagator is

$$\text{~~~~~} = \frac{-i}{p^2 + i\epsilon} \left[g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right] \quad (288)$$

- There are additionally four different vertex interactions which are as follows:

- The first interaction is



$$= ie(-p_\mu^1 - p_\mu^2) \quad (289)$$

- The second interaction is

(290)

- The third interaction is

(291)

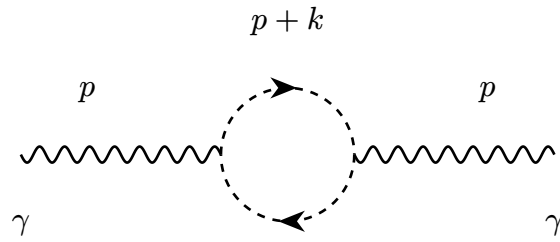
- Finally, the fourth interaction is

(292)

- These are the *only* interactions (at tree level) permitted by QED (scalar QED but similar is true for spinor QED) due to the fact that the interaction term in the lagrangian is

$$ieA_\mu[\phi^* \vec{\partial}_\mu \phi] \quad (293)$$

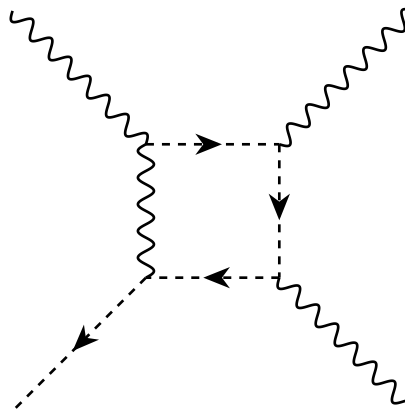
- There are also loops which are possible in any quantum theory.
 - There are an infinite number of such loops, so it is impossible to enumerate them.
 - However, here are a few examples:



$$(294)$$

$$= \varepsilon_\mu(p) \int \frac{d^4 k}{(2\pi)^4} (iek) \frac{i}{k^2 - m^2} \frac{i}{(p+k)^2 - m^2} (-iek) \varepsilon_\nu^*(p)$$

- This yields a divergence which is the reason that renormalization exists (as you will see later)
- There are also other loops, such as



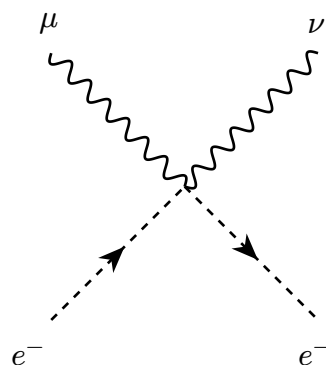
$$(295)$$

(which I have not evaluated here).

- There is an additional vertex which is given by the term

$$\mathcal{L}_{\text{int}} = e^2 A_\mu^2 |\phi|^2 \quad (296)$$

- This is not unphysical but it does not happen in reality since, although it appears in our toy theory, it does not appear in real QED and hence it doesn't actually represent a real interaction
 - It is required in scalar QED due to gauge invariance
- This vertex is known as the *seagull vertex*
- The seagull vertex Feynman Diagram is



$$= 2ie^2 g_{\mu\nu} \quad (297)$$

3.2.3.1. External States

- External States are extremely important for understanding how we go from a diagram (with photons, it will have tensor indices μ, ν) to a squared matrix element which is what we need to calculate the S matrix.
- The idea is that photons that exit and enter the graph (and hence are external) get factors of polarization, which you will see
- Recall the photon quantization:

$$A_\mu(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \sum_{i=1}^2 (\varepsilon_\mu^i(k) a_{k,i} e^{-ikx} + \varepsilon_\mu^{i*}(k) a_{k,i}^\dagger e^{ikx}) \quad (298)$$

- External states have momenta and polarization
- Note that

$$\langle 0 | A_\mu(x) | k, \epsilon \rangle = \epsilon_\mu(k) e^{-ikx} \quad (299)$$

- Hence, the LSZ reduction is modified by adding ε_μ for incoming photons and ε_μ^* for outgoing photons

3.2.4. Scattering In Scalar QED

- Now, we move onto real particle interactions that occur in scalar QED
- It is convenient to utilize three variables known as the mandelstam variables: s, t, u which will be defined soon.
- Consider the t channel defined as

$$\begin{aligned}
 i\mathcal{M}_t = & \quad \begin{array}{c} \text{Diagram: A wavy line (photon) connects two vertices. The top vertex has incoming dashed line p_1 and outgoing dashed line p_3. The bottom vertex has incoming dashed line p_2 and outgoing dashed line p_4.} \end{array} \\
 = & (-ie)(p_1^\mu + p_3^\mu) \frac{-i \left(g_{\mu\nu} - (1-\xi) \frac{k_\mu k_\nu}{k^2} \right)}{k^2} (-ie)(p_2^\nu + p_4^\nu)
 \end{aligned} \quad (300)$$

with $k^\mu = p_3^\mu - p_1^\mu$

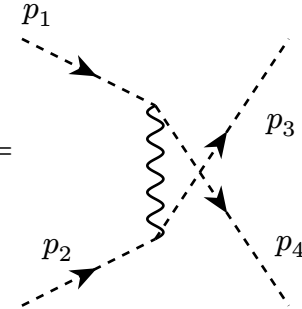
- Notice that

$$k^\mu (p_1^\mu + p_3^\mu) = p_3^2 - p_1^2 = m^2 - m^2 = 0 \quad (301)$$

- Hence,

$$\mathcal{M}_t = \frac{(p_1^\mu + p_3^\mu)(p_2^\mu + p_4^\mu)}{t} \quad (302)$$

- Now consider the u channel:



$$i\mathcal{M}_u =$$

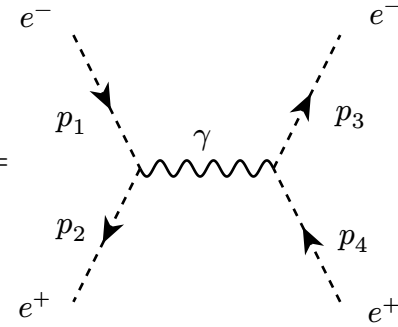
$$= (-ie)(p_1^\mu + p_4^\mu) \frac{-i[g_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2}]}{k^2} (-ie)(p_2^\nu + p_3^\nu)$$
(303)

• With $k^\mu \equiv p_4^\mu - p_1^\mu$

• Now we have similar equations to before

$$\mathcal{M}_u = e^2 \frac{(p_1^\mu + p_4^\mu)(p_2^\mu + p_3^\mu)}{u}$$
(304)

• The final diagram is the s-channel



$$i\mathcal{M}_s =$$

$$= (-ie)(-p_1^\mu + p_2^\mu) \frac{-i\left(g_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2}\right)}{k^2} (-ie)(-p_3^\nu + p_4^\nu)$$
(305)

with $k^\mu = p_1^\mu + p_2^\mu$

• Notice that $k^\mu(p_1^\mu - p_2^\mu) = 0$

• Hence, this diagram evaluates to

$$i\mathcal{M}_s = \frac{(p_1^\mu - p_2^\mu)(p_3^\mu - p_4^\mu)}{s}$$
(306)

• Our Mandelstam variables that we introduces u, s, t evaluate to

$$u = (p_4 - p_1)^2, \quad s = (p_1 - p_2)^2, \quad t = (p_1 + p_2)^2$$
(307)

• s is commonly used to represent the square of the COM energy from a particle accelerator

3.2.4.1. Moller Scattering

• The cross section for $e^-e^- \rightarrow e^-e^-$ moller scattering can be evaluated with u and t channels, providing

$$\begin{aligned}
& \frac{d\sigma}{d\Omega}(e^-e^- \rightarrow e^-e^-) \\
&= \frac{e^4}{64\pi^2 E_{cm}^2} \left[\frac{(p_1^\mu + p_3^\mu)(p_2^\mu + p_4^\mu)}{t} + \frac{(p_1^\mu + p_4^\mu)(p_2^\mu + p_3^\mu)}{u} \right]^2 \\
&= \frac{\alpha^2}{4s} \left[\frac{s-u}{t} + \frac{s-t}{u} \right]^2
\end{aligned} \tag{308}$$

3.2.5. Ward Identity

- Notice how the above matrix elements are all independent of our gauge parameter ξ
 - This relates to the Ward Identity, an extremely important result which can be generalized to the Ward Takahashi identity (as the reader may see later)
- The photon propagator that we had was

$$i\Pi_{\mu\nu} = \frac{-i \left[g_{\mu\nu} - (1-\xi) \frac{p_\mu p_\nu}{p^2} \right]}{p^2 + i\varepsilon} \tag{309}$$

- With an internal photon, the matrix element satisfies

$$M_{\mu\nu} \Pi^{\mu\nu} = 0 \tag{310}$$

- Due to gauge invariance $\Pi^{\mu\nu}$ should be independent of ξ .
- Hence,

$$\mathcal{M}_{\mu\nu} p^\mu p^\nu = 0 \tag{311}$$

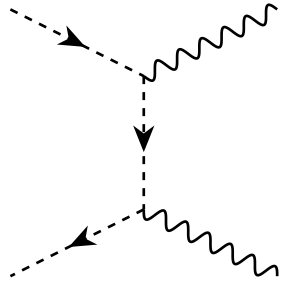
- An alternative way to write this is

$$p_\mu \mathcal{M}_{\mu\nu} = 0 \tag{312}$$

- The proof for the ward identity is difficult perturbatively

Example 3.2.5.1 (Nontrivial Scattering):

- An example of using external photons is the non trivial $e^-e^- \rightarrow \gamma\gamma$
- The t channel diagram for this is



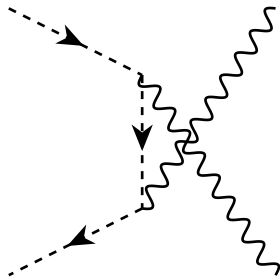
The diagram shows two incoming electrons (dashed lines with arrows) and two outgoing photons (wavy lines). A vertical dashed line represents the t-channel exchange of an electron. The incoming electron from the top-left has momentum p_1 and the outgoing electron to the bottom-left has momentum p_3 . The incoming electron from the top-right has momentum p_2 and the outgoing electron to the bottom-right has momentum p_4 . The vertical dashed line has momentum $p_1 - p_3$.

$$i\mathcal{M}_t = (-ie)^2 \frac{i(2p_1^\mu - p_3^\mu)(p_4^\nu - 2p_2^\nu)}{(p_1 - p_3)^2 - m^2} \epsilon_3^{*\mu} \epsilon_4^{*\nu} \quad (313)$$

- Then,

$$\mathcal{M}_t = e^2 \frac{(p_3 \cdot \epsilon_3^* - 2p_1 \cdot \epsilon_3^*)(p_4 \cdot \epsilon_4^* - 2p_2 \cdot \epsilon_4^*)}{p_3^2 - 2p_3 \cdot p_1} \quad (314)$$

- The u channel diagram for this interaction is



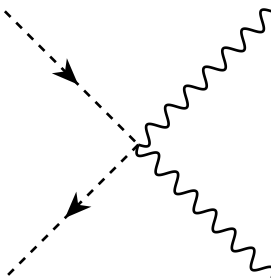
The diagram is similar to the t-channel one, but the vertical dashed line has momentum $p_2 - p_4$. The incoming electron from the top-left has momentum p_1 and the outgoing electron to the bottom-left has momentum p_3 . The incoming electron from the top-right has momentum p_2 and the outgoing electron to the bottom-right has momentum p_4 .

$$i\mathcal{M}_u = ie^2 \frac{(p_3 \cdot \epsilon_3^* - 2p_2 \cdot \epsilon_3^*)(p_4 \cdot \epsilon_4^* - 2p_1 \cdot \epsilon_4^*)}{p_3^2 - 2p_3 \cdot p_2} \quad (315)$$

- Checking the ward identity, we replace $\epsilon_3^{*\nu}$ with p_3^μ .

$$\begin{aligned} \mathcal{M}_1 + \mathcal{M}_u &= e^2 [p_4 \cdot \epsilon_4^* - 2p_2 \cdot \epsilon_4^* + p_4 \cdot \epsilon_4^* - 2p_1 \cdot \epsilon_4^*] \\ &= 2e^2 \epsilon_4^{*\mu} (p_4^\mu - p_2^\mu - p_1^\mu) \neq 0 \end{aligned} \quad (316)$$

- The ward identity isnt true because we don't have the seagull vertex:



The diagram shows two incoming electrons (dashed lines with arrows) and two outgoing photons (wavy lines) meeting at a central vertex. The incoming electron from the top-left has momentum p_1 and the outgoing electron to the bottom-left has momentum p_3 . The incoming electron from the top-right has momentum p_2 and the outgoing electron to the bottom-right has momentum p_4 .

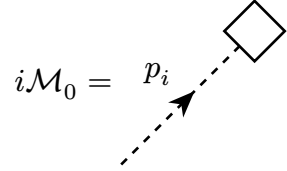
$$i\mathcal{M}_4 = ieg_{\mu\nu} \epsilon_3^{*\mu} \epsilon_4^{*\nu} \quad (317)$$

- Then, replacing polarization with momentum,

$$\begin{aligned} \mathcal{M}_f + \mathcal{M}_u + \mathcal{M}_4 &= 2e^2 \epsilon_\mu (p_4^\mu - p_2^\mu - p_1^\mu + p_3^\mu) \\ &= 0 \end{aligned} \quad (318)$$

Theorem 3.2.5.1 (Massless Spin 1):

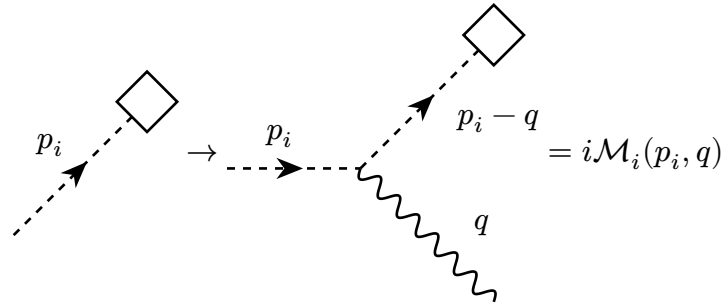
- Massless Spin 1 particles imply conservation of charge
- This is a nontrivial result that has important implications
- Consider



$$i\mathcal{M}_0 = p_i \quad (319)$$

where the diamond is a black box

- Now consider



$$p_i \rightarrow p_i \rightarrow p_i - q = i\mathcal{M}_i(p_i, q) \quad (320)$$

- Then,

$$\mathcal{M}_i(p_i, q) = (-ie) \frac{i[p_i^\mu + (p_i^\mu - q^\mu)]}{(p_i - q)^2 - m^2} \epsilon^\mu \mathcal{M}_0(p_i - q) \quad (321)$$

- If $p_i^2 = m^2$, $q^2 = 0$, $q_\mu \epsilon_\mu = 0$, we have that

$$\mathcal{M}_i(p_i, q) = -e \frac{p_i \cdot \epsilon}{p_i \cdot q} \mathcal{M}_0(p_i, -q) \quad (322)$$

- In the limit $|q \cdot p_i| \ll |p_j \cdot p_k|$,

$$\mathcal{M}_0(p_i - q) \approx \mathcal{M}_0(p_i) \quad (323)$$

- In this case, we would obtain

$$\mathcal{M}_i(p_i, q) \approx -e \frac{p_i \cdot \epsilon}{p_i \cdot q} \mathcal{M}_0(p_i) \quad (324)$$

- In the case of an incoming positron,

$$\mathcal{M}_i(p_i, q) \approx e \frac{p_i \cdot \epsilon}{p_i \cdot q} \mathcal{M}_0(p_i) \quad (325)$$

- Now consider outgoing particles

$$i\mathcal{M}_0 = \text{[Diagram: A dashed line with an arrow labeled } p_i \text{ entering a square loop from the top-left, and a dashed line exiting from the top-right.]} \quad (326)$$

- Adding a photon coming out of the electron provides

$$\text{[Diagram: A dashed line with an arrow labeled } p_i \text{ enters a square loop from the top-left. A dashed line with an arrow labeled } p_i \text{ exits from the top-right. A wavy line labeled } q \text{ exits from the bottom-right.]} \rightarrow \text{[Diagram: A dashed line with an arrow labeled } p_i \text{ enters a square loop from the top-left. A dashed line with an arrow labeled } p_i - q \text{ exits from the top-right. A wavy line labeled } q \text{ exits from the bottom-right.]} = i\mathcal{M}_i \quad (327)$$

- This implies that

$$\begin{aligned} \mathcal{M}_i(p_i, q) &= (-ie) \frac{i(p_i^\mu + (p_i^\mu + q^\mu))}{(p_i + q)^2 - m^2} \varepsilon_\mu \mathcal{M}_0(p_i + q) \\ &\approx e \frac{p_i \cdot \varepsilon}{p_i \cdot q} \mathcal{M}_0(p_i) \end{aligned} \quad (328)$$

- For a positron,

$$M_i(p_i, q) \approx -e \frac{p_i \cdot \varepsilon}{p_i \cdot q} M_0(p_i) \quad (329)$$

- Notice that this implies that

$$\mathcal{M} \simeq e\mathcal{M}_0 \left[\sum_{inc.} Q_i \frac{p_i \cdot \varepsilon}{p_i \cdot q} - \sum_{outgo.} Q_i \frac{p_i \cdot \varepsilon}{p_i \cdot q} \right] \quad (330)$$

- Under a lorentz transformation, $\varepsilon_\mu \rightarrow \varepsilon_\mu + q_\mu$ and hence

$$\mathcal{M} \rightarrow \mathcal{M} + \mathcal{M}_0 \left[\sum_{incoming} Q_i - \sum_{outgoing} Q_i \right] \quad (331)$$

- Therefore,

$$\sum_{incoming} Q_i = \sum_{outgoing} Q_i, \quad (332)$$

Part IV

Spinors

4.1. Lorentz and Spinors

Some Supplementary material for this section is Spinors for Beginners by Eigenchris which is very useful to get a grasp of the actual math that is going on.

4.1.1. Lorentz Transformations

Definition 4.1.1.1 (Metric Tensor): A (or in special relativity, *The*) metric tensor is defined as being the dot product of basis vectors

$$g_{\mu\nu} = \hat{e}_{(\mu)} \cdot \hat{e}_{(\nu)} \quad (333)$$

This definition is all but useless in Quantum Field Theory due to the whole theory entirely using Special Relativity instead of General Relativity⁷

The special relativity metric tensor (which is subject to arguments over convention), is either $g_{\mu\nu} = \text{diag}(+, -, -, -)$ or $g_{\mu\nu} = \text{diag}(-, +, +, +)$ ⁸.

Definition 4.1.1.2 (Lorentz Transformation & Lorentz Invariance): A *Lorentz transformation* Λ is defined as a coordinate transformation

$$\Lambda : (t, x, y, z) \rightarrow (t', x', y', z') \quad (334)$$

such that

$$\Lambda^T g \Lambda = g \quad (335)$$

or

$$\Lambda_{\alpha}^{\mu} \Lambda_{\beta}^{\nu} \eta_{\mu\nu} = \eta_{\alpha\beta} \quad (336)$$

In special relativity (minkowski space), this condition is loosened to

$$s^2 = t^2 - x^2 \quad (337)$$

is conserved but the previous condition is still useful in discussion of group theory and derivation of *Spinors*. Generally, Lorentz transformations are transformations that

4.1.2. Group Theory (cont'd)

Definition 4.1.2.1 (Lorentz Group): This is the set of operations Λ such that

$$\Lambda^T g \Lambda = g \quad (338)$$

- For instance, a Lorentz transformation could follow the 4-vector representation, defined as

⁷Unless string theory, again

⁸Personally I am used to $\text{diag}(+, -, -, -)$.

$$x_\mu \rightarrow \Lambda_{\mu\nu} x_\nu \quad (339)$$

- More generally, a field ϕ transforms under a representation of the lorentz group as

$$\phi_{i(x)} \rightarrow R(\Lambda)_{ij} \phi_j(\Lambda^{-1}x) \quad (340)$$

(Similar to a matrix transformation)

- The continuous Lorentz transformations that exist are *rotations* and *boosts*, which are the most general possible (continuous) Lorentz transformations which can occur as they (+ the discrete Lorentz transformations such as P and T) form the **Lorentz Group**
- Rotations can be represented, again in the 4-vector representation, as rotation matrices. We let $\theta_x, \theta_y, \theta_z$ represent rotations in the x, y, z axes respectively. Then, the rotation matrices can be represent as as

$$\Lambda(\theta_x) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & \cos \theta_x & \sin \theta_x & \\ & -\sin \theta_x & \cos \theta_x & \end{pmatrix}, \Lambda(\theta_y) = \begin{pmatrix} 1 & & & \\ & \cos \theta_y & -\sin \theta_y & \\ & & 1 & \\ & \sin \theta_y & & \cos \theta_y \end{pmatrix} \quad (341)$$

$$\Lambda(\theta_z) = \begin{pmatrix} 1 & & & \\ & \cos \theta_z & \sin \theta_z & \\ & -\sin \theta_z & \cos \theta_z & \\ & & & 1 \end{pmatrix}$$

- In a similar manner, boosts can be represented by matrices in the 4-vector representation. Denoting $\beta_x, \beta_y, \beta_z$ as the boost angles (in the hyperbolic trigonometric functions), we have that

$$\Lambda(\beta_x) = \begin{pmatrix} \cosh \beta_x & \sinh \beta_x & & \\ \sinh \beta_x & \cosh \beta_x & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \Lambda(\beta_y) = \begin{pmatrix} \cosh \beta_y & \sinh \beta_y & & \\ & 1 & & \\ \sinh \beta_y & & \cosh \beta_y & \\ & & & 1 \end{pmatrix} \quad (342)$$

$$\Lambda(\beta_z) = \begin{pmatrix} \cosh \beta_z & & \sinh \beta_z & \\ & 1 & & \\ & & 1 & \\ \sinh \beta_z & & & \cosh \beta_z \end{pmatrix}$$

- Can utilize *infinitesimal* Lorentz transformations in order to extract the group ($O(1, 3)$) (we let β_i parametrize boosts and θ_i parametrize rotations)

►

$$\begin{aligned} \delta x_0 &= \beta_i x_i \\ \delta x_i &= \beta_i x_0 - \varepsilon_{ijk} \theta_j x_k \end{aligned} \quad (343)$$

or

$$\delta x_\mu = i \left[\sum_{i \in \{1,2,3\}} \theta(J_i)_{\mu\nu} + \beta_i (K_i)_{\mu\nu} \right] x_\nu \quad (344)$$

- J_i and K_i matrices:

$$\begin{aligned}
J_1 &= i \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix}, J_2 = i \begin{pmatrix} 0 & & & \\ & 0 & 1 & \\ & & 0 & 0 \\ -1 & & & 0 \end{pmatrix}, J_3 = i \begin{pmatrix} 0 & & & \\ & 0 & -1 & \\ & 1 & 0 & \\ & & & 0 \end{pmatrix} \\
K_1 &= i \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}, K_2 = i \begin{pmatrix} 0 & -1 & & \\ & 0 & & \\ -1 & & 0 & \\ & & & 0 \end{pmatrix}, K_3 = i \begin{pmatrix} 0 & & -1 & \\ & 0 & & \\ & & 0 & \\ -1 & & & 0 \end{pmatrix}
\end{aligned} \tag{345}$$

- Such matrices are *generators of the Lorentz group*
 - Can use to obtain the Lie Algebra

$$\Lambda = \exp(i\theta_i J_i + i\beta_i K_i) \tag{346}$$

4.2. Dirac Spinors (TODO)

4.2.1. Spinors

Definition 4.2.1.1 (Dirac Spinor): An object ψ obeying the transformation properties

$$\psi \rightarrow S[\Lambda]\psi, \psi^\dagger \rightarrow \psi^\dagger S[\Lambda] \tag{347}$$

- What does this mean?
- A lorentz transformation can be written as

$$\Lambda_\nu^\mu = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}(J^{\rho\sigma})_\nu^\mu\right) \tag{348}$$

with new notation introduced for convenience such that $\Omega_{0i} = \beta_i$ boosts, $\Omega_{12} = -\Omega_{21} = \theta_z$ (rotation angles) with $(J^{\rho\sigma})^{\mu\nu} = g^{\rho\mu}g^{\sigma\nu} - g^{\sigma\mu}g^{\rho\nu}$

- These are another way to write generators for the lorentz group
- Notice that, using this tensor, we can write

$$K_i = -i(J^{0i})_\nu^\mu \tag{349}$$

for boosts and

$$J_i = \frac{i}{2}\varepsilon_{ijk}(J^{jk})_\nu^\mu \tag{350}$$

for rotations

- These generators form $\mathfrak{so}(1, 3)$, the Lorentz Algebra
 - Their exponential gives a representation of the Lorentz Group double cover $\text{Spin}(1, 3) \cong \text{SL}(2, \mathbb{C})$

Theorem 4.2.1.1: The Lorentz Algebra, $\mathfrak{so}(1, 3)$, has two commuting sub-algebras represented by the 3d rotation algebras $\mathfrak{su}(2)$

Proof:

- Notice that if we switch back to J_i and K_i , we can derive $SU(2)$, the group for spinors.
- Define $J_i^+ \equiv \frac{1}{2}(J_i + iK_i)$, $J_i^- \equiv \frac{1}{2}(J_i - iK_i)$
- We have then,

$$\begin{aligned} [J_i^+, J_j^+] &= i\varepsilon_{ijk}J_k^+ \\ [J_i^-, J_j^-] &= i\varepsilon_{ijk}J_k^- \\ [J_i^+, J_i^-] &= 0 \end{aligned} \tag{351}$$

hence, $\mathfrak{so}(1, 3)$ has 2 commuting sub algebras, both of which are 3d rotation groups given by the Levi-Cevita connections. Hence, $SO(1, 3) \cong SU(2) \oplus SU(2)$

■

- Hence, we now have the definition of a spinor

Definition 4.2.1.2 (Spinor): A *Spinor* is an object that transforms under $SU(2)$

4.2.2. Clifford Algebra

- To construct Dirac Spinors, one must first construct gamma matrices, which form the basis of a Clifford Algebra $\mathcal{C}\ell_{1,3}(\mathbb{R})$

Definition 4.2.2.1 (Gamma Matrices): Gamma matrices are defined as being anticommuting matrices such that

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \tag{352}$$

and having the property that

$$(\gamma^0)^2 = 1, (\gamma^i)^2 = -1 \tag{353}$$

- As seen in the *math toolkit*, these matrices can be represented as

$$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix} \tag{354}$$

with $\sigma^\mu = (\mathbb{1}, \vec{\sigma})$, $\bar{\sigma}^\mu = (\mathbb{1}, -\vec{\sigma})$ and σ^i being Pauli Matrices

- This representation is known as the Weyl Representation
- An additional piece of useful notation is the idea that $\not{p} = p_\mu \gamma^\mu = p^\mu \gamma_\mu$

Pauli Matrix and Gamma Matrix Identities

$$\sigma^i \sigma^j = \delta^{ij} + i\epsilon^{ijk} \sigma^k, \{\sigma^i, \sigma^j\} = 2\delta^{ij}, [\sigma^i, \sigma^j] = 2i\epsilon^{ijk} \sigma^k \quad (355)$$

For Pauli Matrices

- A useful quantity to define is a representation of the Lorentz Algebra

$$S^{\rho\sigma} = \frac{1}{2}[\gamma^\rho, \gamma^\sigma] = \frac{1}{2}\gamma^\rho\gamma^\sigma - \frac{1}{2}\gamma^\sigma\gamma^\rho \quad (356)$$

4.2.3. Lorentz Group Representation

- We now want to derive why spinors behave the way that they do. That is, why do they have spin $\frac{1}{2}$?
- As seen before, we have our generators written as linear combinations

$$J_i^+ \equiv \frac{1}{2}(J_i + iK_i), \quad J_i^- \equiv \frac{1}{2}(J_i - iK_i) \quad (357)$$

which formed two commuting subgroups $\text{SO}(1, 3) \cong \text{SU}(2) \times \text{SU}(2)$

- There are rotation generators $\vec{J}$ and hence the Lorentz group is a super of $\text{SO}(3)$
 - Any representation of $\text{SO}(3)$ is a representation of the Lorentz Group
- Notice that $\text{SU}(2)$ both have quantum numbers A and B . The spin is their sum.

$\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ REPRESENTATION	(A, B)	$(0, 0)$	$(\frac{1}{2}, 0)$	$(0, \frac{1}{2})$	$(\frac{1}{2}, \frac{1}{2})$	$(1, 1)$
$\mathfrak{so}(3)$ representation	$j = A + B$	0	$\frac{1}{2}$	$\frac{1}{2}$	$1 \oplus 0$	$2 \oplus 1 \oplus 0$

- In this table, you can see that there are two different representations of $j = \frac{1}{2}$. Therefore, there are two different type of spinors (this corresponds to chirality which you will see)
- Now we 2×2 matrices such that

$$\begin{aligned} [J_i^+, J_j^+] &= i\epsilon_{ijk} J_k^+ \\ [J_i^-, J_j^-] &= i\epsilon_{ijk} J_k^- \\ [J_i^+, J_j^-] &= 0 \end{aligned} \quad (358)$$

- Remember our sigma matrix identities?, we had that

$$\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i\epsilon_{ijk} \frac{\sigma_k}{2} \quad (359)$$

which is the perfect $\mathfrak{so}(3)$ algebra that we wanted

- For the $(\frac{1}{2}, 0)$ representation, we can set $J^+ = 0$ and $J_i^- = \frac{1}{2}\sigma_i$ as our generators
- For the $(0, \frac{1}{2})$ representation, we can set $J^- = 0$ and $J_i^+ = \frac{1}{2}\sigma_i$ as our generators
- Now we are able to get our Lorentz transformations.

- For rotations $\vec{J} = \vec{J}^- + \vec{J}^+$
- For our boosts, $\vec{K} = i(\vec{J}^- - \vec{J}^+)$
- Thus,

$$\begin{aligned} \left(\frac{1}{2}, 0\right) : \vec{J} &= \frac{1}{2}\vec{\sigma}, \quad \vec{K} = \frac{i}{2}\vec{\sigma} \\ \left(0, \frac{1}{2}\right) : \vec{J} &= \frac{1}{2}\vec{\sigma}, \quad \vec{K} = -\frac{i}{2}\vec{\sigma} \end{aligned} \quad (360)$$

Note (Hermicity):

- Rotations are hermitian and boosts are anti hermitian
- Now we have an even better way to define spinors in terms of their handedness

Definition 4.2.3.1 (Right Handed Weyl Spinor):

- An object transforming under the $(0, \frac{1}{2})$ representation of the Lorentz group
- Right handed weyl spinors transform as

$$\psi_R \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j + \beta_j\sigma_j)}\psi_R = (1 + \frac{i}{2}\theta_j\sigma_j + \frac{1}{2}\beta_j\sigma_j + \dots)\psi_R \quad (361)$$

Definition 4.2.3.2 (Left Handed Weyl Spinor):

- An object transforming under the $(\frac{1}{2}, 0)$ representation of the Lorentz group
- These transform as

$$\psi_L \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j - \beta_j\sigma_j)}\psi_L = (1 + \frac{i}{2}\theta_j\sigma_j - \frac{1}{2}\beta_j\sigma_j + \dots)\psi_L \quad (362)$$

- There are also infinitesimal transformations which are

$$\begin{aligned} \delta\psi_R &= \frac{1}{2}(i\theta_j + \beta_j)\sigma_j\psi_R \\ \delta\psi_L &= \frac{1}{2}(i\theta_j - \beta_j)\sigma_j\psi_L \\ \delta\psi_R^\dagger &= \frac{1}{2}(-i\theta_j + \beta_j)\psi_R^\dagger\sigma_j \\ \delta\psi_L^\dagger &= \frac{1}{2}(-i\theta_j - \beta_j)\psi_L^\dagger\sigma_j \end{aligned} \quad (363)$$

Note (Unitarity): Since there is no finite dimensional unitary representation of the lorentz group, spinor bases must be built in terms of momentum, much like the photon field

4.2.4. Lorentz Invariant Lagrangian

- How do we get a Lorentz invariant lagrangian with our new spinor valued fields?
- First, let us write
 - $(\frac{1}{2}, 0)$ rep:

$$\psi_R(x) = \begin{pmatrix} \psi_{1R}(x) \\ \psi_{2R}(x) \end{pmatrix} \quad (364)$$

- $(0, \frac{1}{2})$ rep:

$$\psi_L(x) = \begin{pmatrix} \psi_{1L}(x) \\ \psi_{2L}(x) \end{pmatrix} \quad (365)$$

- We want a lagrangian with the right number of DOF, of course
- Let us be naive and write

$$\mathcal{L} = (\psi_R)^\dagger \square \psi_R + m^2 (\psi_R)^\dagger \psi_R \quad (366)$$

- not L.I.:

$$\begin{aligned} \delta(\psi_R^\dagger \psi_R) &= \frac{1}{2} \psi_R^\dagger [i\theta_j + \beta_j \sigma_j] \psi_R + \frac{1}{2} [\psi_R^\dagger (-i\theta_j + \beta_j) \sigma_j] \psi_R \\ &= \beta_i \psi_R^\dagger \sigma_i \psi_R \neq 0 \end{aligned} \quad (367)$$

- Therefore, we must have two fields ψ_R and ψ_L
 - Let's try terms like $\psi_L^\dagger \psi_R$:

$$\delta(\psi_L^\dagger \psi_R) = \left[\psi_L^\dagger \frac{i}{2} (-i\theta_i - \beta_i) \sigma_i \right] \psi_R + \psi_L^\dagger \left[\frac{i}{2} (i\theta_i + \beta_i) \sigma_i \right] \psi_R = 0 \quad (368)$$

- Adding the h.c. will give us a lagrangian:

$$\mathcal{L}_{\text{Dirac Mass}} = m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L) \quad (369)$$

- Now we want kinetic terms to add to the lagrangian
 - we can try adding terms like $\mathcal{L}_{\text{kin}} = \psi_L^\dagger \square \psi_R + \psi_R^\dagger \square \psi_L$ but these are “not interesting” [1]
 - Let's look at a term like $\psi_R^\dagger \sigma_i \psi_R$:

Algebra: calculations for our new term

$$\begin{aligned} \delta(\psi_R^\dagger \sigma_i \psi_R) &= \frac{1}{2} \psi_R^\dagger \sigma_i [(i\theta_j + \beta_j) \sigma_j] \psi_R + \frac{1}{2} [\psi_R^\dagger (-i\theta_j + \beta_j) \sigma_j] \sigma_i \psi_R \\ &= \frac{\beta_j}{2} \psi_R^\dagger (\sigma_i \sigma_j + \sigma_j \sigma_i) \psi_R + \frac{i\theta_j}{2} (\sigma_i \sigma_j - \sigma_j \sigma_i) \psi_R \\ &= \beta_i \psi_R^\dagger \psi_R - \theta_j \epsilon_{ijk} \psi_R^\dagger \sigma_k \psi_R \end{aligned} \quad (370)$$

$$\therefore \delta(\psi_R^\dagger \psi_R, \psi_R^\dagger \sigma_i \psi_R) = (\beta_i \psi_R^\dagger \sigma_i \psi_R, \beta_i \psi_R^\dagger \psi_R - \epsilon_{ijk} B_j \psi_R^\dagger \sigma_k \psi_R) \quad (371)$$

- Notice that this is exactly how a vector transforms

- $$\delta(v_0, v_i) = (\beta_i v_i, \beta_i v_0 - \epsilon_{ijk} \theta_j v_k) \quad (372)$$

Part V

QED (TODO)

Definition 5.1 (Quantum Electrodynamics): Quantum Electrodynamics is a relatively simple theory with only one force involved (as suggested by the name, it is the electromagnetic force), mediated by the *photon* A_μ and one/two type(s) of fermion depending on what you consider a type of particle (an *electron/positron*)

5.0.1. Charge

- The charge in quantum electrodynamics is associated with the U(1) symmetry

$$\phi \rightarrow e^{-i\alpha(x)} \phi \quad (373)$$

(in fact, this applies in spinor QED as well, due to the)

•

5.1. Spinor QED (Standard Quantum Electrodynamics)

- The Spinors in Quantum Electrodynamics are Dirac Spinors due to their presence of charge (Since Majorana spinors are the same as their antiparticle, they of course cannot be a candidate for this, and hence the spinors must be Dirac Spinors).

5.1.1. Solution to the Dirac Equation

- We want to quantize our dirac spinors ψ .
- We follow a similar procedure to what we did with photons (we will introduce basis spinors similar to polarizations ε_μ)
- The dirac equation

$$(i\not{\partial} - m)\psi = 0 \quad (374)$$

implies that, for spinors

$$\left((i\partial_\mu - eA_\mu)^2 - \frac{e}{2} F_{\mu\nu} \sigma^{\mu\nu} - m^2 \right) \psi = 0 \quad (375)$$

- For antispinors,

$$\bar{\psi} \left((i\tilde{\partial}_\mu + eA_\mu)^2 + \frac{e}{2} F_{\mu\nu} \sigma^{\mu\nu} - m^2 \right) = 0 \quad (376)$$

- Similarly to a Scalar field theory, we will quantize particles by writing

$$\psi_s(x) = \int \frac{d^3p}{(2\pi)^3} u_s(p) e^{ipx} \quad (377)$$

for particles and

$$\chi_s(x) = \int \frac{d^3p}{(2\pi)^3} v_s(p) e^{ipx} \quad (378)$$

for antiparticles

- These $u_s(p)$ and $v_s(p)$ are *basis spinors* which satisfy the momentum space dirac equation. That is,

$$\begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u_s(p) = \begin{pmatrix} -m & -p \cdot \sigma \\ -p \cdot \bar{\sigma} & -m \end{pmatrix} v_s(p) = 0 \quad (379)$$

- Again, these $u_s(p)$ and $v_s(p)$ are 4-index Dirac spinors.

Basis Dirac Spinors

In general, the solution to this equation is

$$u_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix}, v_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta_s \\ -\sqrt{p \cdot \bar{\sigma}} \eta_s \end{pmatrix} \quad (380)$$

$$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m \quad (381)$$

$$\sum_s v_s(p) \bar{v}_s(p) = \not{p} - m \quad (382)$$

5.1.2. CPT Theorem

- In order to have the entire Lorentz Group, we must define the following discrete transforms for spinors.
- Time reversal T which maps

$$T : (t, \vec{x}) \rightarrow (-t, \vec{x}) \quad (383)$$

- Parity P which maps

$$P : (t, \vec{x}) \rightarrow (t, -\vec{x}) \quad (384)$$

- Charge conjugation

$$C : \psi \rightarrow -i\gamma^2 \bar{\psi} \quad (385)$$

reverses charge (essentially it is particle $\rightarrow$ antiparticle)

5.1.3. Interacting QED

- The Lagrangian for Quantum Electrodynamics is

$$\begin{aligned} \mathcal{L} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i\bar{\psi} \not{D} \psi - m\bar{\psi} \psi \\ &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi - e\bar{\psi} \gamma^\mu \psi A_\mu \end{aligned} \quad (386)$$

- This gives rise to the interaction $-ie\gamma^\mu$ with the Feynman Diagram

Theorem 5.1.3.1 (Spin-Statistics Theorem): The Spin statistics theorem is the theorem that spinors are governed by Fermi-Dirac statistics and that Vectors are governed by Bose-Einstein statistics. Essentially $\psi(x)\chi(y) = -\chi(x)\psi(y)$ and

- The Feynman propagator for spinors is

$$\langle 0|T\psi(0)\bar{\psi}(x)|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} e^{ipx} \quad (387)$$

with of course the implicit limit of $\varepsilon \rightarrow 0$

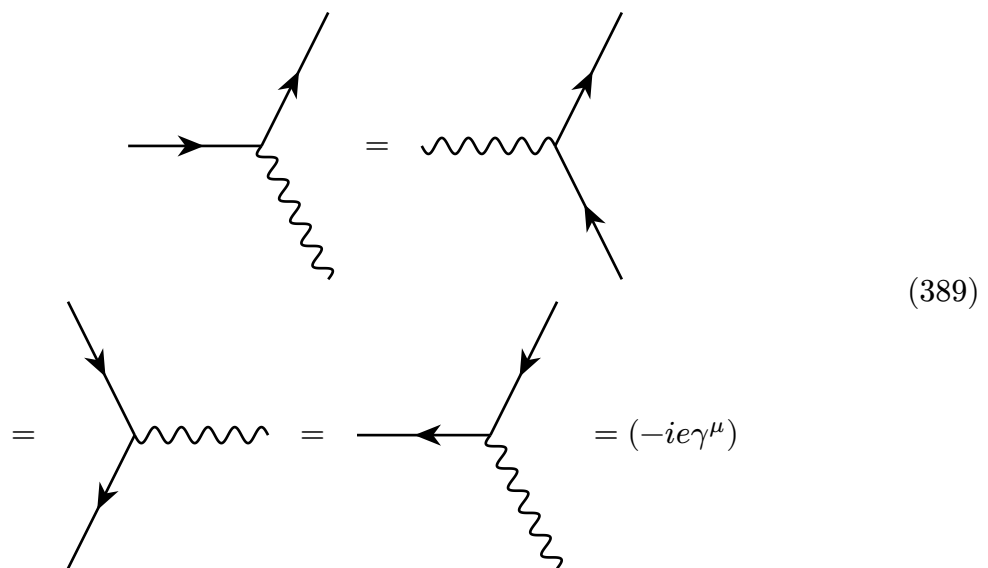
▸ The derivation for this is as follows

▸

- The Feynman propagator clearly gives rise to the electron propagator present on internal electrons

$$\text{feynman diagram} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} \quad (388)$$

- The QED vertex factor is given by the interaction term $-e\bar{\psi}\gamma^\mu A_\mu\psi$



The diagram shows four equivalent representations of the QED vertex factor, connected by equals signs.
 1. Top-left: An incoming electron line (solid arrow) from the left meets a vertex. From this vertex, an outgoing electron line (solid arrow) goes up-right and a photon line (wavy line) goes down-right.
 2. Top-right: A photon line (wavy line) from the left meets a vertex. From this vertex, an outgoing electron line (solid arrow) goes up-right and an incoming electron line (solid arrow) goes down-right.
 3. Bottom-left: An incoming electron line (solid arrow) from the top-left meets a vertex. From this vertex, an outgoing electron line (solid arrow) goes down-left and a photon line (wavy line) goes right.
 4. Bottom-right: A photon line (wavy line) from the left meets a vertex. From this vertex, an outgoing electron line (solid arrow) goes up-right and an incoming electron line (solid arrow) goes left.
 The entire sequence of diagrams is followed by the expression $= (-ie\gamma^\mu)$. The label (389) is on the right.

- Notice how all of these terms are equivalent (as opposed to scalar QED)
 - This is one of the many differences between spinor QED and scalar QED
- Another difference in Spinor QED is the fact that it has no seagull interaction allowed by an $A^2\phi^2$ term in scalar QED, simply because this term doesn't exist in QED

Part VI

QCD

6.1. Non Abelian Gauge Theories

- Non Abelian gauge theories are the fundamentals of QCD (in fact, they ARE Quantum Chromodynamics).
 - We need to establish quite a bit more formal background in order to do actual physics, otherwise nothing would make sense

6.1.1. Lie Groups and Lie Algebras

Definition 6.1.1.1 (Non Abelian Gauge Group):

- Suppose that you have a representation of a Lie Group, written as

$$R(G) = U = \exp(i\theta^a T^a) \quad (390)$$

- θ^a is a number
- T^a are the group generators for the *Lie Algebra* that generates the *Lie Group* G .
- The generators T^a satisfy the commutation

$$[T^a, T^b] = if^{abc}T^c \quad (391)$$

- If $f^{abc} = 0$, the gauge group G is abelian
- If $f^{abc} \neq 0$, the gauge group G is *non abelian*

- The commutation relations $[...]$ are a mapping $\mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$, without having to have a well-defined product
 - In the special case that a product is well defined, $[A, B] = AB - BA$

- Note: The **Jacobi Identity** must be satisfied regardless of the product being well defined

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0 \quad (392)$$

- By the Jacobi Identity,

$$f^{abd}f^{dce} + f^{bcd}f^{dae} + f^{cad}f^{dbe} = 0 \quad (393)$$

Definition 6.1.1.2 (Ideal):

- An *ideal* subalgebra $\mathfrak{I} \subset \mathfrak{g}$ is one that satisfies $[s, i] \in \mathfrak{I} \forall g \in \mathfrak{g}, i \in \mathfrak{I}$

Definition 6.1.1.3 (Simple):

- A simple Lie Algebra $\mathfrak{g}$ is one that has no trivial *ideal* subalgebras

Example 6.1.1.1 (Simple Lie Algebras):

- Very common Simple Lie Groups/Algebras are $\mathfrak{so}(N)$ and $\mathfrak{su}(N)$
- There is also the symplectic group $\mathfrak{sp}(N)$
- All of these you will soon see

Definition 6.1.1.4 (Semisimple):

- A semisimple Lie algebra $\mathfrak{g}$ is one that can be expressed as the direct sum of simple Lie Algebras $\mathfrak{g}_1, \mathfrak{g}_2, \dots, \mathfrak{g}_n$
 - That is,

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3 \oplus \dots \oplus \mathfrak{g}_n \quad (394)$$

- The standard model is a very common example of a Semisimple Lie Group
 - The Lie group for the Standard model is

$$G_{\text{SM}} = \text{SU}(3) \otimes \text{SU}(2) \otimes U(1) \quad (395)$$

- The Lie algebra for the Standard model is

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1) \quad (396)$$

Theorem 6.1.1.1 (Hermicity):

- *All finite dimensional representations of semisimple Lie algebras are hermitian*

Proof Hermicity:

- Unfortunately I have tried and failed to prove it, so at this point i cannot provide a proof

■

Definition 6.1.1.5 (Unitary Groups):

- A unitary group is a group who's representation satisfies

$$\langle \vec{\psi} | \vec{\chi} \rangle = \langle \vec{\psi} | U^\dagger U | \vec{\chi} \rangle \quad (397)$$

- i.e. the inner product is conserved
- This directly implies that $U^\dagger U = \mathbb{1}$

- Elements of $\text{SU}(N)$ have determinant 1
 - In general, matrix representations of unitary groups always have determinant 1 due to their being unitary
- The group $\text{SU}(N)$ is defined by its action upon N dimensional vector spaces

- $$U = \exp(i\theta^a T^a) \quad (398)$$
- In $SU(N)$, T^a are always hermitian
- There are $N^2 - 1$ generators for $SU(N)$
 - This implies that the dimension $d(G)$ is equal to $d(G) = N^2 - 1$

Definition 6.1.1.6 (Orthogonal Group):

- Orthogonal groups preserve a strictly real inner product
- $V \cdot W = V \cdot O^T \cdot O \cdot W$
- This directly implies that $O^T O = \mathbb{1}$
- The dimension of an orthogonal group $SO(N)$ is $d(SO(N)) = \frac{1}{2}d(SU(N)) = \frac{1}{2}N(N - 1)$
 - This is due to the fact that there are half as many generators for orthogonal groups due to their strict realness
- All elements of an orthogonal group (not special orthog.) have determinant $\det(O) = \pm 1$
 - In $SO(N)$, the matrices are the ones with $\det(O) = +1$
 - This has no impact on the dimension of the group

Definition 6.1.1.7 (Symplectic Group):

- Symplectic groups $Sp(N)$ are defined via preserving a quaternionic inner product
- They satisfy

$$\Omega S = -S^T \Omega \quad (399)$$

where $S \in Sp(N)$ and

$$\Omega = \begin{pmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{pmatrix} \quad (400)$$

- There are, finally, the *exceptional* simple groups G_2, F_4, E_6, E_7, E_8 which are of no use to standard model particle physics
- The *Only* finite dimensional simple Lie Algebras are the algebras for the exceptional simple groups and $\mathfrak{so}(N), \mathfrak{su}(N), \mathfrak{sp}(N)$

6.1.2. Representations

- Consider representations of $SU(N)$ groups
- Free theories of N fields are invariant under $U(1) \times SU(N)$
 - Since $SU(N)$ is simply connected, any representation of $SU(N)$ is one to one representation of $\exp(\mathfrak{su}(N))$

Example 6.1.2.1 (Two Fields):

- Consider 2 fields ϕ_1, ϕ_2 with a kinetic lagrangian

$$\mathcal{L}_{kin} = (\partial_\mu \phi_1^*)(\partial_\mu \phi_1) + (\partial_\mu \phi_2^*)(\partial_\mu \phi_2) = (\partial_\mu \vec{\phi})^\dagger (\partial_\mu \vec{\phi}) \quad (401)$$

- We can write

$$\vec{\phi} = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \quad (402)$$

- This is invariant under a global SU(2) symmetry $\vec{\phi} \rightarrow U\vec{\phi}$
- Generally, as we have seen, we can write

$$U = \exp[i(\alpha_1 \tau_1 + \alpha_2 \tau_2 + \alpha_3 \tau_3)] = \exp(ia^a \tau^a) \quad (403)$$

$$w/\tau^a = 1/2\sigma^a$$

- Such τ^a satisfy

$$\tau^a : [\tau^a, \tau^b] = i\epsilon^{abc} \tau^c \quad (404)$$

- This provides $f^{abc} = \epsilon^{abc}$

Definition 6.1.2.1 (Fundamental):

- A fundamental representation is the smallest non trivial representation of a Lie Algebra

- For SU(N), the fundamental representation is the set of $N \times N$ hermitian matricians with determinant 1
- Under the fundamental representation, a set of N fields transform (infinitesimally) as

$$\phi_i \rightarrow \phi_i + i\alpha^a (T_{\text{fund}}^a)_{ij} \phi_j \quad (405)$$

for some $\alpha^a \in \mathbb{R}$

Definition 6.1.2.2 (Anti-fundamental Representation):

- $T_{\text{anti-fund}}^a = -(T_{\text{fund}}^a)^*$ (406)
- The anti-fund representation of a group is the conjugate of the fundamental representation
- Notice that there is a minus sign in order to preserve the law $[T^a, T^b] = if^{abc} T^c$

- Therefore, the infinitesimal transformation law for anti-fundamental is

$$\phi_i^* \rightarrow \phi_i^* + i\alpha^a (T_{\text{anti-fund}}^a)_{ij} \phi_j^* = \phi_i^* - i\alpha^a \phi_j^* (T_{\text{fund}}^a)_{ji} \quad (407)$$

Note (Indices):

- Mid latin alphabet i, j are used to index color
- In order to determine the generators for a theory, we expand around $\mathbb{1}$
 - In $SU(2)$, the generators are σ^μ

$$T^a = \tau^a \equiv \frac{\sigma^a}{2} \quad (408)$$

- In $SU(3)$, the generators are *eight* (not $3 \otimes \bar{3} = 9$) **gellmann matrices** (such that $T^a = \frac{\lambda^a}{2}$) which are as follows:

$$\begin{aligned} \lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & \lambda_2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} & \lambda_4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \\ \lambda_5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} & \lambda_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \\ \lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \end{aligned} \quad (409)$$

- Normalization of f^{abc} is arbitrary in QCD/non abelian gauge theory.
- A commonly used convention for normalization (in physics) is

$$\sum_{c,d} f^{acd} f^{bcd} = N \delta^{ab} \quad (410)$$

- In math, the convention is

$$\sum_{c,d} f^{acd} f^{bcd} = \delta^{ab} \quad (411)$$

- The fixing the normalization of structure constants implies that the normalization of generators is fixed *in all representations*
- For representations,

$$[T^a, T^b] = i f^{abc} T^c \quad (412)$$

must hold when f^{abc} is the same

- Notice that this equation is not invariant under a rescaling of T_R^c
- Normalization of $SU(N)$:
 - It is normalized s.t.

$$\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab} \quad (413)$$

- In a general lie algebra, $[T^a, T^b]$ is well defined and $T^a T^b$ is not (as seen before)
 - In the fundamental representation of $\text{SU}(N)$, such generators, in fact, can be multiplied

$$T^a T^b = \frac{1}{2N} \delta^{ab} + \frac{1}{2} d^{abc} T^c + \frac{1}{2} i f^{abc} T^c \quad (414)$$

$$d^{abc} = 2 \text{tr}[T^a \{T^b, T^c\}] \quad (415)$$

- In $\text{SU}(N)$, there is a unique d^{abc} up to a constant
 - In $\text{SU}(2)$, $d^{abc} = 0$
- Some useful trace identities for generators are the following:

$$\begin{aligned} \text{tr}[T^a T^b T^c] &= \frac{1}{4} (d^{abc} + i f^{abc}) \\ \text{tr}[T^a T^b T^c T^d] &= \frac{1}{4N} \delta^{ab} + \frac{1}{8} (d^{abe} + i f^{abe}) \times (d^{cde} + i f^{cde}) \end{aligned} \quad (416)$$

Definition 6.1.2.3 (Adjoint Representation):

- The adjoint representation is the group acting upon its own elements
- Extremely formally

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}) \quad (417)$$

with G an element of a Lie group, $\mathfrak{g}$ an element of a Lie algebra, and GL being the group of invertible matrices

- If the lie group generated by a lie algebra is e^X , then $\text{Ad}_A(X) = AXA^{-1}$ with $A \in G$.

- The adjoint representation amounts to acting upon the space that is spanned by the generators, as you have seen
- In $\text{SU}(N)$, there are $N^2 - 1$ generators and hence there are $N^2 - 1$ dimensions for the adjoint representation
- Matrices in the adjoint representation are given by $(T_{\text{adj}}^a)^{bc} = -i f^{abc}$
 - For $\text{SU}(2)$, the matrices are

$$T_{adj}^1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, T_{adj}^2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, T_{adj}^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (418)$$

- In $\text{SU}(3)$, they are 8×8 matrices as you may be able to guess
- *Gauge fields transform in the adjoint representation*

Definition 6.1.2.4 (Casimirs):

- Casimirs are used to “characterize representations” [1]

Definition 6.1.2.5 (Quadratic Casimir):

- The Quadratic Casimir is defined via

$$\left(\sum_a \right) T_R^a T_R^a = C_2(R) \mathbb{1} \quad (419)$$

Theorem 6.1.2.1 (Schur's Lemma):

- Group elements that commute with all other elements are proportional to $\mathbb{1}$

- This theorem is extremely nontrivial. You can see a proof in [2]
- I am, however, able to provide a proof on why the $T^a T^a$ commutes with everything

$$[T_R^a T_R^a, T_R^b] = (i f^{abc} T_R^c) T_R^a + T_R^a (i f^{abc} T_R^c) = -i f^{abc} \{T_R^c, T_R^a\} = 0 \quad (420)$$

- Evaluating quadratic Casimirs involve choosing generators such that

$$f_r [T_p^a, T_p^b] = T(R)^{ab} \quad (421)$$

- $T(R)$ is known as the *index*
- In the fundamental representation, the convention implies that
 - $T(\text{fund}) \equiv T_F = \frac{1}{2}$ since $(T^a)^i_j (T^a)^j_k = \frac{1}{2} \delta^i_k$
- In the adjoint representation, $T(\text{adj}) \equiv T_A = N$
 - This is due to the fact that $f^{acd} f^{bcd} = N \delta^{ab}$
- Setting $a = b$ and summing over a for the definition of $T(R)$ yields
 - $d(R) C_2(R) = T(R) d(G)$

(422)

 - $d(R)$ is the dimension of the representation
 - $d(G)$ is the dimension of the group

- Fundamental Casimir:

$$C_F = C_2(\text{fund}) = \frac{N^2 - 1}{2N} \quad (423)$$

- In $SU(2)$, $C_F = \frac{3}{4}$
- In $SU(3)$, $C_F = \frac{4}{3}$
- Adjoint Casimir:
 - $C_A \equiv C_2(\text{adj}) = N$
 - Generally, terms in YM theories will contain at least some factors of C_F and C_A
- For any representation,

$$\text{tr}([T_R^a, T_R^b] T_R^c) = i f^{abc} \text{tr}(T_R^d T_R^d) = i f^{abc} T(R) \quad (424)$$

- This implies that

$$f^{abc} = -\frac{i}{T_F} \text{tr} ([t^a, t^b] t^c) \quad (425)$$

- In $\text{SU}(N)$, the *Fierz Identity* is satisfied:

$$\sum_a T_{ij}^a T_{kl}^a = \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \right) \quad (426)$$

- Since generators in $\text{SU}(N)$ are traceless, the sum over δ^{ij} and δ^{kl} yields zero
- Hence,

$$\text{tr}[T^a A] \text{tr}[T^a B] = \frac{1}{2} [\text{tr}(AB) - \frac{1}{N} \text{tr}(A) \text{tr}(B)] \quad (427)$$

Definition 6.1.2.6 (Anomaly Coefficient):

- An anomaly coefficient $A(R)$ is defined by the equation

$$\text{tr}[T^a \{T^b, T^c\}] = \frac{1}{2} A(R) d^{abc} = A(R) \text{tr}[T^a \{T^b, T^c\}] \quad (428)$$

6.2. Wilson Lines

6.2.1. Abelian Wilson Lines

- The motivation between wilson lines is a want for telling if $\phi(x) = \phi(y)$ (some how)
- Notice that one cannot simply take the difference since

$$\phi(x) - \phi(y) \rightarrow e^{i\alpha(x)} \phi(x) - e^{i\alpha(y)} \phi(y) \quad (429)$$

- It is additionally impossible to take any notion of $\partial_\mu \phi$ due to derivative definition based on differences

Definition 6.2.1.1 (Wilson Line):

- A wilson line $W(x, y)$ is something that transforms as $W(x, y) \rightarrow e^{i\alpha(x)} W(x, y) e^{-i\alpha(y)}$
 - This is a “bi-local transformation” [1]

- A key feature of Wilson lines is that they satisfy the property

$$\begin{aligned} W(x, y) \phi(y) - \phi(x) &\rightarrow e^{i\alpha(x)} W(x, y) e^{-i\alpha(y)} e^{i\alpha(y)} \phi(y) - e^{i\alpha(x)} \phi(x) \\ &= e^{i\alpha(x)} (W(x, y) \phi(y) - \phi(x)) \end{aligned} \quad (430)$$

- Taking $y^\mu = x^\mu + \delta x^\mu$, we can obtain the derivative via

$$D_\mu \phi(x) \equiv \lim_{\delta x^\mu \rightarrow 0} \frac{W(x, x + \delta x) \phi(x + \delta x) - \phi(x)}{\delta x} \quad (431)$$

- This gives a transformation law that we expect

$$D_\mu \phi(x) \longrightarrow e^{i\alpha(x)} D_\mu \phi(x) \quad (432)$$

- Writing the wilson line in terms of a perturbation,

$$W(x, x + \delta x) = 1 - e\delta x^\mu A_\mu(x) + \mathcal{O}(\delta x^2) \quad (433)$$

- This gives, as we expect, the abelian gauge transformation

$$A_\mu(x) \rightarrow A_\mu(x) + \frac{1}{e} \partial_\mu \alpha(x) \quad (434)$$

- The Guage field is a connection
- This provides the abelian covariant derivative

$$D_\mu \phi(x) = \partial_\mu \phi(x) - ie A_\mu \phi \quad (435)$$

- The *closed-form* for the wilson line is written as

$$W_P(x, y) = \exp \left(ie \int_y^x A_\mu(z) dz^\mu \right) \quad (436)$$

- Notice that this is a line integral
 - This is the true Wilson Line
- Parametrizing the wilson line with $z^\mu(\lambda)$ such that $0 \leq \lambda \leq 1$ and $z^\mu(0) = y^\mu, z^\mu(1) = x^\mu$, we obtain that

$$W_P(x, y) = \exp \left(ie \int_0^1 \frac{dz^\mu(\lambda)}{d\lambda} A_\mu(z(\lambda)) d\lambda \right) \quad (437)$$

- We can test how this behaves under a gauge transformation

$$\begin{aligned} W_P(x, y) &\rightarrow \exp \left[ie \int_y^x A_\mu(z) dz^\mu + i \int_y^x \partial_\mu \alpha(z) dz^\mu \right] \\ &= e^{i\alpha(x)} W_P(x, y) e^{-i\alpha(y)} \end{aligned} \quad (438)$$

- This behaves correctly from our definition of a Wilson line
- Wilson Loops:
 - Setting $x = y$, we obtain that

$$W_P^{\text{loop}} = \exp \left(ie \oint_P A_\mu dx^\mu \right) \quad (439)$$

- Via stokes' theorem,

$$W_p^{\text{loop}} = \exp \left(i \frac{e}{2} \int_\Sigma F_{\mu\nu} d\sigma^{\mu\nu} \right) = 1 + i \frac{e}{2} \int_\Sigma F_{\mu\nu} d\sigma^{\mu\nu} + \mathcal{O}(e^2) \quad (440)$$

where $\sigma^{\mu\nu}$ is a surface element

- Notice that now the wilson line only depends on the field tensor $F^{\mu\nu}$ (this should not be surprising since it depends on the gauge field)

- Checking the transformation for the commutator $[D_\mu, D_\nu]$, we obtain

$$[D_\mu, D_\nu]\phi(x) \rightarrow e^{i\alpha(x)}[D_\mu, D_\nu]\phi(x) \quad (441)$$

- therefore, we obtain

$$[D_\mu, D_\nu]\phi(x) = ([\partial_\mu, \partial_\nu] - ie[\partial_\mu, A_\nu] + ie[\partial_\nu, A_\mu])\phi(x) = -ieF_{\mu\nu}(x) \quad (442)$$

- Hence, we can define the EM field tensor as

$$F_{\mu\nu}(x) = \frac{i}{e}[D_\mu, D_\nu] \quad (443)$$

6.2.2. Non Abelian Wilson Lines

- We would like to repeat the same process as we did with abelian wilson lines
 - Lagrangians have more symmetry than just phase rotation in non abelian theories (by definition)
- Consider the kinetic lagrangian with N particles

$$\mathcal{L}_{kin} = \sum_{j=1}^n \bar{\psi}_j(i\not{\partial} - m)\psi_j \quad (444)$$

- This has a global $SU(N)$ symmetry
- Such particles transform as

$$\psi_i \rightarrow (e^{i\alpha^a T^a})_{ij}\psi_j \quad (445)$$

- α^a is not a function of x for global symmetries

Definition 6.2.2.1 (Non Abelian Wilson Line):

- We are now motivated to write

$$W_P(x, y) = P \left\{ \exp \left(ig \int_y^x A_\mu^a(z) T^a dz^\mu \right) \right\} \quad (446)$$

where P is path ordering

- The Taylor expansion of the wilson line is

$$\begin{aligned} W_P(z, y) &= 1 + ig \int_0^1 \frac{dz^\mu(\lambda)}{d\lambda} A_\mu^a(z(\lambda)) T^a d\lambda \\ &\quad - \frac{1}{2} g^2 \int_0^1 d\lambda \int_0^1 d\tau \frac{dz^\mu(\lambda)}{d\lambda} \frac{dz^\nu(\tau)}{d\tau} A_\mu^a(z(\lambda)) A_\nu^b(z(\tau)) (T^a T^b \theta(\lambda - \tau) + T^b T^a \theta(\tau - \lambda)) + \dots \end{aligned} \quad (447)$$

- Under a gauge transformation

$$W_P(x, y) = e^{i\alpha^a(x)T^a} W_P(x, y) e^{-i\alpha^a(y)T^a} \quad (448)$$

which definitely looks correct

- We are motivated to define a variable $\underline{A}_\mu \equiv A_\mu^a T^a$
 - This gives

$$W_P(x, y) = P \left\{ \exp \left(ig \int_{y-x}^x \underline{A}_\mu(z) dz^\mu \right) \right\} \quad (449)$$

- An infinitesimal transformation for W is

$$W(x^\mu, x^\mu + \delta x^\mu) = 1 + i\epsilon \underline{A}_\mu \delta x^\mu \quad (450)$$

- Local transformations can be expressed in terms of

$$U(x) = e^{i\alpha^a(x)T^a} \quad (451)$$

- In $SU(N)$, $U^\dagger(y) = U(y)$
- Then,

$$\begin{aligned} \vec{\psi} &= U(x) \cdot \vec{\psi}(x) \\ W(x, y) &\rightarrow U(x)W(x, y)U^\dagger(y) \end{aligned} \quad (452)$$

- We would like to analyze what happens to the gauge field under such a transformation
 - Consider how the covariant derivative transforms:

$$D_\mu \vec{A} \rightarrow U \cdot D_\mu \vec{A} \quad (453)$$

- Therefore, we have that

$$(\partial_\mu - ig \underline{A}'_\mu)U\psi = U(\partial_\mu - ig \underline{A}_\mu)\psi \quad (454)$$

where $\underline{A}_\mu$ is the transformed A_μ

- Therefore, the transformation for A_μ^a is

$$A_\mu^a(x) \rightarrow A_\mu^a(x) + \frac{1}{g} \partial_\mu \alpha^a(x) - f^{abc} \alpha^b(x) A_\mu^c + O(\alpha^2) \quad (455)$$

- The covariant derivative commutator transforms as

$$[D_\mu, D_\nu] \psi(x) = (-ig(\partial_\mu \underline{A}_\nu - \partial_\nu \underline{A}_\mu) - g^2 [\underline{A}_\mu, \underline{A}_\nu]) \psi(x) \quad (456)$$

- This allows for a definition of the *gluon field tensor*:

$$\begin{aligned} \underline{F}_{\mu\nu} &\equiv \frac{i}{g} [D_\mu, D_\nu] = (\partial_\mu \underline{A}_\nu - \partial_\nu \underline{A}_\mu) - ig [\underline{A}_\mu, \underline{A}_\nu] \\ \underline{F}_{\mu\nu} &= F_{\mu\nu}^a T^a \end{aligned} \quad (457)$$

-

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \quad (458)$$

- The transformation law for the field tensor is

$$F_{\mu\nu}^a \rightarrow F_{\mu\nu}^a - f^{abc} \alpha^b F_{\mu\nu}^c \quad (459)$$

which is the same whether or not α is global or local

- This is because

$$F_{\mu\nu}^\alpha = -F_{\nu\mu}^\alpha \quad (460)$$

- Kinetic terms in the lagrangian only depend on $F_{\mu\nu}^a$
 - The $SU(N)$ lagrangian is

$$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu}^a)^2 + \sum_{i,j=1}^N \bar{\psi}_i (i\delta_{ij} \not{D} + g A_\mu^a T_{ij}^a - m\delta_{ij}) \psi_j \quad (461)$$

- In this lagrangian, g acts as $-e$
- There is an extra term that is consistent with gauge invariance that can be added to this lagrangian:

$$\mathcal{L}_\theta = \theta \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu}^\alpha F_{\alpha\beta}^\alpha = 2\theta \partial_\mu (\varepsilon^{\mu\nu\alpha\beta} A_\nu^\alpha F_{\alpha\beta}^\alpha) \quad (462)$$

- Notice that this is a total derivative, hence it does not contribute perturbatively
- There should be non perturbative effects that this term adds (the lack of them is known as the *strong CP problem*)
- I claim that there is no conserved current for this lagrangian
- Consider the expansion of the lagrangian

$$\mathcal{L} = -\frac{1}{4}(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c)^2 + \bar{\psi}_i (i\delta_{ij} \gamma^\mu \partial_\mu + g \gamma^\mu A_\mu^a T_{ij}^a - m\delta_{ij}) \psi_j \quad (463)$$

- The EoM for this lagrangian are

$$\partial_\mu F_{\mu\nu}^a + g f^{abc} A_\mu^b F_{\mu\nu}^c = -g \bar{\psi}_i \gamma_\nu T_{ij}^a \psi_j \quad (\text{Gauge}) \quad (464)$$

and

$$-(i\not{D} - m)\psi_i = -g \not{A}^a T_{ij}^a \psi_j \quad (\text{Ferm.}) \quad (465)$$

- The global symmetries for this are

$$\begin{aligned} \psi_i &\rightarrow \psi_i + i\alpha^a T_{ij}^a \psi_j \\ A_\mu^a &\rightarrow A_\mu^a - f^{abc} \alpha^b A_\mu^c \end{aligned} \quad (466)$$

- By Noether's theorem,

$$J_\mu = \sum_n \frac{\partial L}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \quad (467)$$

- In a non abelian theory, there are $N^2 - 1$ currents
 - Summing over ψ_i and A_μ^a provides

$$J_m^\mu = -\bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j + f^{abc} A_\nu^b F_c^{\mu\nu} \quad (468)$$

- Notice, however, that this current is gauge invariant.
 - Hence, it is not physical
- We can try and create a matter current

$$j_\mu^a = -\bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j \quad (469)$$

- This satisfies

$$D_\mu j_\mu^a = 0 \quad (470)$$

- It Still doesnt satisfy $\partial_\mu j_\mu^a = 0$

Theorem 6.2.2.1 (Weinberg Witten theorem):

- A non abelian symmetry for massless spin 1 implies that there is no gauge invariant conserved charge

6.2.3. The Gluon Propagator

6.2.3.1. F-P ghosts

Summary

- There is a lot of math and a lot of unseen math in this section so it may be impossible to read
- The summary for this section is tha the Lagrangian for QCD involves ghosts which are unphysical and not gauge invariant
- The QCD lagrangian that we derive is

- To Start, let us consider the equations of motion for a photon with an external current

$$(g_{\mu\nu} \square - \partial_\mu \partial_\nu) A_m = J_\nu \quad (471)$$

- This is not invertible, similar to the gluon equation of motion

Trick (Mildly Useful Insight)

We can introduce (and have introduced) a term into the lagrangian of the form

$$\frac{1}{2\xi} (\partial_\mu A_\mu)^2 \quad (472)$$

- This gives us, in the path integral approach of QFT,

$$f(\xi) = \int \mathcal{D}\pi e^{-i \int d^4x \frac{1}{2\xi} (D\pi)^2} = \int \mathcal{D}\pi e^{-i \int d^4x \frac{1}{2\xi} (D\pi - \partial_\mu A_\mu)^2} \quad (473)$$

- This is independed of A_μ since we can take $\pi \rightarrow \pi - \frac{1}{\square} \partial_\mu A_\mu$
- Therefore, we obtained

$$\begin{aligned}
\int \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x L[A, \phi_i]} &= \frac{1}{f(\xi)} \int \mathcal{D}\pi \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x (L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A^\mu)^2)} \\
&= \left[\frac{1}{f(\xi)} \int \mathcal{D}\pi \right] \int \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x (L[A, \phi_i] - \frac{1}{2\xi} (\partial_\mu A^\mu)^2)}
\end{aligned} \tag{474}$$

in QED

- We want to do a similar thing for QCD
 - The gauge transformation is

$$A_\mu^a \rightarrow A_\mu^a + \frac{1}{g} \partial_\mu \alpha^a + f^{abc} A_\mu^b \pi^c \tag{475}$$

- Notice that there are $N^2 - 1$ π^a fields due to them transforming in the adjoint representation
- Since π^a transforms in the adjoint rep we can write

$$A_\mu^a \rightarrow A_\mu^a + \frac{1}{g} D_\mu \pi^a \tag{476}$$

where the adjoint covar. derivative is

$$D_\mu \pi^a = \partial_\mu \pi^a + g f^{abc} A_\mu^b \pi^c \tag{477}$$

- We now have the machinery to write a functional

$$f[A] = \int \mathcal{D}\pi \exp \left[-i \int d^4x \frac{1}{2\xi} (\partial_\mu D_\mu \pi^a)^2 \right] \tag{478}$$

Algebra: Tickle Tip

- Define $\alpha^a[A]$ s.t.

$$\partial_\mu A_\mu^a = \frac{1}{g} \partial_\mu D_\mu \alpha^a[A] \tag{479}$$

- Therefore,

$$f[A] = \int \mathcal{D}\pi \exp \left[-i \int d^4x \frac{1}{2\xi} (\partial_\mu A_\mu^\alpha - \partial_\mu A_\mu^\alpha)^2 \right] \tag{480}$$

- This allows us to obtain the full path integral

$$\begin{aligned}
&\int \mathcal{D}A \mathcal{D}\phi_i e^{i \int d^4x L(A, \phi_i)} \\
&= \int \mathcal{D}\pi \mathcal{D}A \mathcal{D}\phi_i \frac{1}{f[A]} \exp \left(i \int d^4x L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A^\mu - \partial_\mu A^\mu)^2 \right) \\
&= \left(\int \mathcal{D}\pi \right) \int \mathcal{D}A \mathcal{D}\phi_i \frac{1}{f[A]} \exp \left(i \int d^4x L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right)
\end{aligned} \tag{481}$$

- We now wish to rewrite $f[A]$ (?)

- $$f = \sqrt{\frac{1}{\det(\partial_\mu D_\mu)^2}} \times \text{const.} \tag{482}$$

- Then,

$$Z[0] = \text{const} \times \int \mathcal{D}A_m \mathcal{D}\phi_i (\det(\mathcal{M}_{mn})) \times \exp \left\{ i \int d^4x \left[\mathcal{L}(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A_m^a)^2 \right] \right\} \quad (483)$$

Algebra: Gaussians

- We can use identities for gaussians to write

$$\det(\partial^\mu D_\mu) = \int \mathcal{D}\bar{c} \mathcal{D}c \exp \left(i \int d^4x \bar{c} (-\partial^\mu D_\mu) c \right) \quad (484)$$

- The identity involved in this is

$$\det(\mathcal{O}) = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(-i \int \bar{\psi} \mathcal{O} \psi \right) \quad (485)$$

- Hence, we obtain the result that

$$Z[c] = \text{const.} \times \int \mathcal{D}A_\mu \mathcal{D}\phi \mathcal{D}\bar{c} \mathcal{D}c \times \exp \left\{ i \int d^4x \left[L(A, \phi) - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 - \bar{c}^a \partial^\mu D_\mu c^a \right] \right\} \quad (486)$$

- We have obtained a very bad result:

Key Result

- In QCD, as a result of trying to find a gluon propagator, we find these annoying things called Fadeev-Popov ghosts and antighosts
- They pop up out of nowhere and are not gauge invariant
 - They are aphysical clearly

- The Lagrangian must (and I mean must unless you are willing to work in a obnoxious gauge) be written with these in mind

$$\mathcal{L}_{R_\xi} = -\frac{1}{4} (F_{\mu\nu}^a)^2 - \frac{1}{2\xi} (\partial_\mu A^{\mu a})^2 + (\partial_\mu \bar{c}^a) (\delta^{ac} \partial^\mu + g f^{abc} A^{\mu b}) c^c \quad (487)$$

- This is called the Fadeev-Popov lagrangian
- Luckily, we also have the gluon propagator

$$\text{Gluon propagator} = i \frac{-g^{\mu\nu} + (1-\xi) \frac{p^\mu p^\nu}{p^2}}{p^2 + i\varepsilon} \delta^{ab} \quad (488)$$

Algebra: Generic gauge

- Observe that

$$1 = \int \mathcal{D}\pi \delta(G(A_\mu^a - D_\mu \pi^a)) \det \left(\frac{\delta G[A_\mu^a - D_\mu \pi^a]}{\delta \pi^a} \right) \quad (489)$$

- $G[A]$ is a functional which represents our choice of gauge

- Then,

$$\det \left[\frac{\delta G \{A_m^a - D_m \sigma^a\}}{\delta \sigma^b} \right] = \det(\partial^m D_m) = \frac{1}{f[A]} = \int \mathcal{D}\bar{c} \mathcal{D}c \exp \left(i \int d^4x \bar{c}^a (-\partial^m D_m) c^a \right) \quad (490)$$

- Multiplying $Z[0]$ by 1 yields

$$\begin{aligned} Z[0] &= \text{const} \times \int \mathcal{D}\pi \int \mathcal{D}A_m \mathcal{D}\phi_i \delta(G[A_m^a - D_m \pi^a]) \\ &\times \det \left(\frac{\delta G[A_m^a - D_m \pi^a]}{\delta \pi^b} \right) \exp \left(i \int d^4x \mathcal{L}[A, \phi] \right) \end{aligned} \quad (491)$$

- We use the trick of shifting

$$A_\mu^a \rightarrow A_\mu^a + D_\mu \pi^a \quad (492)$$

due to gauge invariance

- Our result is that

$$\begin{aligned} \Rightarrow Z[0] &= \text{const.} \left(\int \mathcal{D}\pi \right) \int \mathcal{D}A_\mu \mathcal{D}\phi_i \delta(G[A_\mu^a]) \\ &\times \det \left(\frac{\delta G[A_\mu^a - D_\mu \pi^a]}{\delta \pi^b} \right) \times \exp \left(i \int d^4x L[A, \phi_i] \right) \end{aligned} \quad (493)$$

$\pi \rightarrow 0$

- Shifting G by a constant keeps the determinant the same. Therefore, we can average over a gaussian which yields

$$\int \mathcal{D}\chi \exp \left(-i \int d^4x \frac{X^2}{2\xi} \right) \delta(G[A_m^a] - \chi) = \exp \left(-i \int d^4x \frac{1}{2\xi} G[A_m^a]^2 \right) \quad (494)$$

- This yields our final form of $Z[0]$

$$Z[0] = \text{const} \times \int \mathcal{D}A_\mu \mathcal{D}\phi_i \det \left(\frac{\delta G[A_\mu, \phi_i]}{\delta \phi_j} \right) \times \exp \left[i \int d^4x \left(L[A, \phi_i] - \frac{1}{2g} G[A_\mu]^2 \right) \right] \quad (495)$$

6.2.3.2. BRST Invariance

- There is another symmetry that the lagrangian is invariant under.
 - This symmetry involves grassmann numbers and qcd ghosts
- Consider the lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 + (D_\mu \phi_i^*)(D^\mu \phi_i) - m^2 \phi_i^* \phi_i - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 - \bar{c} \square c \quad (496)$$

- If we define a shift

$$\begin{aligned} \phi_i(x) &\rightarrow e^{i\theta(x)} \phi_i(x) \approx \phi_i(x) + i\theta(x) \phi_i(x) \\ A_\mu(x) &\rightarrow A_\mu(x) + \frac{1}{e} \partial_\mu \theta(x) \end{aligned} \quad (497)$$

and either use the EOM or take

$$\begin{aligned}
(\partial_\mu A_\mu)^2 &\rightarrow (\partial_\mu A_\mu)^2 + \frac{2}{e}(\partial_\mu A_\mu)(\theta \square c) + \frac{1}{e^2}(\theta \square c)(\theta \square c) \\
\bar{c}(x) &\rightarrow \bar{c}(x) - \frac{1}{e}\theta \frac{1}{\xi} \partial_\mu A^\mu(x)
\end{aligned} \tag{498}$$

, this lagrangian is invariant

- This gives us **BRST Invariance**

Definition 6.2.3.2.1 (BRST Invariance):

- BRST Invariance is a generalization of guage invariance for the Fadeev Popov lagrangian

- This works in the non-abelian case similarly, where the lagrangian is

$$\mathcal{L}_{FP} = \mathcal{L}[A_\mu^a, \phi_i] - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + (\partial_\mu \bar{c}^a)(D_\mu c^a) \tag{499}$$

- One can define transformations

$$\begin{aligned}
\phi_i &\rightarrow \phi_i + i\theta^a T_{ij}^a \phi_j \\
A_\mu^a &\rightarrow A_\mu^a + \frac{1}{g}\theta D_\mu c^a \\
\bar{c}^a &\rightarrow \bar{c}^a - \frac{1}{g}\theta \frac{1}{\xi} \partial_\mu A_\mu^a
\end{aligned} \tag{500}$$

which leave everything but D_μ invariant. Hence. we need a final transformation which, after a lot of algebra, results in

$$\begin{aligned}
D_\mu c^a &\rightarrow D_\mu c^a + \theta f^{abc}(D_\mu c^b)c^c \\
c^a &\rightarrow c^a - \frac{1}{2}\theta f^{abc}c^b c^c
\end{aligned} \tag{501}$$

- or alternatively, after more algebra,

$$\begin{aligned}
D_\mu c^a &\rightarrow D_\mu c^a + g f^{abc}(D_\mu c^b)c^c - g f^{abc} \left[\frac{1}{2}(\partial_\mu c^b)c^c + \right. \\
&\quad \left. \frac{1}{2}c^b(\partial_\mu c^c) + \frac{g}{2}A_\mu^b f^{cde}c^d c^e \right]
\end{aligned} \tag{502}$$

6.2.3.3. Axial gauges

Summary

- Axial gauges are not very useful to work with but they remove the ghosts that are inconvenient to work with
- The gluon propagator in an axial gauge is

$$i\Pi_{\text{lightcone}}^{\mu\nu ab} = \frac{i}{p^2 + i\epsilon} \left[-ig^{\mu\nu} + \frac{r^\mu p^\nu + p^\mu r^\nu}{rp} \right] \delta^{ab} \quad (503)$$

which, as you may be able to see, is not the most fun to work with

Algebra: Axial Gauges

- The most general lagrangian that we can write to gauge fix our other one is

$$-\frac{1}{2\lambda}(\partial^\mu A_\mu^a)^2 + \bar{c}^a \partial^\mu (\delta^{ab} \partial_\mu + g f^{abc} A_\mu^b) c^c \quad (504)$$

with two parameters: λ , a scalar, and r^μ , a 4-vector.

- The general propagator in this axial gauge is

$$\Pi_{axial}^{\mu\nu ab} = \frac{i}{p^2 + i\epsilon} \left[-g^{\mu\nu} + \frac{r^\mu p^\nu + r^\nu p^\mu}{r \cdot p} - \frac{(r^2 + \lambda p^2) p^\mu p^\nu}{p^2} \right] \delta^{ab} \quad (505)$$

- We of course want this to satisfy the ward identity, but we do not have to change anything since it already does.
- In the *Lightcone gauge*, we force that $r^2 = 0$, $\lambda = 0$, which yields our propagator

6.3. Lattice QCD

Summary

- Lattice QCD works by discretizing standard QCD into plaquettes and evaluating wilson loops on them
- Fields are at specific lattice sites

Definition 6.3.1 (Discrete Wilson line):

- First, lets consider some notation. $\hat{\mu}$ and $\hat{\nu}$ will be unit vectors (of length a since in lattice qcd, everything is scaled by such a factor) in the μ and ν directions. $W_\mu(n)$ will denote a wilson line at some site n and some other site $n + \hat{\mu}$
- These wilson lines will transform under the law

$$W_\mu(n) \rightarrow U(n) W_\mu(n) U^\dagger(n + \hat{\mu}) \quad (506)$$

because $\phi(n) \rightarrow U(n) \phi(n)$

Algebra: Verification

- Just testing that our wilson lines behave correctly (as they should),

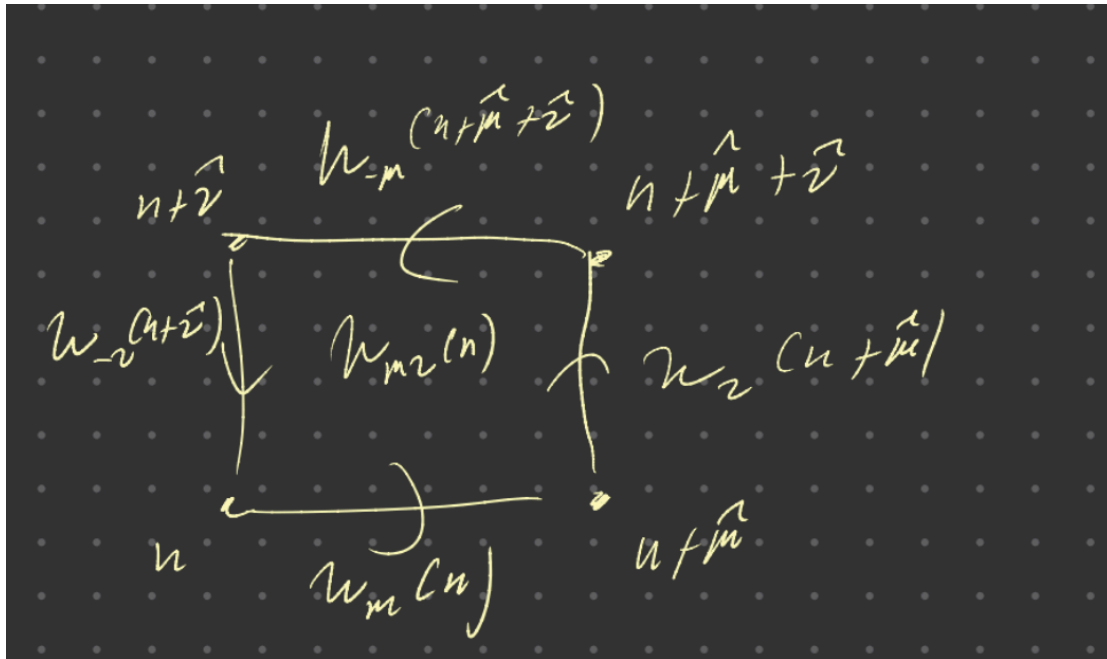
$$\begin{aligned} \vec{\phi}^\dagger(n) W_\mu(n) \vec{\phi}(n + \hat{\mu}) &\rightarrow \vec{\phi}^\dagger(n) U^\dagger(n) U(n) W_\mu(n) \\ &\times U^\dagger(n + \hat{\mu}) U(n + \hat{\mu}) \vec{\phi}(n + \hat{\mu}) = \vec{\phi}^\dagger(n) W_\mu(n) \vec{\phi}(n + \hat{\mu}) \end{aligned} \quad (507)$$

Definition 6.3.2 (Backwards Wilson Line):

- $W_{-\mu\nu}(n) = W_{\mu}^{\dagger}(n - \hat{\mu})$

Definition 6.3.3 (plaquette):

- something of the form in this picture



- It is evaluated as

$$W_{\mu\nu}(N) \equiv W_{-\nu}(n + \hat{\nu}) W_{-\mu}(n + \hat{\mu} + \hat{\nu}) W_{\nu}(n + \hat{\mu}) W_{\mu}(N) \quad (508)$$

- If we write

$$W_{\mu}(n) = \exp(ia \underline{A}_{\mu}(n)) \quad (509)$$

, we can see how this would be related to the continuous version

Algebra: Evaluating Plaquettes

- How do we evaluate plaquettes?
 - Consider the *Campbell Baker Hausdorff* formula:

$$\exp(A) \exp(B) = \exp\left(A + B + \frac{1}{2}[A, B] + \dots\right) \quad (510)$$

- Applying this to our definition of $W_{\mu\nu}$,

$$\begin{aligned}
\ln W_{m\nu}(N) &= ia(\underline{A}_\mu(n) + \underline{A}_\nu(n + \hat{\mu}) - \underline{A}_\mu(n + \hat{\nu}) - \underline{A}_\nu(n)) \\
&+ \frac{a^2}{2} \{ [\underline{A}_\nu(N) + \underline{A}_\mu(n + \hat{\nu}), \underline{A}_\nu(n + \hat{\mu}) - \underline{A}_\mu(n)] \\
&- [\underline{A}_\nu(N), \underline{A}_\mu(n + \hat{\nu})] - [\underline{A}_\nu(n + \hat{\mu}), \underline{A}_\mu(n)] \} \\
&+ \mathcal{O}(a^3)
\end{aligned} \tag{511}$$

- Via a taylor expansion,

$$\underline{A}_\nu(n + \hat{\mu}) = \underline{A}_\nu(n) + a\partial_\mu \underline{A}_\nu(n) + \mathcal{O}(a^2) \tag{512}$$

,

$$\begin{aligned}
W_{\mu\nu}(n) &= \exp\{ia^2(\partial_\mu \underline{A}_\nu(n) - \partial_\nu \underline{A}_\mu(n)) + a^2[\underline{A}_\mu(n), \underline{A}_\nu(n)] + \mathcal{O}(a^3)\} \\
&= \exp\{ia^2 \underline{F}_{\mu\nu}(n) + \mathcal{O}(a^3)\}
\end{aligned} \tag{513}$$

- Consider the Yang Mills action:

$$S_{YM}[F_{\mu\nu}] = i \int d^4x \left(-\frac{1}{4g^2} (F_{\mu\nu}^a)^2 \right) \tag{514}$$

- We would like to discretize this

- At small a ,

$$W_{\mu\nu}(N) = \mathbb{I} + ia^2 \underline{F}_{\mu\nu}(N) - \frac{a^4}{2} \underline{F}_{\mu\nu}^2(N) + \mathcal{O}(a^6) \tag{515}$$

- Additionally, we can discretize the YM action by writing

$$S_{YM}[F_{\mu\nu}] = i \int d^4x \left(-\frac{1}{4g^2} (F_{\mu\nu}^a)^2 \right) = \frac{-ia^4}{4Ng^2} \sum_{n,\mu\nu} \text{tr}(F_{\mu\nu}^2) \tag{516}$$

- We finally obtain the lattice action:

$$S_{\text{lattice}}[W_{\mu\nu}] = \frac{-i}{2g^2N} \sum_{\nu,\mu} \text{Re}(\text{tr}(\mathbb{I} - W_{\mu\nu}(N))) \tag{517}$$

- For corr. functions, you would use this [action](#) in a path integral like

$$C(x) \langle \Omega | \mathcal{O}(0) \mathcal{O}(x) | \Omega \rangle = \int \mathcal{D}A_\mu \mathcal{D}\bar{u} \mathcal{D}u e^{iS} \mathcal{O}(0) \mathcal{O}(x) \tag{518}$$

However, it would give a rapid phase shift. Instead, you must perform a **wick rotation**

6.4. QCD Feynman Rules

- Now, we have the full lagrangian for QCD (the F-P) one allowing us to read off feynman rules

$$\begin{aligned}
\mathcal{L} &= -\frac{1}{4} (F_{\mu\nu}^a)^2 - \frac{1}{2\xi} (\partial_\mu A_\mu^a)^2 + \\
&(\partial_\mu \bar{c}^a)(\delta^{ac} \partial_\mu + g f^{abc} A_\mu^b) c^c + \bar{\psi}_i (i \not{\partial} \delta_{ij} + g A_\mu^a T_{ij}^a - m \delta_{ij}) \psi_j + \\
&[(\delta_{ki} \partial_\mu - ig A_\mu^a T_{ki}^a) \phi_i]^\dagger [(\delta_{kj} \partial_\mu - ig A_\mu^a T_{kj}^a) \phi_j] - m^2 \phi_i^\dagger \phi_i
\end{aligned} \tag{519}$$

- The kinetic terms for the lagrangian are

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4}(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \bar{\psi}_i(i\not{D} - m)\psi_i - \phi_i^*(\square + m^2)\phi_i - \bar{c}^a \square c^a \quad (520)$$

- The rules are as follows:

$$\mu; a \text{ --- } \nu; b = \frac{-g^{\mu\nu} + (1 - \xi)\frac{p^\mu p^\nu}{p^2}}{p^2} \delta^{ab} \quad (521)$$

- For ghosts,

$$\text{---} \Rightarrow \text{---} = \frac{\delta^{ab}}{p^2 + i\epsilon} \quad (522)$$

- For colored fermions,

$$i \longrightarrow j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon} \quad (523)$$

- For colored scalars,

$$i \text{ ----} \Rightarrow \text{----} j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon} \quad (524)$$

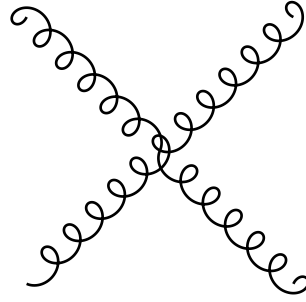
- The interaction terms (which are where this gets interesting) are the following:

$$\begin{aligned} \mathcal{L}_{\text{int}} = & -gf^{abc}(\partial_\mu A_\nu^a)A_\mu^b A_\nu^c - \frac{1}{4}g^2(f^{eab}A_\mu^a A_\nu^b)(f^{ecd}A_\mu^c A_\nu^d) \\ & + gf^{abc}(\partial_\mu \bar{c}^a)A_\mu^b c^c + gA_\mu^a \bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j \\ & + igA_\mu^a T_{ij}^a (\phi_i \overleftrightarrow{\partial}^\mu \phi_j) + g^2 \phi_i^* A_\mu^a T_{ik}^a T_{kj}^b A_\mu^b \phi_j \end{aligned} \quad (525)$$

- Trips g:

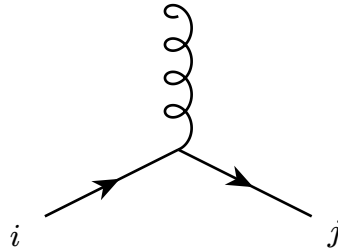
$$\begin{array}{c} \mu; a \\ k \\ \nu; b \text{ --- } p \text{ --- } q \\ \rho; c \end{array} = gA^{abc}[g^{\mu\nu}(k-p)^\mu + g^{\nu\rho}(p-q)^\mu + g^{\rho\mu}(q-k)^\nu] \quad (526)$$

- Quads g:



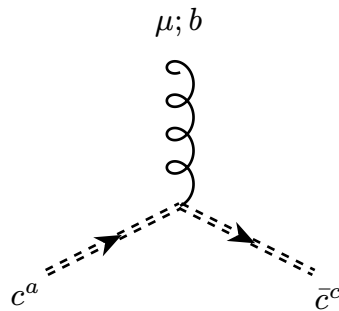
(527)

$$= -ig^2[f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ade}f^{bce}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma})]$$



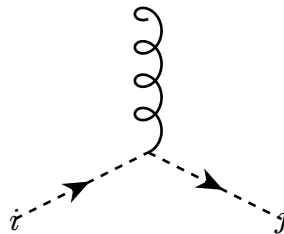
$$= ig\gamma^\mu T_{ij}^a$$

(528)



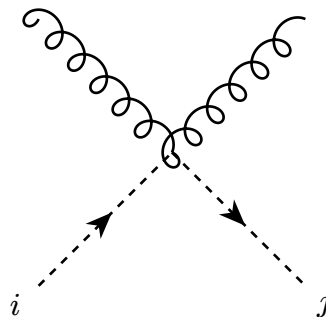
$$= -gf^{abc}p^\mu$$

(529)



$$= ig(k^\mu + q^\mu)T_{ij}^a$$

(530)



$$= ig^2 T_{ik}^a T_{kj}^b g^{\mu\nu}$$

(531)

6.5. Useful Math formalism (Appendix): Helicity Spinors (TODO)

Summary

- Helicity spinors are a way to write polarizations ε_μ and momentum p_μ in terms of a more natural way (bispinors)
 - This is more convenient specifically for calculating gluon scattering

- They are written in the weyl representation where ψ_α and $\tilde{\chi}^{\dot{\alpha}}$ are left and right handed spinors

- Notice that spin 1 fields transform in the $(\frac{1}{2}, \frac{1}{2})$ representation of the Lorentz group
 - This entails that we can write them as bispinors instead of 4-vectors
- Remember the mathematical relations for spinors in the weyl representation:
 - $\epsilon^{\alpha\beta}\psi_\alpha(x)\psi_\beta(x)$ and $\epsilon_{\dot{\alpha}\dot{\beta}}\tilde{\psi}^{\dot{\alpha}}\tilde{\psi}^{\dot{\beta}}$ are L.I.
- Some useful relations with these (and the pauli matrices) are

$$\begin{aligned} g^{\mu\nu}\sigma_\mu^{\alpha\dot{\alpha}}\sigma_\nu^{\beta\dot{\beta}} &= 2\epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}} \\ \epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}}\sigma^{\mu\dot{\beta}\beta} &= \bar{\sigma}^\mu_{\dot{\alpha}\alpha} \end{aligned} \quad (532)$$

- For the inner product, defined as $\psi\chi \equiv \epsilon^{\alpha\beta}\psi_\alpha(x)\chi_\beta(x)$
 - $\psi\chi = \chi\psi$
- All of these spinors are fields. What happens when we use constant spinors?

Definition 6.5.1 (Helicity Spinor):

- $\mathbb{R}$ or $\mathbb{C}$ doublets that transform in the $(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$ representation of the Lorentz Group

- The inner products for these are

$$\langle\lambda\chi\rangle = \epsilon^{\alpha\beta}\lambda_\alpha\chi_\beta = \lambda_\alpha\chi^\alpha = -\lambda^\alpha\chi_\alpha \quad (533)$$

$$[\lambda\chi] = \epsilon_{\alpha\beta}\tilde{\lambda}^\alpha\tilde{\chi}^\beta = \tilde{\lambda}^\alpha\tilde{\chi}_\alpha = -\tilde{\chi}_\alpha\tilde{\lambda}^\alpha \quad (534)$$

- These products have some identities due to them commuting:

$$\begin{aligned} \langle\lambda\chi\rangle &= -\langle\chi\lambda\rangle, [\lambda\chi] = -[\chi\lambda] \\ [\lambda\lambda] &= \langle\lambda\lambda\rangle = 0 \end{aligned} \quad (535)$$

6.5.1. Momentum

- For bispinor representation of momentum, we have:

$$p_{\alpha\dot{\alpha}} \equiv \sigma_{\alpha\dot{\alpha}}^\mu p_\mu = \begin{pmatrix} p^0 - p^3 & -p^1 + ip^2 \\ -p^1 - ip^2 & p^0 + p^3 \end{pmatrix} \quad (536)$$

- **Relations** with these spinors are:

$$\begin{aligned} p^{\alpha\dot{\alpha}} &\equiv \sigma_\mu^{\alpha\dot{\alpha}} p^\mu, \quad p_{\alpha\dot{\alpha}} = \bar{\sigma}_{\alpha\dot{\alpha}}^\mu p_\mu, \\ p^\mu &= \frac{1}{2}\bar{\sigma}^{\mu\alpha\dot{\alpha}} p_{\alpha\dot{\alpha}}, \quad p^\mu = \frac{1}{2}\bar{\sigma}_{\alpha\dot{\alpha}}^\mu p^{\alpha\dot{\alpha}}. \end{aligned} \quad (537)$$

- If $p^{\alpha\dot{\alpha}}$ is lightlike (that is, $\det(p^{\alpha\dot{\alpha}}) = 0$), we can express it in terms of an outer product

$$p^{\alpha\dot{\alpha}} = \lambda^\alpha \tilde{\chi}^{\dot{\alpha}} \quad (538)$$

$$\begin{aligned}\lambda_\alpha &= \frac{z}{\sqrt{p^0 - p^3}} \begin{pmatrix} p^0 - p^3 \\ -p^1 - ip^2 \end{pmatrix}, \\ \tilde{\lambda}^{\dot{\alpha}} &= \frac{z^1}{\sqrt{p^0 - p^3}} (p^0 - p^3, -p^1 + ip^2)\end{aligned}\tag{539}$$

6.5.1.1. Little Group

- These momenta bispinors are invariant under $\lambda_\alpha \rightarrow z\lambda_\alpha$ and $\tilde{\lambda}^{\dot{\alpha}} \rightarrow \frac{1}{z}\tilde{\lambda}^{\dot{\alpha}}$
 - This is equivalent to a phase shift $z = e^{i\phi}$ if the momenta are real

6.5.1.2. Connecting Momentum and Helicity Spinors

- Notice that the dot product

$$\begin{aligned}p \cdot q &= \frac{1}{4} g_{\mu\nu} \sigma_{\alpha\dot{\alpha}}^\mu \sigma_{\beta\dot{\beta}}^\nu \lambda^\alpha \tilde{\lambda}^{\dot{\alpha}} \chi^\beta \tilde{\chi}^{\dot{\beta}} \\ &= \frac{1}{2} \varepsilon_{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} \lambda^\alpha \tilde{\lambda}^{\dot{\alpha}} \chi^\beta \tilde{\chi}^{\dot{\beta}} = \frac{1}{2} \langle \chi \lambda \rangle [\chi \lambda]\end{aligned}\tag{540}$$

- Therefore, we obtain that $\langle \chi \lambda \rangle$ and $[\chi \lambda]$ are the square roots of inner products up to a phase shift

$$\begin{aligned}\langle \lambda \chi \rangle &= \sqrt{2p \cdot q} e^{i\phi} \\ [\chi \lambda] &= \sqrt{2p \cdot q} e^{-i\phi}\end{aligned}\tag{541}$$

- This allows us to write

$$\lambda_\alpha = p\rangle, \quad \lambda_\alpha = \langle p, \quad \tilde{\lambda}^{\dot{\alpha}} = [p, \quad \tilde{\lambda}_{\dot{\alpha}} = p]\tag{542}$$

As for our momentum,

$$p^{\alpha\alpha} = p\rangle[p, p_{\alpha\alpha} = p]\langle p\tag{543}$$

- The dot products of momenta are

$$\begin{aligned}q \cdot p &= q^\mu p_\mu = \frac{1}{2} q_{\alpha\dot{\alpha}} p^{\dot{\alpha}\alpha} \\ &= \frac{1}{2} \text{tr}\{q\} \langle qp \rangle [p] \\ &= \frac{1}{2} \langle qp \rangle [qp]\end{aligned}\tag{544}$$

- Here is another identity that is useful:

$$\begin{aligned}\sum_i p_i &= 0 \\ \sum_j \lambda_\alpha^j \lambda_\alpha^j &= 0 \\ \sum_{j=1}^n j\rangle [j=1\rangle [1+2\rangle [2+\dots+n\rangle [n=0\end{aligned}\tag{545}$$

- To help us with being terse, we write $i\rangle \equiv p_i\rangle$, etc.

- Our conservation of momentum

$$\sum_j \langle ij \rangle [jk] = 0 \quad (546)$$

- Since the spinors are two dimensions, we can write them in terms of two other spinors (a *linear combination*)

$$1\rangle = \frac{\langle 13 \rangle}{\langle 23 \rangle} 2\rangle - \frac{\langle 12 \rangle}{\langle 23 \rangle} 3\rangle \quad (547)$$

6.5.2. Polarizations

- All polarizations satisfy the relation that

$$\varepsilon_\mu^* \varepsilon^\mu = -1, \quad p_\mu \varepsilon^\mu = 0 \quad (548)$$

- Notice that $\varepsilon_\mu \varepsilon^\mu = 0$

Example 6.5.2.1 (Helicity Polarizations):

- If the momentum is $p^\mu = (E, 0, 0, E)$ then we obtain polarizations

$$\begin{aligned} \epsilon_+^\mu &= \frac{1}{\sqrt{2}}(0, 1, i, 0), \\ \epsilon_-^\mu &= \frac{1}{\sqrt{2}}(0, 1, -i, 0) \end{aligned} \quad (549)$$

- as seen before, they are the *helicity polarizations*

Definition 6.5.2.1 (Reference Momentum):

- In order to write the polarizations in terms of the momentum, we need a reference momentum. In the case of gluon scattering which this is a very useful tool for, we take the reference momentum to be the momentum of the other gluons.
- This is denoted by r^μ

- Writing

$$p^{\alpha\dot{\alpha}} = p\rangle[p], \quad r^{\alpha\dot{\alpha}} = r\rangle[r] \quad (550)$$

- Thus, the polarizations in the bispinor format is written as

$$\begin{aligned} [\varepsilon_p^-(v)]^{\alpha\dot{\alpha}} &= \sqrt{2} \frac{p\rangle[r]}{[pr]} \\ [\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} &= \sqrt{2} \frac{r\rangle[p]}{\langle rp} \end{aligned} \quad (551)$$

$$\begin{aligned}
\varepsilon_p^-(v) \cdot \varepsilon_p^+(v) &= \frac{1}{2} [\varepsilon_p^-(v)]_{\dot{\alpha}\alpha} [\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} \\
&= \frac{1}{2} \frac{2}{[pr]\langle rp\rangle} \text{tr}\{r\}\langle pr\rangle[p] = -1
\end{aligned} \tag{552}$$

- since $\langle pp\rangle[pp] = 0$, we have that $\varepsilon^+\varepsilon^+ = \varepsilon^-\cdot\varepsilon^- = 0$

Part VII

Electroweak (TODO)

Part VIII

Renormalization (TODO)

“The shell game that we play ... is technically called ‘renormalization’. But no matter how clever the word, it is still what I would call a dippy process! Having to resort to such hocus-pocus has prevented us from proving that the theory of quantum electrodynamics is mathematically self-consistent. It’s surprising that the theory still hasn’t been proved self-consistent one way or the other by now; I suspect that renormalization is not mathematically legitimate.”

— Richard Feynman [3]

8.1. Stereotypical Examples of (Counterterm) Renormalization

- Renormalization is the process of removing infinities by putting them in “bare” variables
- The three most stereotypical examples of renormalization of quantum electrodynamics (and *in general*) are the following
 1. Vacuum Polarization
 2. Electron Self Interaction
 3. The anomalous magnetic moment
- There are two types of renormalization:
 1. Counterterm renormalization
 2. Wilsonian renormalization
 - Arguably the coolest type of renormalization but it seems like it’s completely different than traditional counterterm renormalization

8.1.1. Mass Renormalization

- This is equivalent to the Electron self interaction

Note (Why Renormalize?): *In general, divergences form due to loop diagrams which give infinite integrals*

Definition 8.1.1.1 (Superficial Degree of Divergence):

- This quantity is a very standard quantity which counts essentially how badly something diverges.
 - In QED, define the following quantities :

$$\begin{aligned}
 N_e &= \# \text{ elec} \\
 N_\gamma &= \# \text{ phot} \\
 P_e &= \# \text{ electron prop.} \\
 P_\gamma &= \# \text{ photon prop.} \\
 V &= \# \text{ vertices} \\
 L &= \# \text{ loops} \\
 D &= 4L - P_e - 2P_\gamma
 \end{aligned} \tag{553}$$

- The degree of divergence is $= 2(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma$

- Now we will make sense of this quantity

Definition 8.1.1.2 (PV Regulation):

- Consider a divergent integral, such as

$$\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2} \quad (554)$$

. The superficial degree of divergence is the difference in powers of k in the numerator and denominator.

▸ That is, $D_{\text{ex.}} = 2$.

- PV regulation works by giving the integral a cutoff

$$\int^\Lambda \frac{d^4k}{(2\pi)^4} \frac{1}{k^2} \propto \Lambda^2 \quad (555)$$

- We would then enter counterterms into the lagrangian which would precisely (or not so precisely in some schemes like $\overline{\text{MS}}$ as you will see) cancel them out

Note (Degree of Div.):

- Now we see that, in fact, the degree of divergence counts the power of Λ . If $D = 0$, it is proportional to $\ln \Lambda$

- There is also another way to evaluate the integrals which shows us a divergence

Definition 8.1.1.3 (DimReg):

- DimReg (dimension regulation) is a method of scanning for divergences by setting $d = 4 - \varepsilon$ and taking $\lim_{\varepsilon \rightarrow 0}$ of the integral
 - You only take this limit **after** you subtract the divergence

- Lets look at the actual electron self energy to see what we can analyze
- At 1 loop,

$$\langle \Omega | T \bar{\psi}(x) \psi(x) | \Omega \rangle = \quad (556)$$

8.1.2. Vacuum Polarization

- This is the photon splitting into an electron and a positron and then them combining again into a photon
- The vertex evaluates to

$$\begin{aligned} \text{feyn diag} &= -(-ie)^2 \int \frac{d^4k}{(2\pi)^4} \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2} \\ &\times \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] \end{aligned} \quad (557)$$

Algebra: Trace

$$\text{tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] = 4[-p^\mu k^\nu - k^\mu p^\nu + 2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + m^2)] \quad (558)$$

- We define the loop in this diagram as

$$i\Pi_2^{\mu\nu} \equiv \int \frac{d^4 k}{(2\pi)^4} \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2} \times \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] \quad (559)$$

- With our trace, we get

$$i\Pi_2^{\mu\nu} = -4e^2 \int d^4 k \frac{2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + m^2)}{[(p-k)^2 - m^2 + i\epsilon][k^2 - m^2 + i\epsilon]} \quad (560)$$

Trick (Mildly Useful Insight)

- We can parametrize this using a trick called feynman parameterization where we take $k^\mu \rightarrow k^\mu + p^\mu(1-x)$ and integrate

Algebra: Pi tensor

$$\begin{aligned} \Pi_2^{\mu\nu} &= 4ie^2 \int d^4 k \int_0^1 dx \frac{2k^\mu k^\nu - g^{\mu\nu}[k^2 - x(1-x)p^2 - m^2]}{[k^2 + p^2 x(1-x) - m^2]^2} \\ &= -8p^2 g^{\mu\nu} \frac{e^2}{(4\pi)^{d/2}} \Gamma(2-d/2) \mu^{4-d} \\ &\quad \times \int_0^1 dx x(1-x) \left(\frac{1}{m^2 - p^2 x(1-x)} \right)^{2-d/2} \end{aligned} \quad (561)$$

- This yields

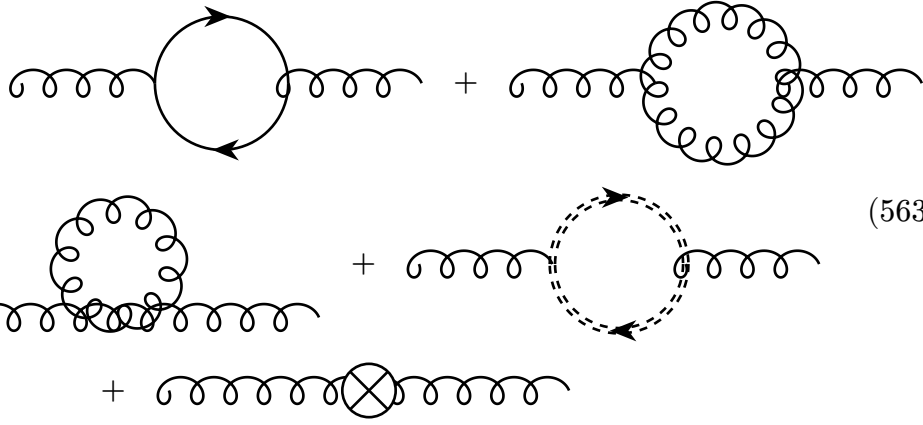
$$\Pi_2^{\mu\nu} = -\frac{e^2}{2\pi^2} p^2 g^{\mu\nu} \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln \left(\frac{\tilde{\mu}^2}{m^2 - p^2 x(1-x)} \right) + O(\epsilon) \right] \quad (562)$$

8.2. Running Coupling of QCD

- This in no way is the order that you should learn QFT but it is a convenient way to keep notes
- How does the strong force coupling constant α_s
 - Spoiler alert, it runs the opposite way that α_{QED} does.
- In order to get the coupling, we need to renormalize it
- In order to renormalize QCD, we need to calculate the loops
- ∴ we should calculate loops lmao

8.2.1. Vac pol

- There are 4 + 1 counterterm graphs for the QCD vacuum polarization

1 loop vac pol =  (563)

- First, consider the gluon $\rightarrow$ quarks $\rightarrow$ gluon diagram

$$i\mathcal{M}_p^{ab\mu\nu} = \text{diagram} \quad (564)$$

$$= \text{tr}(T^a T^b) (ig)^2 \int d^4k \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2} \text{tr} [\gamma^\mu (\not{p} - \not{k} + m) \gamma^\nu (\not{k} + m)]$$

- This is the exact same as QED except there is a color factor $\text{tr}[T^a T^b] = T_F \delta^{ab} = \frac{1}{2} \delta^{ab}$
- The answer is of the form

$$\mathcal{M}_F^{ab\mu\nu} = -g^2 (g^{\mu\nu} p^2 - p^\mu p^\nu) \delta^{ab} \Pi_2(p^2) \quad (565)$$

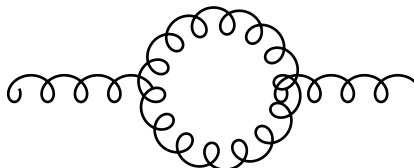
- Then,

$$\mathcal{M}_F^{ab\mu\nu} = -\delta^{ab} T_F \frac{g^2}{2\pi^2} (p^2 g^{\mu\nu} - p^\mu p^\nu) \times \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln \frac{\tilde{\mu}^2}{m^2 - p^2 x(1-x)} + O(\epsilon) \right] \quad (566)$$

- Notice how the pole and the coefficient of the log is indep. of the quark mass
- In the massless quark limit,

$$\mathcal{M}_F^{ab\mu\nu} = \delta^{ab} T_F \left(\frac{g^2}{16\pi^2} \right) (p^2 g^{\mu\nu} - p^\mu p^\nu) \left[-\frac{8}{3\epsilon} - \frac{20}{9} - \frac{4}{3} \ln \frac{\mu^2}{-p^2} \right] \quad (567)$$

- Now let's look at the double gluon bubble (the vertices are triple g)

•  (568)

$$= \frac{g^2}{2} \int d^4k \frac{-i}{k^2} \frac{-i}{(k-p)^2} f^{ace} f^{bdf} g^{cf} \delta^{ed} N^{\mu\nu}$$

- The numerator $N^{\mu\nu}$ is equivalent to

$$\begin{aligned}
N^{\mu\nu} &= [g^{\mu\alpha}(p+k)^\rho + g^{\alpha\rho}(p-2k)^\mu + g^{\rho\mu}(k-2p)^\alpha] \\
&\times g^{\alpha\beta}g^{\rho\sigma}[g^{\nu\beta}(p+n)^\sigma - g^{\beta\sigma}(2k-p)^\nu - g^{\sigma\nu}(2p-k)^\beta]
\end{aligned} \tag{569}$$

Algebra: Feynman Parameterization

- Parameterizing with feynman parameters

$$\frac{1}{k^2(p-k)^2} = \int_0^1 dx \frac{1}{[(1-x)k^2 + x(p-k)^2]^2} \tag{570}$$

- Taking $k \rightarrow k + xp$ and completing the square, we obtain that

$$i\mathcal{M}_3^{ab\mu\nu} = \frac{g^2}{2} \int_0^1 dx \int d^4k \frac{1}{(k^2 - \Delta)^2} f^{acd} f^{bcd} N^{\mu\nu} \tag{571}$$

where $\Delta = x(x-1)p^2$

- Now, our numerator is

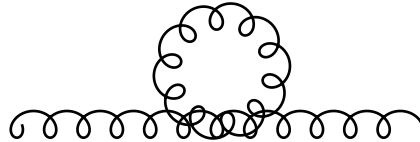
$$\begin{aligned}
N^{\mu\nu} &= 2k^2g^{\mu\nu} - (6-4d)k^\mu k^\nu - (6(x^2-x+1) - d(1+2x)^2)p^\mu p^\nu \\
&\quad - (2x^2-2x+5)p^2g^{\mu\nu} \\
&\quad - \underbrace{(2-4x)g^{\mu\nu}(k \cdot p) + (2d-3)(2x-1)(k^\mu p^\nu + k^\nu p^\mu)}_{\text{vanishes since are odd in } k \rightarrow -k}
\end{aligned} \tag{572}$$

- In DimReg, we can replace the $k^\mu k^\nu$ terms with $\frac{1}{d}k^2g^{\mu\nu}$

- Now, we have

$$\begin{aligned}
\mathcal{M}^{ab\mu\nu} &= -\frac{g^2\mu^{4-d}}{(4\pi)^{d/2}}\delta^{ab} \int_0^1 dx \left(\frac{1}{\Delta}\right)^{2-d/2} \{g^{\mu\nu}3(d-1)\Gamma(1-d/2)\Delta \\
&\quad + p^\mu p^\nu [6(x^2-x+1) - d(1-2x)^2]\Gamma(2-d/2) \\
&\quad + g^{\mu\nu}p^2[(-2x^2+2x-5)\Gamma(2-d/2)]\}
\end{aligned} \tag{573}$$

- Our next diagram that we have to evaluate is



$$\equiv i\mathcal{M}_4^{ab\mu\nu} \tag{574}$$

- This is proportional to $\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2}$ which vanishes in DimReg but it is still fairly reasonable to actually evaluate the diagram just to see (in PV the divergence is quadratic)

$$\begin{aligned}
i\mathcal{M}_4^{ab\mu\nu} &= -\frac{ig^2}{2}\mu^{4-d} \int d^dk \frac{-ig^{\rho\sigma}\delta^{cd}}{k^2+i\epsilon} \times [f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}) \\
&\quad + f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma}) + f^{ade}f^{bce}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho})] \\
&= -g^2\delta^{ab}g^{\mu\nu}C_A(d-1)\mu^{4-d} \int d^dk \frac{1}{k^2+i\epsilon}
\end{aligned} \tag{575}$$

- Multiplying by $\frac{(p-k)^2}{(p-k)^2}$ yields $N^{\mu\nu} = g^{\mu\nu}(p-k)^2$
- Parametrizing gives

$$\begin{aligned}
\mathcal{M}_\mu^{ab\mu\nu} &= -g^2 \delta^{ab} C_A \frac{m^{4-d}}{(4\pi)^{d/2}} g^{\mu\nu} \int_0^1 dx \left(\frac{1}{\Delta}\right)^{2-d/2} \\
&\times (d-1) \left[-\frac{d}{2} \Gamma(1-d/2) \Delta + (1-x)^2 p^2 \Gamma(2-d/2) \right] \\
&= 0
\end{aligned} \tag{576}$$

- Now, we evaluate the diagram for the ghost

$$\begin{aligned}
i\mathcal{M}_{\text{gh}}^{ab\mu\nu} &= \text{Diagram: A ghost loop (dashed line) with two external gluon lines (curly lines).} \\
&= (-1) g^2 \int d^4 k \frac{i}{(k-p)^2} \frac{i}{k^2} f^{cad} k^\mu f^{dbc} (k-p)^\nu
\end{aligned} \tag{577}$$

- This implies that

$$\begin{aligned}
\mathcal{M}_{\mu\nu}^{ab} &= g^2 \frac{\mu^{4-d}}{(2\pi)^{d/2}} \delta^{ab} C_A \int_0^1 dx \left(\frac{1}{\Delta}\right)^{2-d/2} \\
&\times \left\{ g_{\mu\nu} \left[\frac{1}{2} \Gamma(1-d/2) \Delta \right] + p_\mu p_\nu [x(1-x)] \Gamma(2-d/2) \right\}
\end{aligned} \tag{578}$$

- Now, we can evaluate the full vacuum polarization that we have wanted to evaluate:

Algebra: Summing Diagrams + Tricks

$$\begin{aligned}
\mathcal{M}_{\text{glue}}^{ab\mu\nu} &= \mathcal{M}_3^{ab\mu\nu} + \mathcal{M}_4^{ab\mu\nu} + \mathcal{M}_{gh}^{ab\mu\nu} \\
&= \delta^{ab} C_A g^2 \int_0^1 dx \left(\frac{1}{\Delta}\right)^{2-d/2} \left\{ g^{\mu\nu} \Delta \left(\frac{3-3d}{d} + (d-1) + \frac{1}{d} \right) \frac{d}{2} \times \Gamma(1-d/2) \right\} \\
&\quad + p^\mu p^\nu [-3(x^2 - x + 1) + d/2(1-2x)^2 + x(1-x)] \Gamma(2-d/2) \\
&\quad + g^{\mu\nu} p^2 [(x^2 - x + 5/2) - (1-x)^2(d-1)] \Gamma(2-d/2)
\end{aligned} \tag{579}$$

- There is a quadratic divergence from $\Gamma(1 - \frac{d}{2})$
 - it has a coefficient of $\frac{1}{d}(d-2)^2$ which cancels
- Via Gamma function factorial identities (i.e. $\Gamma(1 - \frac{d}{2})(d-2) = 2\Gamma(2 - \frac{d}{2})$), we obtain

$$\begin{aligned}
\mathcal{M}_{\text{glue}}^{ab\mu\nu} &= \delta^{ab} C_A g^2 \frac{\mu^{4-d}}{(4\pi)^{d/2}} \int_0^1 dx \left(\frac{1}{\Delta}\right)^{2-d/2} \Gamma(2-d/2) \left\{ g^{\mu\nu} p^2 \left[(1-2x+3x-1)d + x(4x-5) + \frac{7}{2} \right] \right. \\
&\quad \left. + p^\mu p^\nu [d/2(1-2x)^2 - 4x^2 + 4x - 3] \right\}
\end{aligned} \tag{580}$$

- Setting $d = 4 - \varepsilon$

$$\mathcal{M}_{\text{glue}}^{ab\mu\nu} = C_A \delta^{ab} \frac{g^2}{16\pi^2} (g^{\mu\nu} p^2 - p^\mu p^\nu) \times \left(\frac{10}{3\varepsilon} + \frac{31}{9} + \frac{5}{3} \ln \frac{\tilde{\mu}^2}{p^2} + O(\varepsilon) \right) \tag{581}$$

- Adding in fermion flavor and colors,

$$\begin{aligned}\mathcal{M}^{ab\mu\nu} &= \delta^{ab} \frac{g^2}{16\pi^2} (g^{\mu\nu} p^2 - p^\mu p^\nu) \\ &\times \left[C_A \left(\frac{10}{3\epsilon} + \frac{5}{3} \ln \frac{\tilde{\mu}^2}{-p^2} \right) - n_f T_F \left(\frac{8}{3\epsilon} + \frac{4}{3} \ln \frac{\tilde{\mu}^2}{-p^2} \right) \right]\end{aligned}\quad (582)$$

- We can simply introduce a counterterm to remove this, which you will see

8.2.2. Other Renormalization

- The renormalized QCD lagrangina is

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4} Z_3 (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)^2 \\ &+ Z_2 \bar{\psi}_i (i \not{\partial} - Z_m m_p) \psi_i - Z_3 \bar{c}^a \square c^a \\ &- g_R Z_{A^3} f^{abc} (\partial_\mu A_\nu^a) A_\mu^b A_\nu^c \\ &- \frac{1}{4} g_R^2 Z_{A^4} (f^{eab} A_\mu^a A_\nu^b) (f^{ecd} A_\mu^c A_\nu^d) \\ &+ g_R Z_1 A_\mu^a \bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j \\ &+ g_R Z_{1c} f^{abc} (\partial_\mu \bar{c}^a) A_\mu^b c^c\end{aligned}\quad (583)$$

Trick (Mildly Useful Insight)

$$\bullet \quad \left[\int d^d k \frac{1}{k^4} \right]_{\text{UV div}} = \frac{i}{8\pi^2} \frac{1}{\epsilon} \quad (584)$$

extracts the UV div. from a $\frac{1}{k^4}$ integral.

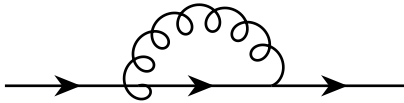
8.2.2.1. 2 Point Functions

- First, let's renormalize 2 point functions.
- For vacuum polarization,

$$\mathcal{M}^{ab\mu\nu} = \delta^{ab} (g^{\mu\nu} p^2 - p^\mu p^\nu) \left\{ \frac{g^2}{16\pi^2} \left[C_A \left(\frac{10}{3\epsilon} \right) - n_f T_F \left(\frac{8}{3\epsilon} \right) \right] - \delta_3 \right\} + \text{finite} \quad (585)$$

$$\therefore \delta_3 = \frac{1}{\epsilon} \frac{g^2}{16\pi^2} \left[\frac{10}{3} C_A - \frac{8}{3} n_f T_F \right] \quad (586)$$

- Now, let's look at the quark self energy.


(587)

Part VIII

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